STRAND

Planet Earth and Beyond

20 The Earth as a system

Key questions

- What are the different parts of the Earth?
- How do these parts interact?
- Why can we refer to the Earth as a system?

Keywords

- biosphere
- hydrosphere
- lithosphere
- atmosphere
- yield
- pollution

In Grade 7 you learnt about the relationship between the Earth and the Sun and the importance of the Sun for life on Earth. In Grade 8 you looked at the relationship between the Earth and other planets in our solar system. This year we will look at the Earth as a system and all the parts of this system.

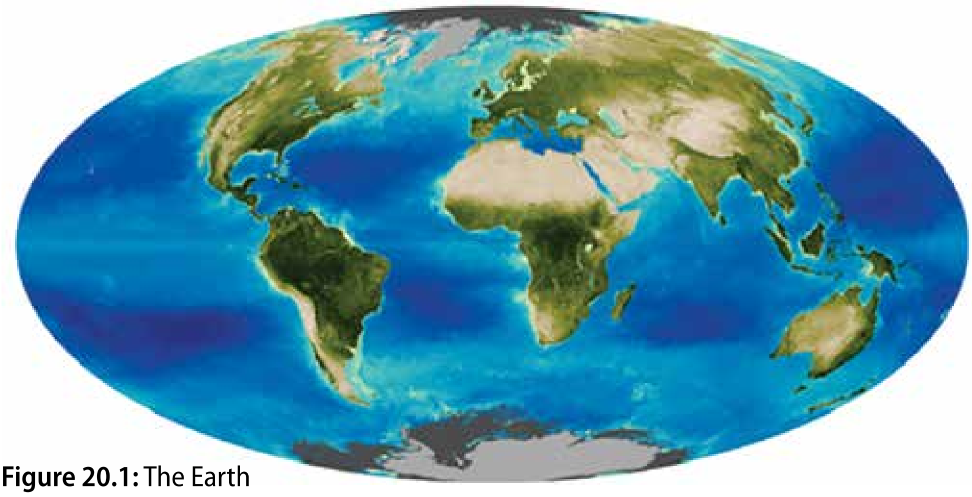


Figure 20.1: The Earth

20.1 Spheres of the Earth

You have learnt about systems and cycles throughout your studies of Natural Sciences over the past 5 years. For example, you have learnt about the life cycle of a butterfly, energy systems in food webs or electric circuits, and the solar system. Much of what we observe in nature is part of one or many systems or cycles. In this unit we are going to learn about the Earth as a closed system and the four different parts (spheres) of this system.

Did you know?

Scientists built a self-contained facility called Biosphere 2, to study the interactions between living things and the environment.

Earth's four spheres

The Earth is made up of four systems, or spheres. The biosphere (life), the lithosphere (land), the hydrosphere (water) and the atmosphere (air). On Earth land, water, air and life interact with one another. As humans, we are also part of this interaction. There is a fine balance between these four systems – if the one becomes altered, it has an effect on all the others.

The Biosphere

The biosphere includes all life on Earth – plants, animals and humans. Most of what is studied in *Life and Living* is about the biosphere. The biosphere also includes life in the oceans, and under the soil.

For example, bacteria living on decaying plant material and the smallest sea creatures and plants are part of the biosphere. Almost all the life on the planet is found between 3 metres below the surface of the Earth, up to 30 metres above the ground, and in the top 200 metres of the oceans.

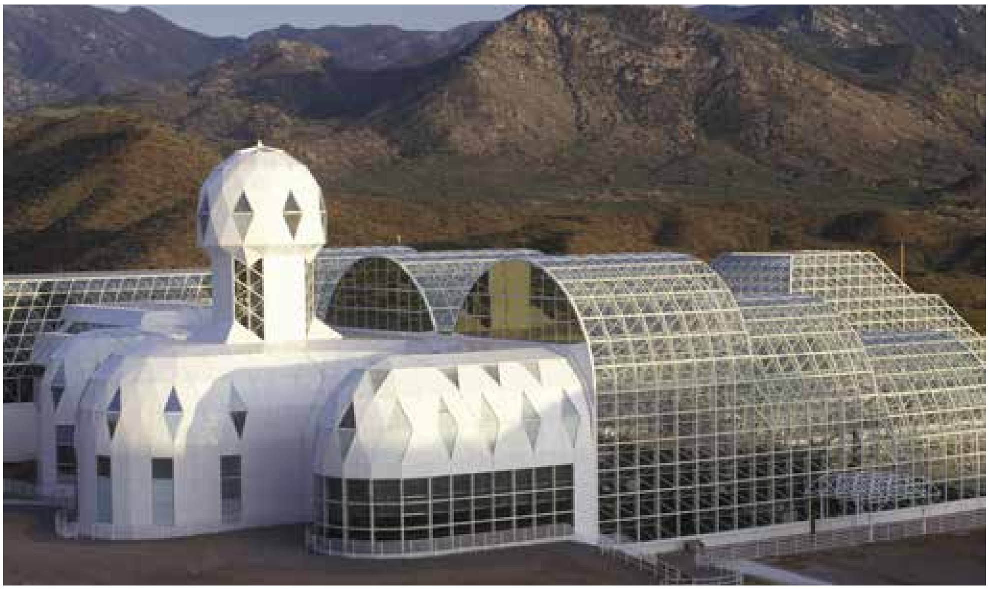
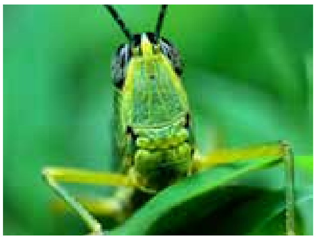


Figure 20.2: Biosphere 2 is a man-made research centre in America, in the Arizona Desert, where scientists have built a large enclosed artificial biosphere.

All living things and their habitats form part of the biosphere. The following photographs provide examples of different organisms in their habitats, living in the biosphere.



Grasshopper.

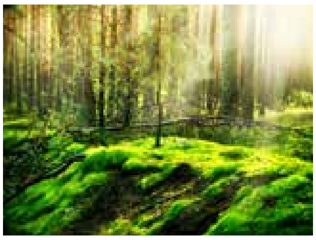
Dolphins in the sea.

Escherichia coli bacteria.

Sugarcane fields

An earthworm in the soil.

Limpets in a rock pool.



Moss in a forest.

Mushrooms in a field.

Phytoplankton in the sea.

Children at school.

A blue crane on the river's edge.

Figure 20.3: Organisms in their habitats.

The hydrosphere

The hydrosphere includes all water on the planet – the oceans, lakes, rivers, groundwater, rain, clouds, glaciers and ice caps. About 70% of the surface of the Earth is covered with water. The oceans contain most of this water, with only a small portion of it being fresh water. All the water on Earth forms part of the hydrosphere.

An iceberg.

A river.

Clouds.

The sea.

Figure 20.4: All the water on Earth forms part of the hydrosphere.

> **Did you know?**
>
> The names of the four spheres are derived from the Greek words for stone (*litho*), air (*atmo*), water (*hydro*), and life (*bio*).

The atmosphere

The atmosphere includes all the air above the surface of the Earth all the way to space. All the gases that are present in the air are included in the atmosphere. Most of the atmosphere is found close to the surface of the Earth, where the air is most dense. The air contains 79% nitrogen, less than 21% oxygen, and a small amount of carbon dioxide and other gases. We will look more closely at the atmosphere later on in this unit.

> **Did you know?**
>
> The first person to use the term 'biosphere' was the geologist Eduard Suess in 1875 when he wrote a definition for the biosphere as 'the place on Earth's surface where life dwells'.

Figure 20.5: The top of Earth's atmosphere.

Figure 20.6: The region of space occupied by Earth's atmosphere.

The lithosphere

The lithosphere includes the Earth's crust and the upper part of the mantle. All mountains, rocks, soil and minerals included in the Earth's crust are part of the lithosphere. Even the seafloor is part of the lithosphere, because it is also made up of sediments of sand and rock. We will look more closely at the lithosphere in the next section.

All the rocks, soil and sand on Earth form part of the lithosphere, as shown in the following photographs.

Rock formations.

Sand dunes.

Soil.

Mountains.

Minerals.

Seafloor.

Figure 20.7: All the rocks, soil and sand on Earth form part of the lithosphere.

The following collage shows the four spheres of the Earth.

Figure 20.8: The four spheres of the Earth.

Did you know?

The total mass of the hydrosphere is approximately $1,4 \times 10$ tonnes! The volume of one tonne of water is approximately 1 cubic metre (this is about 900 A4 textbooks).

Take note

The word 'sphere' is used in Mathematics to describe a round shape. The Earth has the shape of a sphere. When we talk about the four spheres of the Earth, we do not mean a ball shape, but rather we refer to the touching and overlapping layers within Earth.

ACTIVITY Exploring the spheres of the Earth

Instructions

1. Find an example on your school ground or at home where all four spheres are present, for example a tree growing in your garden.
2. Describe the location and what you have included in your example.
3. Identify each of the spheres in a short paragraph.
4. Your teacher might also ask you to present your example as a poster with illustrations and short descriptions, identifying each sphere.

Interaction between the spheres

The different spheres of the Earth are closely linked and interact with each other. Let's investigate this in the following activity.

ACTIVITY Interaction between the spheres

Instructions

1. Study the photo of thorn trees on the savannah.
2. Answer the questions that follow.

Figure 20.9: Thorn trees.

Questions

1. Identify the four different spheres of the Earth in the example.
2. What will happen if the trees do not get enough water?
3. Describe the interaction between the hydrosphere and the biosphere in this example.
4. What will happen if the carbon dioxide levels change dramatically?
5. Describe the interaction between the atmosphere and the biosphere in this example.

6. Is there any interaction between the lithosphere and the hydrosphere in this example?
7. Use the example you have chosen in the previous activity (Exploring the spheres of the Earth) and describe three different interactions between the different spheres.

There is an interaction between the tree and other plants (biosphere) and the air (atmosphere) as they use carbon dioxide from the air during photosynthesis and give off oxygen. There is an interaction between the plants (biosphere) and the soil (lithosphere) when they absorb water (hydrosphere) and minerals (lithosphere) from the soil (lithosphere). The soil is also used to anchor the plants. The tree (biosphere) produces flowers and then fruit. Animals eat the fruit and the leaves of the trees and other plants.

ACTIVITY Identifying the interactions of the spheres on Earth

Instructions

1. The following picture is of the dam wall that was built for the Gariep Dam on the border between the Free State and the Eastern Cape. The wall is used to generate hydroelectric power, as we learnt in *Energy and Change*.
2. Answer the questions below.

Figure 20.10: Gariep Dam in the Orange River.

Questions

1. Discuss in pairs all the possible interactions between the spheres of the Earth.
2. Work on your own to complete the following map. Write a description of the interaction on each of the arrows. One example has been done for you:

Take note

Think back to the definition of a system which we discussed in Energy and Change. A system is a set of parts working together where a change in one part affects other parts. This also applies when talking about the Earth as a whole system.

There is an interaction between the lithosphere and the atmosphere in that the wind (moving air) will cause erosion of the rocks surrounding the dam. Where possible include more than one interaction on the arrow linking the spheres.

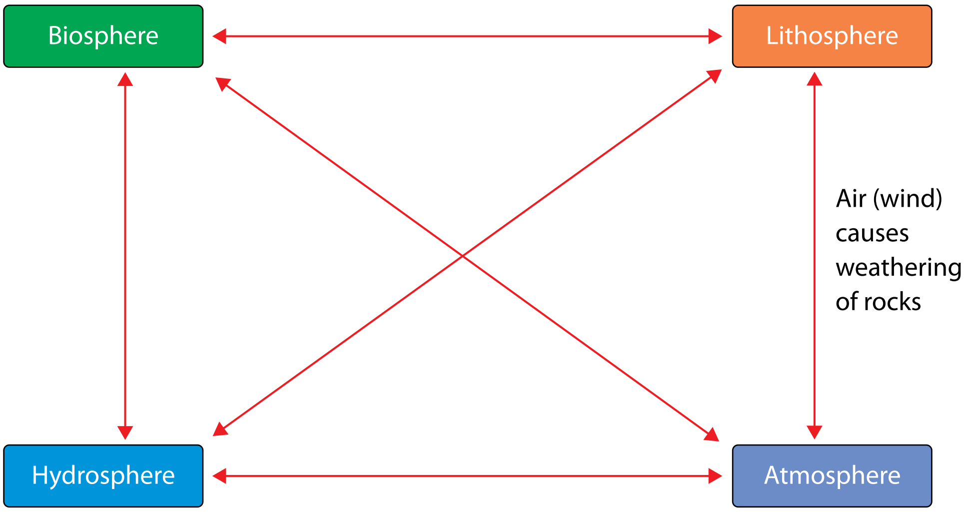


Figure 20.11: Interaction between the lithosphere and the atmosphere

The pictures below show how crops are harvested. The process of growing and harvesting crops are good examples of how the different spheres of the Earth interact. The hydrosphere provides water for the crops to grow. The soil provides minerals for the crops to give a good yield. The air provides carbon dioxide to the crops for photosynthesis and in return the plants give off oxygen to the air. The people (biosphere) make use of the materials from the lithosphere to build machinery or make sharp tools (metal from the lithosphere) for cutting wheat for example. Many interactions play a role in ensuring a healthy crop.

Figure 20.12: Growing wheat in fields.

Figure 20.13: A harvester cutting and gathering the wheat.

Upsetting the balance in the ecosystem

Let's look at our example of the thorn trees in a savannah ecosystem again. If the balance in any part of the system is changed, it affects the whole system. For example, if there is not enough water, the tree won't flourish and produce fruit (in this case, seed pods). If the air is polluted, it affects the availability of carbon dioxide to the tree. If there are not sufficient minerals in the soil, the plants cannot grow.

ACTIVITY Upsetting the balance

Figure 20.14: Interactions between the spheres of Earth.

Questions

1. Identify the four spheres of the Earth from the photograph.
2. Predict what the influence of the following scenarios on each sphere would be:
   a) Large deposits of coal are found here.
   b) The average temperature rises considerably as a result of global warming.
3. What is our responsibility as humans in the two scenarios in the previous question?

As you have seen in the activity, all the spheres on Earth interact closely with each other. When there is a change in one of the spheres, the others are also affected. The changes can be due to natural causes, for example floods or earthquakes, but more often these changes are due to human influence. As humans we have a responsibility to understand the interactions on Earth and to look after the planet so that future generations will be able to live on Earth.

Summary

Key concepts

- The Earth is a complex system where all the parts (spheres) interact.
- The Earth consists of four spheres: the lithosphere, the hydrosphere, the atmosphere and the biosphere.
- The lithosphere consists of solid rock, soil and minerals.
- The hydrosphere consists of water in all its forms.
- The atmosphere is the layer of gases around the Earth.
- The biosphere consists of all living plants and animals and their
- interactions with the rocks, soil, air and water in their habitats.

Concept map

Draw your own concept map for this unit.

Spheres of the Earth

Revision

1. Identify the four spheres of the Earth. What is each sphere composed of? List only three components for each. [8]

2. How does the lithosphere interact with the hydrosphere, biosphere and atmosphere in the photographs below? [6]

A large open copper mine.

A sand dune in the Namib Desert

3. You have a wet towel which you hang outside to dry. Describe and compare the interaction between the hydrosphere (water droplets on your towel) and the atmosphere (temperature and air movement around you), if you live in a dry area, and if you live in a humid area. [2]

Total [16 marks]

21 The lithosphere

Key questions

- What does the centre of the Earth look like?
- Why is it important to know about the structure of the Earth?
- Why is there so much variety in the rocks you see around you?
- How do rocks form?
- Why do we need to know about rocks?
- Why are rocks important?

Keywords

- geosphere
- lithosphere
- continental crust
- oceanic crust
- crust
- mantle
- core
- composition

The Earth is a system consisting of many parts. In the previous unit we looked at how these parts or spheres interact. In this section we are going to look more closely at one of these spheres, namely the **lithosphere**.

21.1 What is the lithosphere?

On Earth, water, air, stone and life interact. Let's think about the stone part of this statement. Where on Earth do we find stone? What are the different forms of stone that are found on Earth? Why is it important to know about this part of the Earth? Let's investigate these questions.

Sand.

Pebbles.

Stones.

Large boulders.

Figure 21.1: Examples of different forms of stone.

ACTIVITY Investigating stones

Materials

- magnifying glasses
- hammers
- paper towel
- samples collected, as described below

Questions

1. Collect the following items and bring them to school: sand, pebbles, a small stone/rock, a larger rock.
2. When you collect sand, stones or rock, look for the samples that look interesting and different and bring these to class.
3. Find at least four different items from different locations.
4. Study the different samples and complete the following table. If you have magnifying glasses available, use these to study the detail of the different samples.
5. Wrap some of the samples in paper towel and see if you can crush them with a hammer. Your teacher might instruct you to do this outside.

	Location Describe where you have found your sample.	Shape and colour Describe the size, shape and colour	Texture Describe the texture and hardness.	Composition Is it made of more than one material? Describe what it is made up of.
Sand				
Pebble				
Small stone/rock				
Larger rock				

Take note

All the 'Keywords' listed in the boxes in the margin are defined in the Glossary at the end of this strand.

In the last investigation you would have seen a lot of variety amongst the types of stone that are found in the area around your school. There is variation in shape, colour and texture amongst the different rocks on Earth.

The lithosphere consists of all the mountains, rocks, stones, top soil and sand found on the planet. In fact, it also includes all the rocks under the sea and under the surface of the Earth. The lithosphere is found all around us and we interact quite closely with it every day.

Inside the Earth

The lithosphere is considered the outer layer of the Earth. The Earth is made up of concentric layers called the crust, the mantle and the core.

Take note

Concentric objects share the same point as their centre.

Imagine that we cut away a slice of the Earth, as shown here:

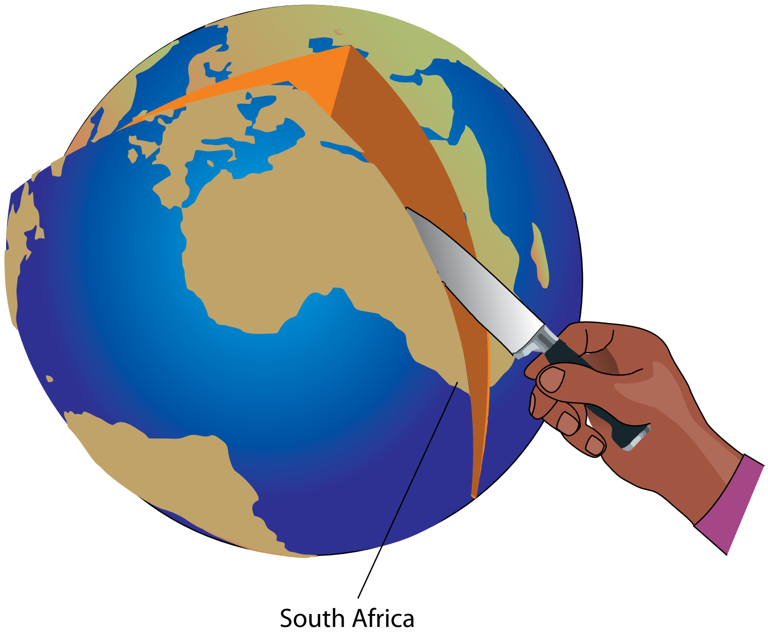


Figure 21.2

Inside, we would then be able to see the layers of the Earth, as shown in the next diagram:

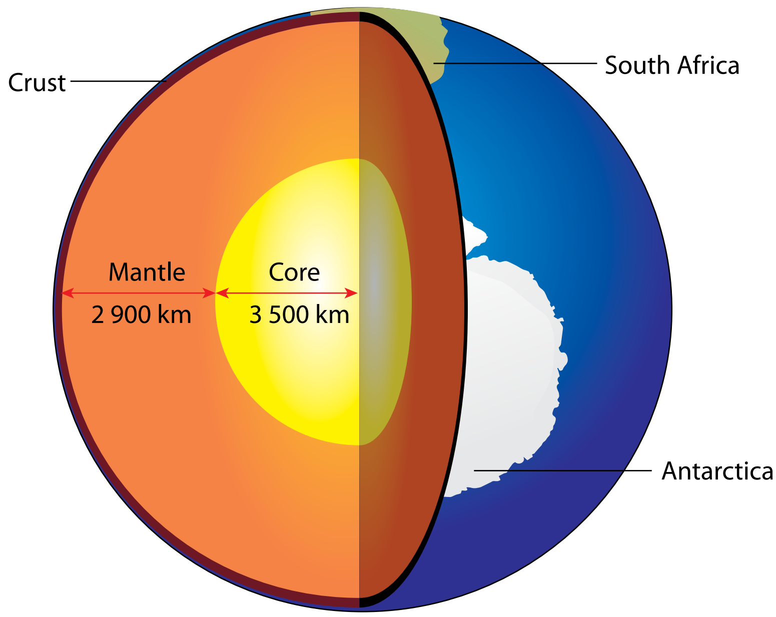


Figure 21.3

Did you know?

The centre of the Earth is 6 371 km deep.

The core has two parts, the **inner core** which is solid and the **outer core** which is liquid. The mantle can also be divided into two parts, the **lower mantle** and the upper mantle. Some parts of the crust are found under the oceans. This is called the **oceanic crust**. Other parts of the crust form part of the continents and is called **continental crust**.

The brittle upper part of the mantle and the crust form the **lithosphere**. The lithosphere, the mantle and the core are sometimes called the **geosphere**. The geosphere is also one of the parts of the Earth, just like the **hydrosphere**, **atmosphere** and **biosphere** that you learnt about in the previous unit.

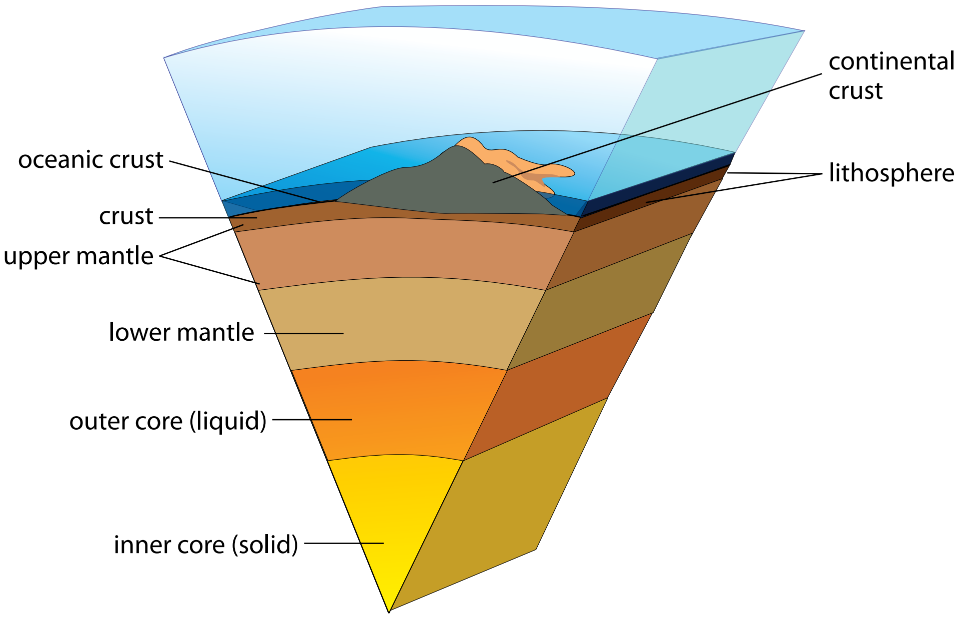


Figure 21.4: The layers inside the Earth.

ACTIVITY Build a 3D-model of the Earth

Instructions

1. Use recycled material and modelling clay to build a three-dimensional model of the inside of the Earth.
2. All the layers of the Earth need to be included and accurately labelled on your model.
3. Write a one-page summary on the layers of the Earth. Read about each of the layers to be able to answer the following questions in your write-up. You can use the internet and library books, or ask knowledgeable people in your community.
   a) Thickness of each layer
   b) State of matter
   c) Temperature
   d) Composition (what it is made up of).

Did you know?

The temperature of the Earth's inner core is about the same temperature as the surface of the Sun, more than 5 000 °C.

ACTIVITY The layers inside the Earth

Instructions

Use all the words in bold in the previous section to complete the following map:

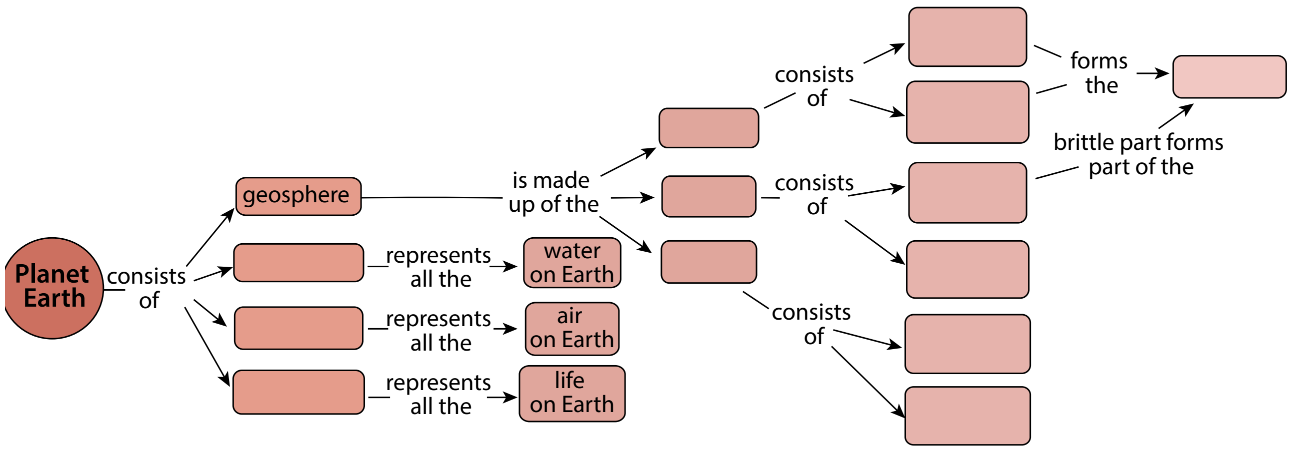


Figure 21.5

In the first activity of this unit we collected different rocks. Why are there different types of rocks, and why do they look different? These are the questions we will answer in the next section.

21.2 The rock cycle

Keywords

- cycle
- weathering
- compaction
- erosion
- deposition
- melting
- cooling
- solidify
- sedimentary rock
- metamorphic rock
- igneous rock
- sediment
- sedimentation
- cementation

In previous grades you learnt about the water cycle. A cycle is a combination of processes that take place in a certain sequence and which repeat over and over again from the beginning. Processes in a cycle do not stop and are therefore said to be continuous. For example, the **water cycle** which is part of the biosphere describes how water forms clouds, rain, rivers and clouds again.

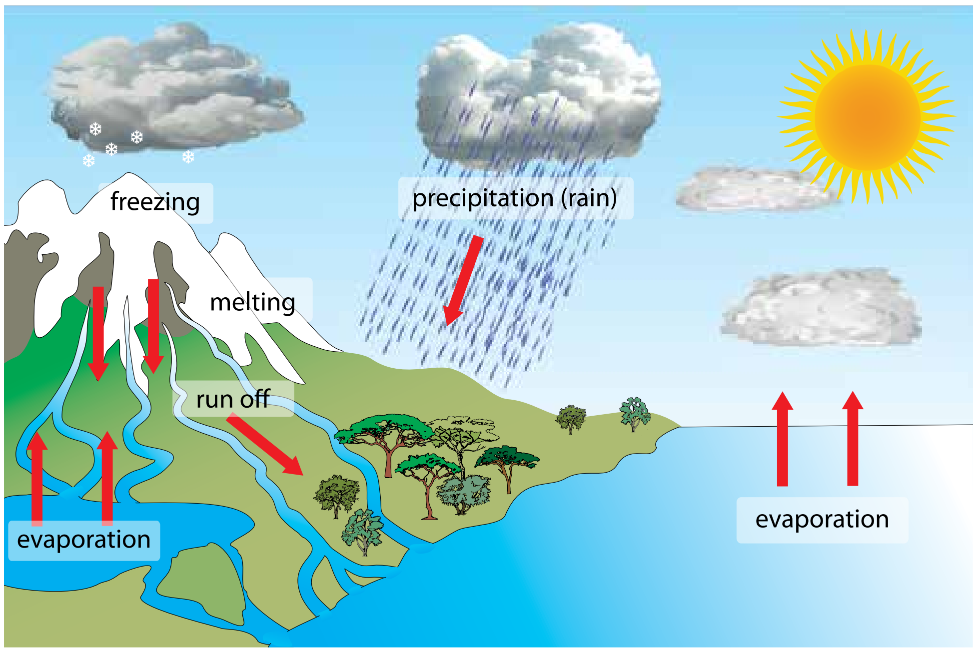


Figure 21.6: An example of a cycle with which you are familiar is the water cycle.

The **carbon cycle**, which takes place as part of the biosphere, describes the movement of carbon through carbon dioxide, fossil fuels and carbohydrates.

The **rock cycle** is part of the lithosphere and it describes how rocks change from one form into another and eventually back into the first form.

How does the rock cycle work?

Rocks on Earth are divided into three broad categories:
1. sedimentary rocks
2. metamorphic rock
3. igneous rock

This classification is based on where the rocks were formed. The following diagram summarises the rock cycle.

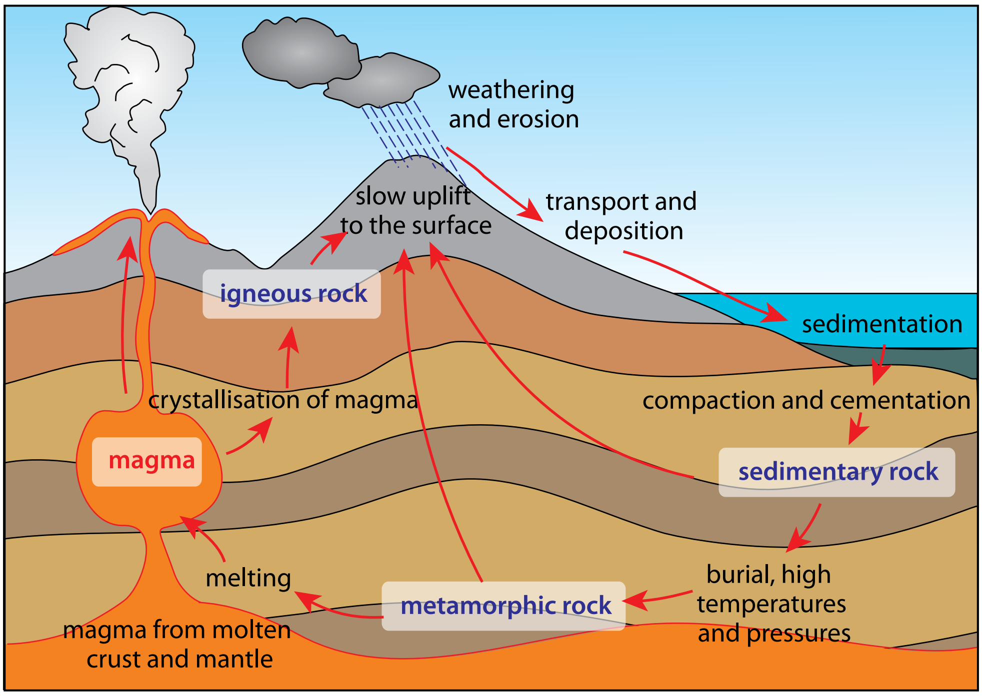


Figure 21.7: The rock cycle.

> **Take note**
>
> Not all rock on Earth is recycled. Thousands of tons of rock fall to Earth from space every year.

The rock cycle is a natural continuous process in which rocks form, are broken down, and re-form again over long periods of time. The process can be described as follows:
- Wind, water, heat and cold cause the weathering of rocks on the surface of the Earth. The rocks are broken up into smaller and smaller pieces and form sand.
- Wind and water wash the sand and small stones away and deposit them as **sediments** into lakes and the ocean. This process is called **deposition**.
- The sediments settle at the bottom of the oceans, lakes and rivers. Over time they get covered with more layers of sediment. The pressure from the additional layers causes the sediments to compact and solidify to form sedimentary rock.
- The sedimentary rock may be buried deeper and deeper beneath the surface of the Earth through movement in the Earth's crust (where oceanic plates and continental plates meet). The rocks can also be pushed deeper (subducted) into the Earth. As the rocks move deeper into the Earth, temperature and pressure increase.

Take note

Heat causes expansion of rocks and cold causes contraction.

Did you know?

A geologist is a scientist who studies the Earth, the rocks from which it is made, and the processes and history that have shaped it.

Take note

Magma and lava are both molten rock, but refer to different locations. Magma is molten rock that forms beneath the Earth's surface. When magma erupts out of a volcano onto the surface, it is referred to as lava.

- Rocks become more compact as processes of compaction and cementation occur. As the chemical compounds in the rocks change, due to heat and pressure, metamorphic rock is formed.
- Over time the metamorphic rock can move deeper into the Earth, melt, and become **magma**.
- Magma moves towards the surface of the Earth through volcanic pipes. The hot magma cools slowly on its way to the surface and forms **igneous rock**. Magma can also break through the surface as lava in volcanoes. In this case, the lava will solidify quickly on the surface to form igneous rocks. Igneous rock can form in the crust or on the surface.
- Igneous rocks get eroded by wind and water and the whole process starts again.

Metamorphic rock is formed deeper under the surface and becomes exposed to the surface only when the layers above it are removed by erosion. Igneous rock, just like sedimentary rock, can move deeper into the Earth and form metamorphic rock because of the increase in pressure and temperatures. As you can see in the previous diagram, rocks of all types may move down through the mantle, melt, and mix with magma. The Earth's crust is continually recycled. This is why we refer to the process as the **rock cycle**.

ACTIVITY Summarising the rock cycle

Questions

1. Copy and complete the diagram in your exercise books by filling in which type of rock belongs where: Sedimentary rock, Metamorphic rock, Igneous rock.

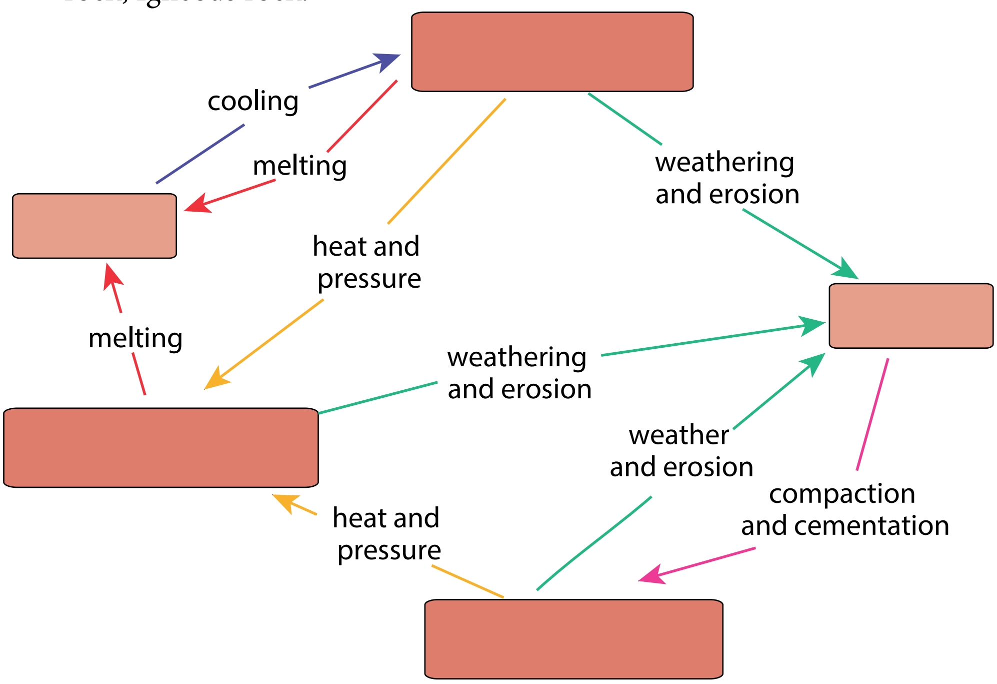


Figure 21.8

2. Name the process by which igneous rock is formed.
3. Which type(s) of rock form sediment?
4. What conditions are needed for metamorphic rock to form?
5. Explain what 'weathering and erosion' of rock mean.

6. Explain what 'compaction' means.
7. What type of rock is formed through compaction?
8. What is magma? Explain the role of magma in the rock cycle.

Keywords

- melting
- deposition
- erosion
- cooling
- compact
- temperature
- pressure
- metamorphic rock
- igneous rock
- sedimentary rock

ACTIVITY Explaining the rock cycle

Instructions

Write a paragraph to explain the rock cycle in your own words. Start the explanation with the formation of igneous rock. Use full sentences and include the following key words in your write-up.

The words can be used more than once, and you should add your own keywords as well. You should also include a labelled diagram in your write-up.

We are now going to take a closer look at each of the three main types of rocks.

Sedimentary rock

Sedimentary rocks are formed when layers of sediment solidify over time. Sediments are layers of particles from pre-existing rock or once-living organisms, for example, shells. Rocks on the surface of the Earth are weathered by expansion and contraction due to changes in temperature, wind and water, and also by erosion caused by animals. Bigger rocks break up into smaller and smaller particles through the process of erosion.

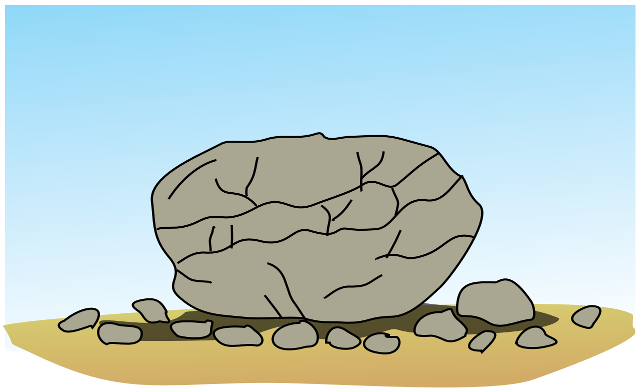


Figure 21.9: Changes in temperature cause rocks to crack and break up. Plants may grow in the cracks, causing them to break up further.

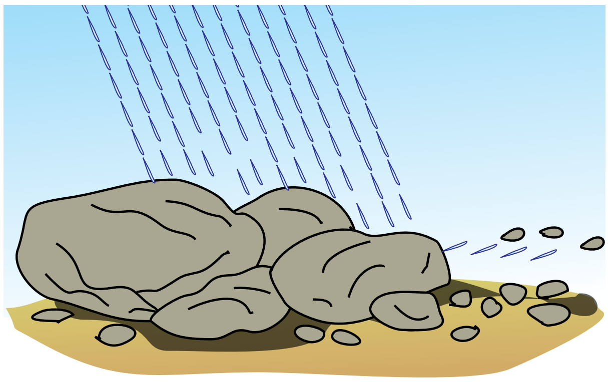


Figure 21.10: Rain wears down rocks and causes smaller pieces to break off.

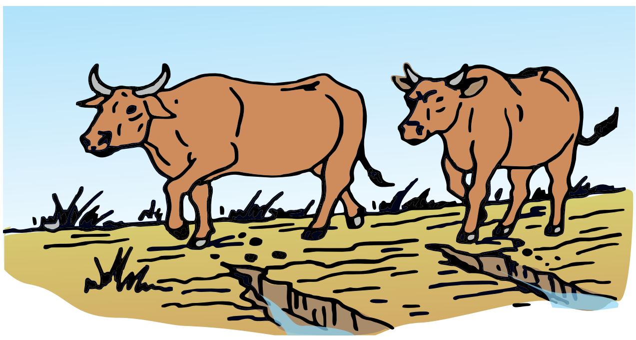


Figure 21.11: Animals break up rocks into smaller particles as they walk along.

Did you know?

The Hoba Meteorite landed in what is now Namibia around 80 000 years ago. It is the largest known meteorite still in a single piece, and the most massive naturally occurring piece of iron known at the Earth's surface. It has a mass of over 60 tons.

Wind and water transport the loose, smaller particles, along with debris from living organisms, and some large stones, eventually depositing them on flood plains and in the sea. This is called erosion.

The material accumulates at the bottom of oceans, rivers, lakes and swamps. The sediment settles and forms layers. These layers build up on each other and cause the compaction of the lower layers.

Figure 21.12: Soil erosion caused by water.

Over time, the bottom layers eventually solidify and form layers of sedimentary rock, as is shown in the following diagram.

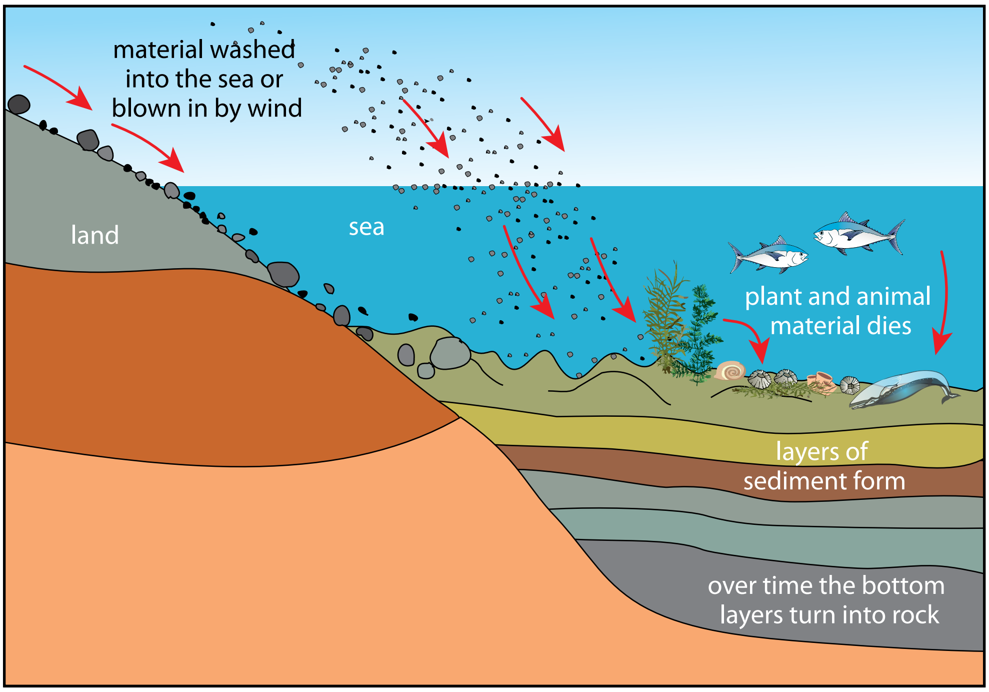


Figure 21.13: The formation of sedimentary rock.

Did you know?

The oldest layers of sedimentary rock visible in the Grand Canyon are believed to be nearly 2 billion years old.

Although sedimentary rock is found in most places on Earth, these rocks make up only 8% of the Earth's crust. Different layers of sedimentary rock may be seen in the mountains and rocks around us on a daily basis.

In the photograph you can clearly see the layers of sediment which have solidified over millions of years to form the sedimentary rocks of the Grand Canyon.

In the photograph below, you can see the layers in the sedimentary rock making up Table Mountain in Cape Town.

Figure 21.14: The layers in the sedimentary rock in the Grand Canyon.

Figure 21.15: The sandstone layers of Table Mountain.

There are different types of sedimentary rock, including sandstone, limestone, dolomite, coal, shale and conglomerate.

Figure 21.16: Sandstone rock in the Cederberg in the Western Cape.

Figure 21.17: Layers of limestone sedimentary rock.

Did you know?

Lime is the word used for calcium-containing compounds like calcium oxide (CaO), calcium hydroxide ($Ca(OH)_2$) and calcium carbonate ($CaCO_3$).

Limestone

Limestone is a sedimentary rock made from the mineral calcium carbonate ($CaCO_3$), often formed from the remains of the skeletons of marine animals. We use limestone as building stone in the manufacture of lime (calcium carbonate), and cement.

Dolomite

Dolomite is a sedimentary rock made from calcium magnesium carbonate ($CaMg(CO_3)_2$). Coal is another example of sedimentary rock formed from the solidified remains of ancient plants at the bottom of swamps. Shale is a very fine-grained sedimentary rock formed from the deposition of mud and silt. It is made up of very thin layers all stuck together. Conglomerate is a sedimentary rock made up of small stones, shells and other pieces of sediment. **Cementation** is the process whereby sand and associated shells, pebbles, and other sediment become cemented together to form sedimentary rock.

Sedimentary rock is softer than the other types of rock. It is eroded by the actions of wind, water or ice (glaciers). Fossils, especially of sea creatures, are often found in sedimentary rocks, lying cemented in the sediments in which they fell when they died. When plants or animals die, they are often covered in sand, which later becomes rock, capturing the fossil inside.

Dolomite mountains.

Fossils in sedimentary rock.

Limestone (creamy-brown) on top of shale (dark grey).

Conglomerate showing the layers with small pebbles embedded in the rock.

Figure 21.18: Examples of sedimentary rock.

Let's take a look at how the layers of sediment are compressed and become harder over time as a result of pressure.

ACTIVITY Modelling the formation of sedimentary rock

Materials

- three slices of white bread
- three slices of brown bread
- heavy books or object

Instructions

1. Cut off the crust from all the sides.
2. Layer the slices on top of each other, alternating the white and brown slices. Each slice represents a different layer of sediment.
3. Draw a labelled diagram of what the stack looks like.
4. Place a piece of plastic on top of the bread stack to protect the bottom book in your bookstack, then place a pile of books on top of the bread stack. Observe what happens to the layers. Write your observations here.
5. Add more books to the pile and observe. What happens to the layers?
6. Remove the books from the bread pile. Can you distinguish the different layers now?
7. Draw a labelled diagram of the bread layer.
8. Explain how this model demonstrates the formation of sedimentary rock.

Figure 21.19

Metamorphic rock

Metamorphic rock makes up a large part of the Earth's crust. Metamorphic rocks are formed when sedimentary or igneous rocks are exposed to heat and pressure. Metamorphic rocks do not form on the surface of the Earth, but rather deeper underneath the surface where the temperatures and pressures are much higher. When other types of rock experience higher pressures and temperatures, the rock crystals are squashed together. They undergo changes in crystal structure to form metamorphic rock.

Metamorphic rocks may move deeper into the Earth where they melt, forming magma. The magma may then cool and form igneous rock.

Some examples of metamorphic rocks are slate, marble, soapstone, and quartzite.

Slate is a metamorphic rock formed from shale (sedimentary rock) that was metamorphosed. Slate is often used for roofing or flooring. Since it can be cut into shapes and does not absorb moisture, it makes a good material for tiles.

Figure 21.20: Examples of metamporphic rock

Did you know?

'Metamorphic' refers to metamorphosis – a process where one thing is transformed into a completely different thing, like a pupa becoming a butterfly.

Marble

Marble is a metamorphic rock that is produced from the metamorphosis of limestone. It is used for countertops, flooring and tombstones, and is a very durable building material.

Roof tiles are made from slate, which was formed from shale (a sedimentary rock).

Figure 21.21: Marble blocks in a wall.

Figure 21.22: A marble arch in London.

Soapstone

Keywords

- extrusive rock
- intrusive rock

Soapstone is a relatively soft metamorphic rock. It is often used as an alternative natural stone countertop instead of granite or marble, for example in kitchens and laboratories. In laboratories it is unaffected by acids and alkalis. In kitchens it is not stained or altered by tomatoes, wine, vinegar, grape juice and other common food items. Soapstone is unaffected by heat. That means that hotpots can be placed directly on it without fear of its melting, burning or other damage. Many statues and carvings are also made from soapstone.

Figure 21.23: Soapstone carvings.

Figure 21.24: A pot made from soapstone.

Quartzite is formed through the actions of heat and temperature on sandstone. If you compare the texture of sandstone with quartzite in the pictures shown here, you will see that the process of metamorphism changes the texture from sandy to more glossy.

The crystals in the quartzite are bigger and the layers have disappeared. Quartzite is much harder than sandstone.

Figure 21.25: Sandstone.

Figure 21.26: Quartzite.

Igneous rock

Igneous rock is formed when magma cools down. Three factors play a role when igneous rocks are formed:

1. **Where it is formed**: If rocks are formed on the surface, they are called intrusive rocks. If they are formed under the surface they are called extrusive rocks.
2. **How quickly it cools**: When magma cools quickly, small crystals are formed and the resulting rock has a fine-grained texture. When it cools slowly, larger crystals form, resulting in a more coarse-grained rock. Sometimes the individual crystals can be seen with the naked eye.
3. **How much gas is trapped**: Magma contains molten rock and lots of gas. The gas is under pressure deep in the Earth. When the magma breaks through the surface, the gas is released. Depending on how quickly the magma cools down, some of the gas may have time to escape. When the magma cools down very quickly, lots of gas is trapped, resulting in cavities and openings forming in the rock.

Take note

Molten refers to liquids which are extremely hot, and whose usual form is a solid (e.g. rock). However, something melted need neither be hot, nor a complete liquid (e.g. butter).

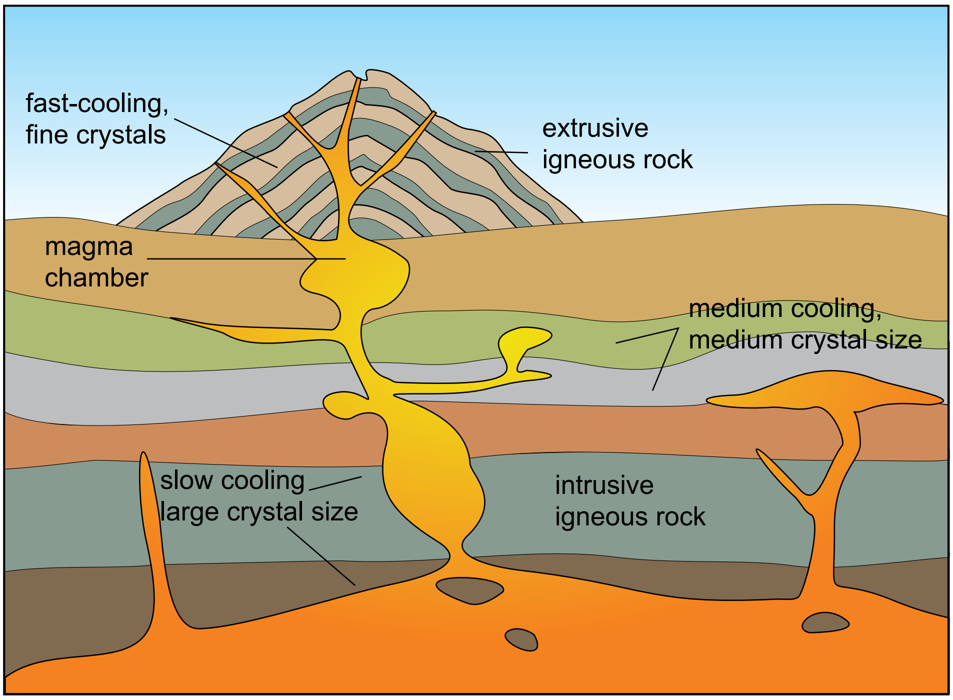


Figure 21.27: The structure of a volcano.

A volcano is an opening or rupture in the surface of the Earth's crust (or another planet) which allows hot lava and volcanic ash to escape in an eruption from the magma chamber below.

Did you know?

Pompeii was an ancient Roman city that was completely destroyed and buried under as hand pumice in the eruption of Mount Vesuvius in 27 AD. The town and objects were preserved for thousands of years and have now been excavated. Today it is visited by millions of tourists annually.

Figure 21.28: The Cleveland Volcano eruption in 2006 in Alaska, photographed from the International Space Station.

Figure 21.29: An eruption from Mount Etna in Italy in 2007.

Examples of igneous rock are basalt, granite and pumice.

Basalt is the most common igneous rock and makes up a large part of the rocks just under the surface of the Earth. Most of the oceanic crust is basalt rock. It is a dark-coloured rock and is used as building material, particularly in building stone walls.

Basalt is found not only on Earth, but also on the Moon and Mars! The highest mountain on Mars, and also the biggest-known volcano in our solar system, Olympus Mons, was formed from basaltic lava flows.

Figure 21.30: Basalt.

Figure 21.31: Olympus Mons, a volcano on Mars.

Granite is an igneous rock with large grains. It was formed from magma which slowly crystallised below the surface of the Earth. Granite is one of the best-known types of rock. It is used to make numerous objects such as tabletops, floor tiles and paving stone.

Figure 21.32: Various colours and patterns of granite rock.

Pumice rock is an example of extrusive igneous rock. It is formed from the lava emitted during volcanic explosions. Because the lava cools down very quickly, a lot of gas is trapped in the rock. As a result, pumice is a very porous rock, with lots of holes in it, making it the only rock that can float on water. Pumice stones are used in lightweight concrete and as an abrasive in industries and in homes.

Figure 21.33: Pumice stone used as an exfoliator.

ACTIVITY Comparing the properties of igneous rocks

Questions

Study the following igneous rocks and compare their similarities and differences in the following table.

Figure 21.34: Sample 1.

Figure 21.35: Sample 2.

Figure 21.36: Sample 3.

Figure 21.37: Sample 4.

Sample	Where was the sample formed? Extrusively or intrusively?	How quickly did it cool? What evidence do you have for your answer?	Was air trapped when it was formed? What evidence do you have for your answer?	Describe the colour
Sample 1				
Sample 2				
Sample 3				
Sample 4				

Keywords

- mineral

ACTIVITY Classifying rocks

Instructions

1. In this project you will be working in pairs. You need to collect rocks from your neighbourhood, or borrow some rocks from someone's rock collection.
2. You will need at least 12 different samples of rock.
3. Try to find as much variety as possible, applying what you now know about the three different rock types.
4. You could also ask a geologist to provide you with a variety of rock samples to identify.
5. Go to the website provided in the Visit margin box and follow the flow diagram to identify all your rock samples.
6. You need to create a display of the rocks and how you have identified them using the flow diagram and their properties.

Rocks contain minerals

We started this unit by collecting rocks and looking closely at their characteristics. We then looked at how rocks were formed. The questions now are why do we need to know about rocks, and why are rocks important. Let's look at what makes rocks so valuable.

Rocks contain minerals. A mineral is a chemical compound which occurs naturally, for example in rocks. There are several thousand types of minerals which are found in different combinations in rocks. They consist of metal and non-metal atoms combined in various ratios.

Let's look at some examples. Copper is a valuable metal because it is a good conductor. It is used in electrical cabling and other electrical applications. There are about 15 different types of rock which contain copper compounds. One such compound is copper(I)sulfide or Cu_2S. When this compound is found in rocks it is called a mineral, named chalcocite. Copper can also be found as the compound $CuFeS_2$ or chalcopyrite. The minerals chalcocite and chalcopyrite can be found in many different types of sedimentary, metamorphic or igneous rocks.

If we would like to use the copper from these rocks, we need to find a way to get it out of the rock and into the metal form. This we will discuss in the next unit.

Figure 21.38: Chalcocite crystals.

Figure 21.39: Chalcopyrite ore.

Figure 21.40: Chalcopyrite crystals.

Figure 21.41: Chalcocite ore.

Quartz and feldspar are the two most abundant minerals in the crust. Quartz is the mineral form of silicon dioxide (SiO_2). Potassium feldspar has the formula KAl_3SiO_8. A rock may be composed almost entirely of one mineral or could be made of a combination of different minerals. Different combinations of different minerals in rock will result in a different types of rock.

ACTIVITY What minerals are found on Earth?

Instructions

In this unit you have learnt about the Earth's crust and minerals that occur in rocks. Read more about the Earth's crust and answer the following questions:

1. What are the most abundant elements in the Earth's crust?
2. Why are these elements so abundant?
3. How did the elements get into the Earth's crust?
4. Why are the elements important?
5. What do you think are the most important element(s) in the Earth's crust? Give a reason for your answer.

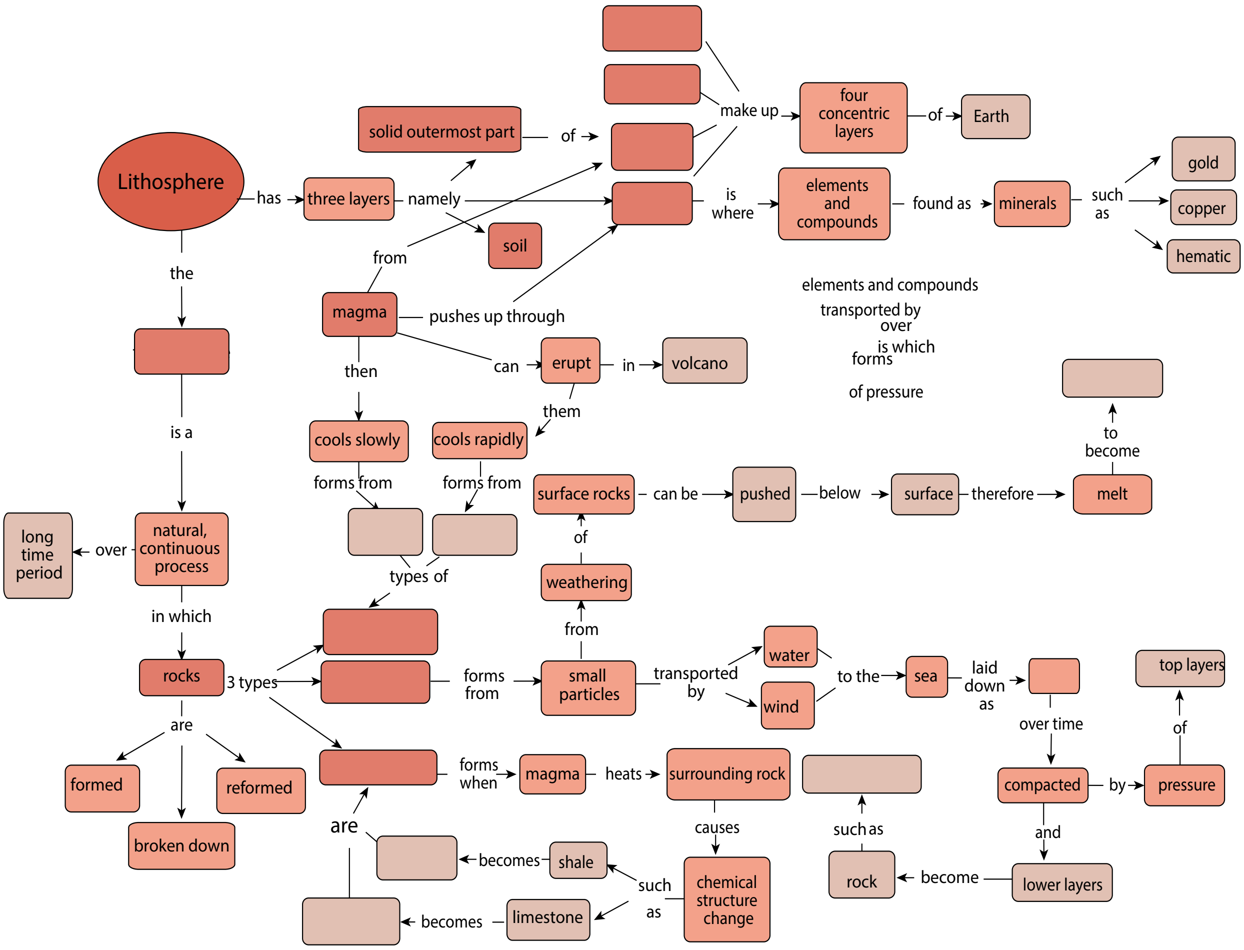
Summary

Key concepts

- The Earth consists of four concentric layers called the inner core, the outer core, the mantle, and the crust.
- The lithosphere consists of the solid outermost part of the mantle, the crust, and sediments covering it.
- The rock cycle is the natural continuous process in which rocks form, are broken down and re-form over long periods of time. The rock cycle has a number of steps.
- There are three rock types: igneous rock, sedimentary rock and metamorphic rock.
- Sedimentary rock is formed when rocks on the surface are weathered and the small particles, along with plant and animal material, are deposited in sediments at the bottom of lakes, oceans and rivers. Over time, more and more layers of sediment are deposited. The resulting increase in pressure causes compaction and the formation of hard layers of sedimentary rock.
- Fossils are often found in sedimentary rock as when some organisms die, they become incorporated into the layers of sediment.
- Hot magma is found deep below the surface of the Earth. When magma cools slowly, below the surface of the Earth, it forms intrusive igneous rock. When the magma pushes up through the crust (for example in a volcano), it cools rapidly and forms extrusive igneous rock.
- Hot magma can heat the surrounding rock and change other types of rock into metamorphic rock.
- Different combinations of elements and compounds form the minerals in the crust.

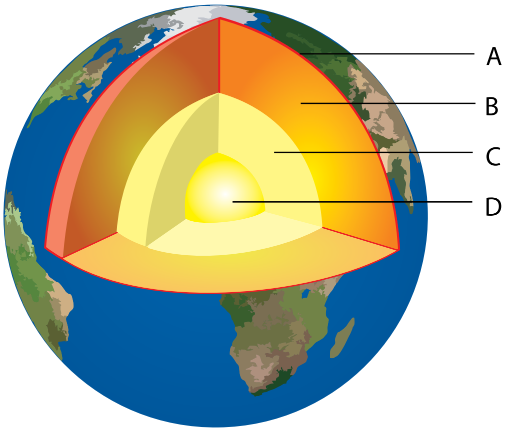
Concept map

Use the concept map on the next page to summarise what you learnt about the lithosphere and rock cycle in this unit. If you want to add more links or information into the concept map, you should do so.



Revision

1. The Earth consists of different layers.
   a) Label the following diagram: [4]

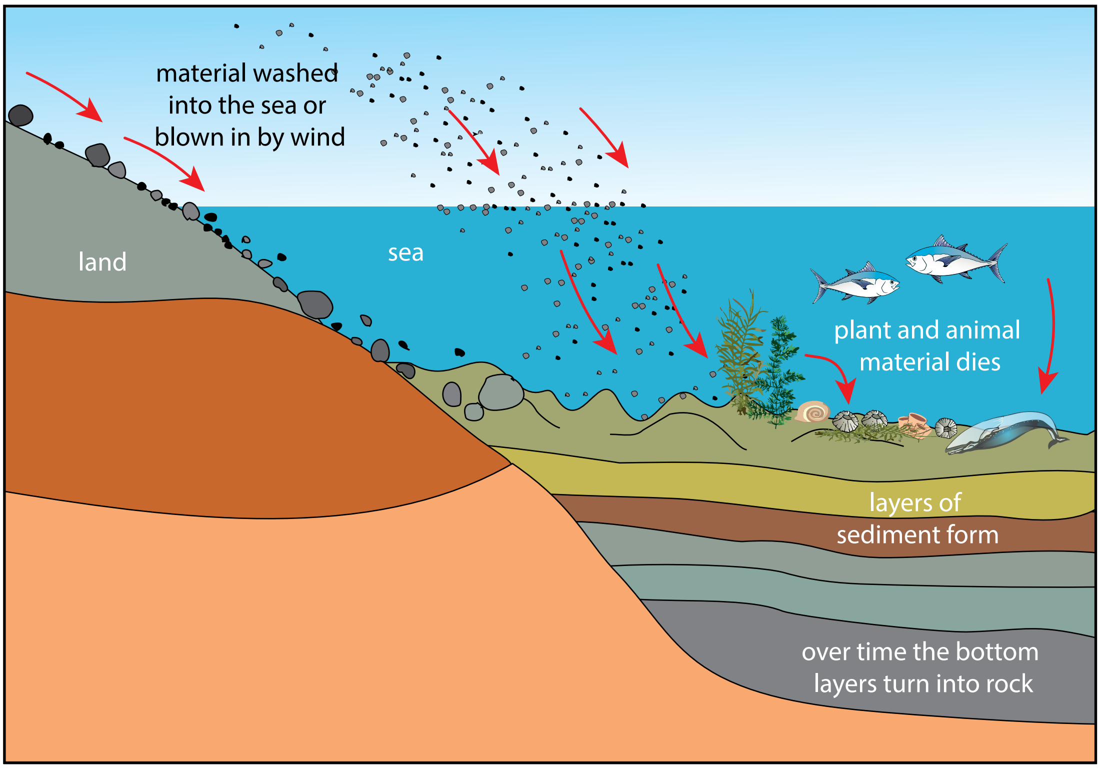


   b) What is the difference between parts C and D? [2]
   c) What does part B consist of? [1]
   d) Give three examples of things found around your school that form part of part A. [3]

2. Why are there so many different rocks found on Earth? [2]

Beyond

3. The diagram below shows the formation of one of the rock types. Study the diagram and answer the question that follow.



a) What type of rock formation is shown in the diagram? Give a reason for your answer. [2]
b) What processes are involved in the formation of this type of rock? [2]
c) What will happen if the rocks formed here move deeper into the Earth? [3]

2. Fossils are often found in sedimentary rocks. Explain why this is the case. [4]

3. Explain the difference between the formation of igneous rock, such as granite, and igneous rock, such as pumice. [4]

4. Iron is an element found abundantly on Earth, especially in the core of the Earth. Iron combines with oxygen to form haematite, the mineral form of iron(III)oxide. Haematite is present in sedimentary rocks, for example in the Sishen area in the Northern Cape.
a) What is the formula for iron(III)oxide? [1]
b) How does iron end up in sedimentary rock? [3]
c) Why is hematite an important mineral? [1]

Total [32 marks]

22 Mining of mineral resources

Key questions

- How do we know where to mine?
- How do we get the valuable ore-rich rocks out of the ground?
- How do we get the minerals or metals out of the ore?
- How do we separate minerals from waste rock?
- How do we refine minerals?
- Where in South Africa are the mineral-rich deposits suitable for mining?
- What do we mine in South Africa?
- What is the impact of mining?

Keywords

- mineral
- ore
- PGM

In the previous two units you have learnt about the spheres of the Earth especially the lithosphere. The lithosphere consists of rocks, which contain minerals. Minerals are natural compounds formed through geological processes.

A mineral could be a pure element, but more often minerals are made up of many different elements combined. Minerals are useful chemical compounds for making new materials that we can use in our daily lives. In this unit we are going to look at how to get the minerals out of the rocks and in a form that we can use. This is what the mining industry is all about. Mining is a very important industry in South Africa. We have a lot of mineral resources in our country and a lot of people depend on mining for a living.

You already know that minerals in rocks cannot be used. Many processes are used to make minerals available for our use. We need to locate the minerals. We must determine whether these concentrations are economically viable to mine. Rocks with large concentrations of minerals are called **ores**. Mining depends on finding good quality ore, preferably within a small area.

The next step is to get the rocks which contain the mineral out of the ground. Once the ore is on the surface, the process of getting the mineral you want out of the rock can start. Once the mineral is separated from the rest of the rock, the mineral needs to be cleaned so that it can be used.

This process can be represented by the flowchart on the left.

In this unit we will look at each of the steps in more detail. You will also apply what you learn about mining to one specific mining industry. This is explained in the research project below.

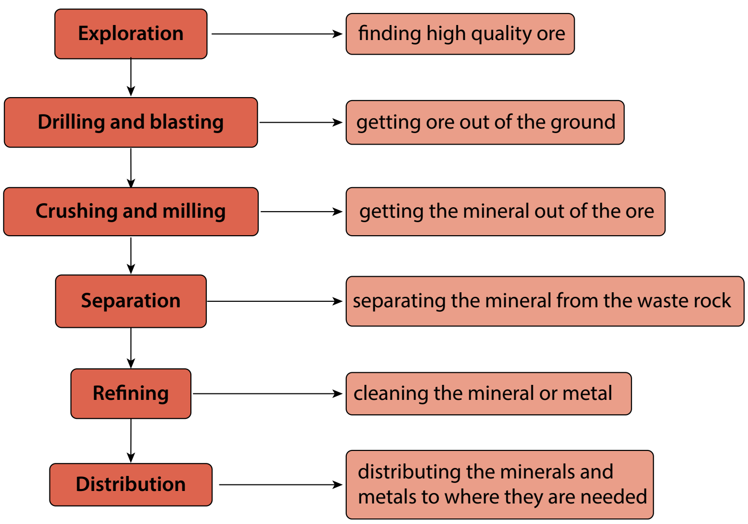


Figure 22.1

ACTIVITY Mining in South Africa

Instructions

1. Work in groups of three.
2. Choose one mining industry in South Africa and find information about the industry of your choice.
3. Choose from the following list: gold, iron, coal, phosphate, manganese, diamond, chromium, copper and the platinum group metals (PGMs).
4. Present your findings to the class in an oral and a poster.
    a) Use the following questions to guide your research:
    b) How do geologists and engineers know where to mine for the mineral of your choice?
    c) What type of mining method is used in this industry?
    d) What processes are used to get the rock out of the ground?
    e) What processes are used to reduce the size of the rocks?
    f) How is the mineral removed from the ore?
    g) How is the mineral separated from its compound?
    h) How is the mineral refined?
    i) Where in South Africa is this mineral mined?
    j) What has the impact of this mining industry been on South Africa?
    k) What has the impact of this mining industry been on the environment?
    l) What careers are involved in this mining industry?

Keywords

- exploration
- remote sensing
- geophysical methods
- geochemical methods

22.1 Exploration: Finding minerals

One of the most important steps in mining is to find the minerals. Most minerals are found everywhere in the lithosphere, but in very, very low concentrations, too low to make mining profitable. For mining to be profitable, high-quality ore needs to be found in a small area. Mining **exploration** is the term we use for finding out where the valuable minerals are.

Today technology helps mining geologists and surveyors to find high-quality ore without having to do any digging. When the geologists and surveyors are quite sure where the right minerals are, only then do they dig test shafts to confirm what their surveying techniques have suggested.

Methods of exploration

In all these methods we use the properties of the minerals and our knowledge of the lithosphere to locate them underground, without going underground ourselves. For example, iron is magnetic, so instruments measuring the changes in the magnetic field can give us clues as to where pockets of iron could be. Exploration methods are used to find and assess the quality of mineral deposits, prior to mining. Generally, a number of explorative techniques are used, and the results are then compared to see if a location seems suitable for mining.

Did you know?

The rare earth elements are a set of 17 elements on the Periodic Table, including the fifteen lanthanides and scandium and yttrium. Despite their names, they are found in relatively plentiful amounts in the Earth's crust.

> **Did you know?**
>
> Kimberlite pipes are the most important source of diamonds in the world. They are named after the town of Kimberley, where a where a 16,7 g diamond was found in 1871, starting the diamond rush.

Remote sensing is the term used to gain information from a distance. For example, by using radar, sonar and satellite images, we can obtain images of the Earth's surface. These images help us to locate possible mining sites, as well as study existing mining sites for possible expansion.

Figure 22.2: This image covers an area of 15 × 19 km and was taken from the NASA research satellite, Terra. It shows the mine at Baiyun Ebo, China, which is the site of almost half the world's rare earth element production.

Geophysical methods make use of geology and the physical properties of the minerals to detect them underground. For example, diamonds are formed deep in the Earth at very high temperatures, in kimberlite pipes of igneous rock. The kimberlite pipe is a carrot shape. The first kimberlite pipe to be detected was in Kimberley in South Africa. The pipe was mined, eventually creating the Big Hole.

Figure 22.3: The Kimberley Big Hole was a diamond mine until 1914 when it closed down, and is now a tourist attraction.

Geochemical methods combine the knowledge of the chemistry of the minerals with the geology of an area to help identify which compounds are present in the ore and how much of it is present. For example, when an ore body is identified, samples are taken to analyse the mineral content of the ore.

ACTIVITY Minerals and the right to own them

Many indigenous people, such as the San, share the same central belief that the land and all it produces are for all the people to use equally.

When colonialists arrived, they realised the potential mineral wealth of South Africa as gold, and later diamonds, were discovered. They ruthlessly took land from the local people wherever minerals were found, completely ignoring their right to ownership and access.

De Beers purchased the mining rights and closed all access to diamond mining areas. Anyone entering the area would be prosecuted and the sale of so-called 'illegal' diamonds was heavily punished. Other large mining companies have tried to claim the right to the minerals that they mine.

In groups or as a class discuss the following:

1. Should a few select people hold the right to the land and the minerals in it?
2. Who owns the minerals?
3. Should big corporations hold these rights?
4. What role should government play in allocating/administering mining rights?
5. Write down some of the main points of your discussion.

Did you know?

The Kimberley Big Hole is considered to be the largest hand-dug excavation on Earth at 463 m wide and 240 m deep. About 22 million tons of earth were excavated, which yielded 3 000 kg of diamonds.

22.1 Extracting ores

Once the ore body has been identified, the process of getting the ore out of the ground begins. There are two main methods of mining – surface mining and underground mining. In some locations a combination of these methods is used.

Keywords

- topsoil
- overburden
- excavation
- rehabilitation
- slurry

Surface mining

Surface mining is exactly what the word says – digging rocks out from the surface, forming a hole or pit. In South Africa, this method is used to mine for iron, copper, chromium, manganese, phosphate and coal. Surface mining is also known as open pit or open cast mining.

Let's look at coal as an example. For surface mining, the minerals need to be close to the surface of the Earth. Most of the coal found in South Africa is shallow enough for surface mining. Usually the rocks are present in layers.

Figure 22.4: An open pit coal mine.

Did you know?

South Africa is one of the seven largest coal-producing countries in the world. A quarter of the coal mined in South Africa is exported, mostly through Richards Bay.

To expose the coal layer, the layers above it need to be removed. The vegetation and soil, called the **topsoil**, is removed and kept aside so that it can be re-deposited in the area after mining.

If there is a layer of rock above the coal face, called the **overburden**, this is also removed before the coal can be **excavated**. Once all the coal has been removed, the overburden and topsoil are replaced to help in restoring the natural vegetation of the area. This is called **rehabilitation**. There is a growing emphasis on the need to rehabilitate old mine sites that are no longer in use. If it is too difficult to restore the site to what it was before, then a new type of land use might be decided for that area.

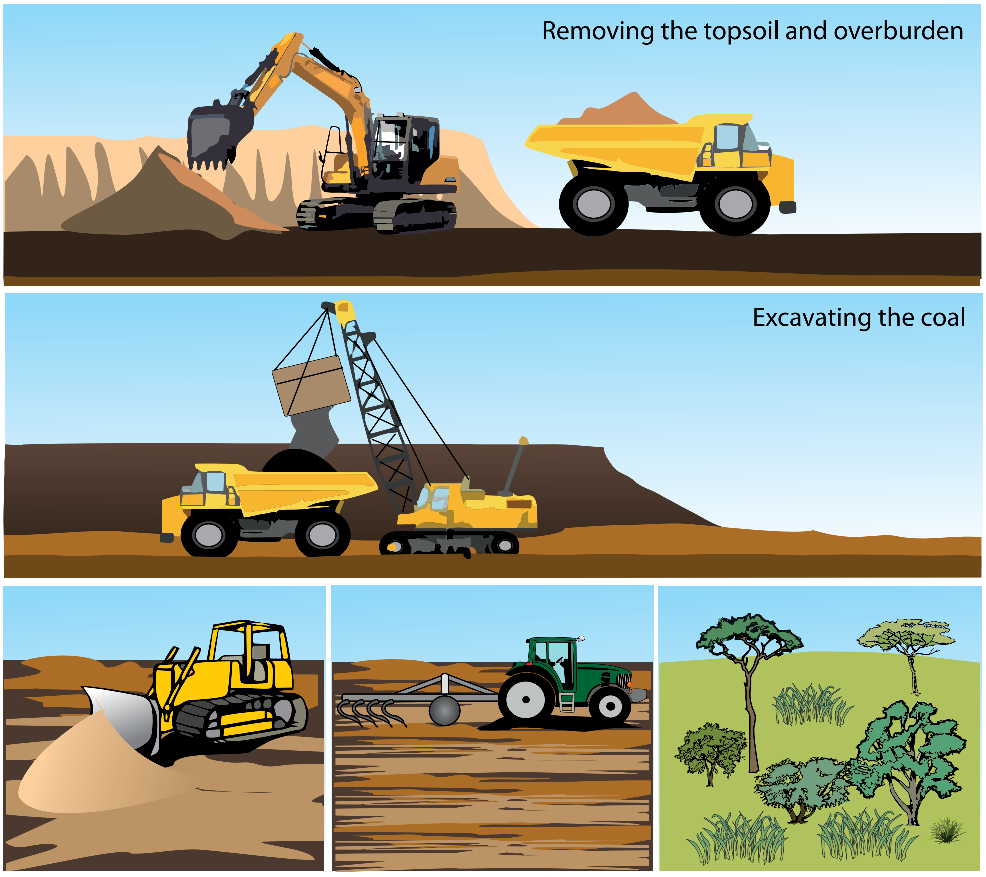


Figure 22.5: Surface mining coal and mine rehabilitation

Figure 22.6: Copper ore from Phalaborwa Mine.

In Phalaborwa in Limpopo province, copper ore is mined using open-pit mining.

The Phalaborwa open pit is one of the world's largest open pit mines. It is 2 km across and is the largest man-made hole in Africa.

When you mine you are digging into solid rock. The rock needs to be broken up into smaller pieces before it can be removed. Holes are drilled in the rock and explosives, like dynamite, are placed inside the holes to blast the rock into pieces. The pieces are still very large and extremely heavy. The rocks are loaded onto very large haul trucks and removed. Sometimes the rocks (ore) are crushed at the mining site to make them easier to transport.

Do you remember learning about how coal is formed? What is coal made from?

Figure 22.7: A mining haul truck being loaded with coal.

Did you know?

Mining trucks are enormous. They are up to 6 metres tall. That's higher than most houses. These trucks can carry 300 tons of material and their engines have an output 10 – 20 times more powerful than a car engine.

Underground mining

Shaft mining

Often the minerals are not found close to the surface of the Earth, but deeper down. In these cases, underground mining, also called shaft mining, is used. Examples of underground mining in South Africa are mining for diamonds, gold, and sometimes the platinum group metals (PGM).

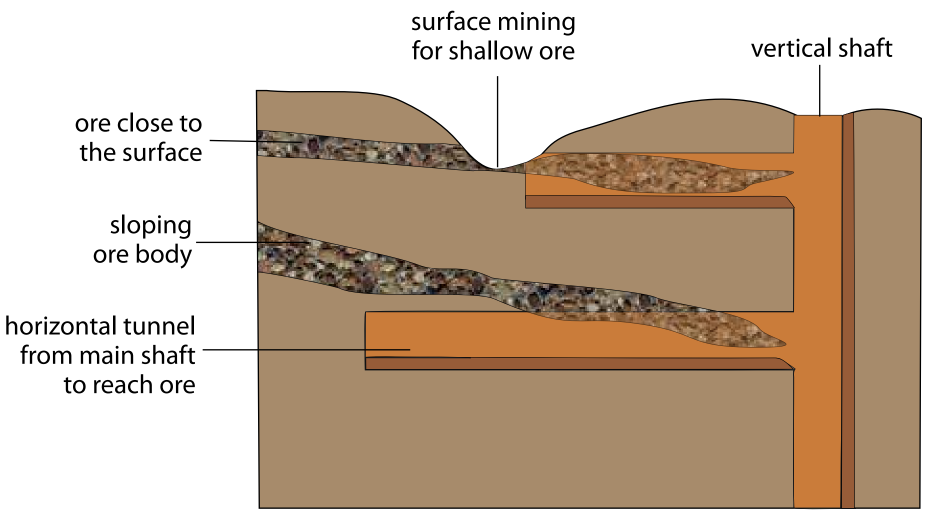


Figure 22.8: Extracting ore.

Take note

Igneous and metamorphic rocks are found where gold is mined.

Did you know?

South Africa is a leader in the field of deep underground mining as we have several mines deeper than 3 km.

Sometimes the ore is very deep, which is often the case with diamonds or gold ore. In these cases mine shafts go vertically down and side tunnels make it possible for the miners and equipment to reach the ore.

A structure called the headgear is constructed above the shaft and controls the lift system into the vertical shaft. Using the lift, it can take miners up to an hour to reach the bottom of the shaft.

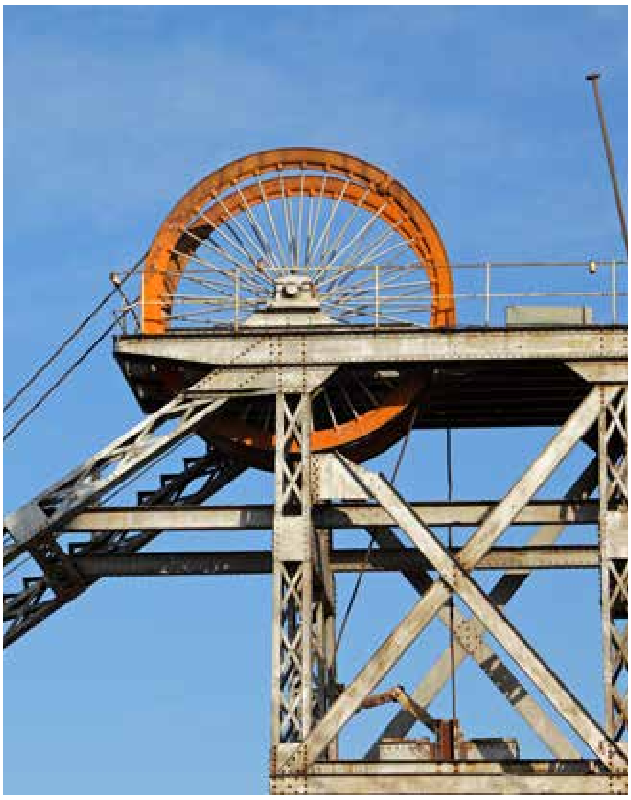


Figure 22.9: Old headgear at the Kimberley Mine in the Northern Cape

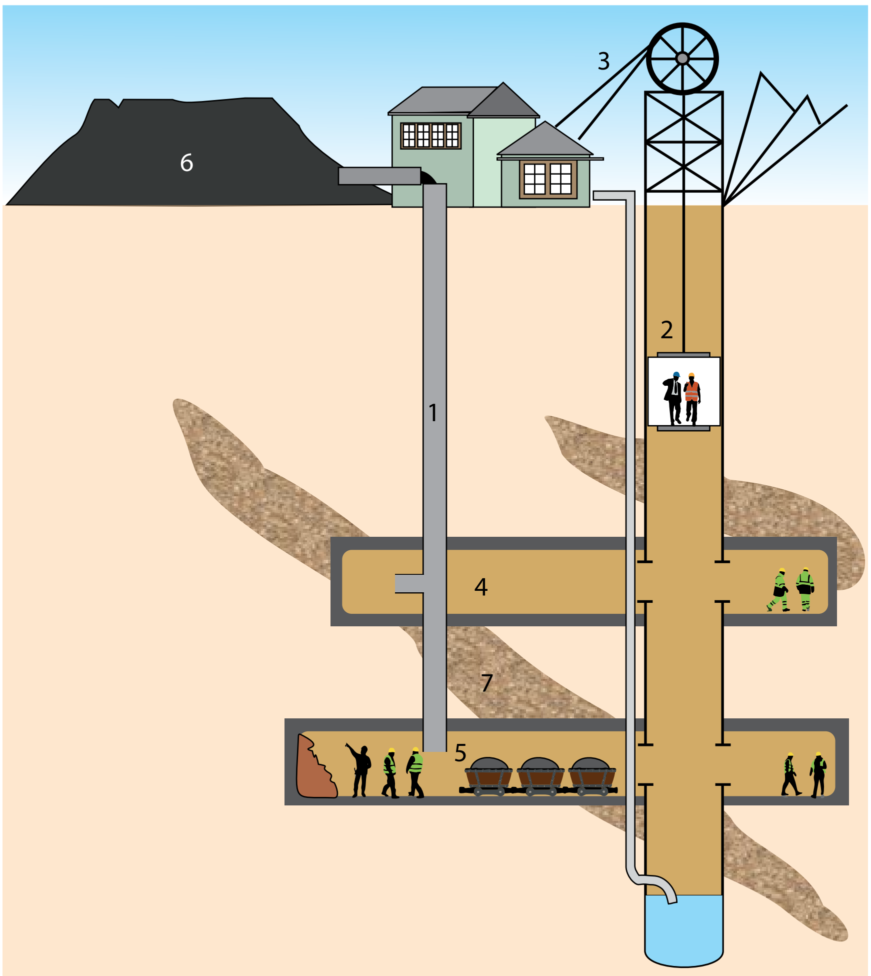
The TauTona Mine in Carletonville, Gauteng is the world's deepest mine. It is 3,9 km deep and has 800 km of tunnels. Working this deep underground is very dangerous. It is very hot, up to 55 °C. To be able to work there, the air is constantly cooled to about 28 °C using air-conditioning vents.

ACTIVITY Gold mining in South Africa

South Africa is a world leader in the gold mining industry. We have been mining gold for more than a century and our mines are the deepest in the world. Until 2010 we were the leading producer of gold in the world. Gold is a lustrous, precious metal, which has very high conductivity.

Questions

1. What mining method is used to mine for gold?
2. What type of rock is found where gold is mined?
3. What is gold used for?
4. Do you think gold mining is dangerous? Why do you say so?
5. Provide labels for numbers 1 – 7 in the following diagram.



Did you know?

The PGMs are six transition metals usually found together in ore. They are ruthenium (Ru), rhodium (Rh), palladium (Pd), osmium (Os), iridium (Ir) and platinum (Pt). South Africa has the highest known reserves of PGMs in the world.

Figure 22.10: Gold mining

Room-and-pillar method

One of the methods used in underground mining is called room and pillar, and is often used for mining coal. Part of the mine is open to the surface and part of it is underground. The coal face is dug out, but pillars of coal are left behind to keep the tunnels open and support the roof. Machines called continuous miners are used to remove the coal. The coal is loaded onto conveyor belts and taken up to the surface for further crushing.

Figure 22.11: A machine called a continuous miner at work at the coal face.

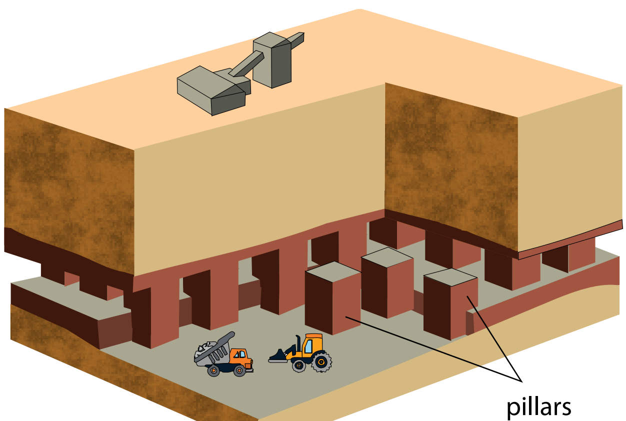


Figure 22.12: Room-and-pillar mining

What happens once the ore has been removed from the crust by mining?

Did you know?

Coal miners used to take a canary with them down the mines. If the canary died, they knew that oxygen levels were being dangerously depleted and that it was not safe to remain underground.

22.2 Crushing and milling


Figure 22.13: Ore is like choc chip biscuits where the minerals are spread through the rocks.

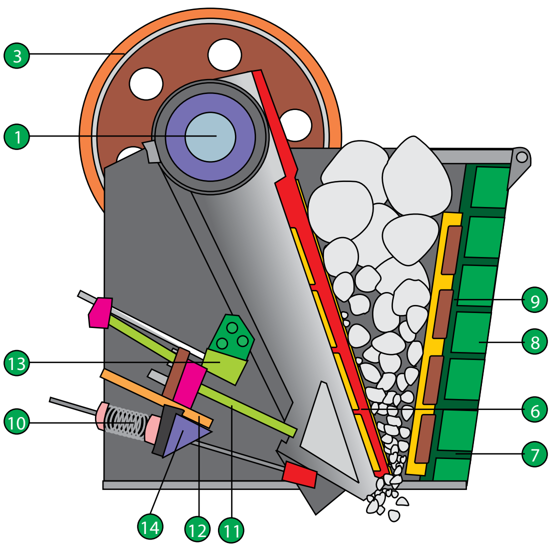
Mineral crystals are spread throughout rocks, just as chocolate chips are spread throughout a choc chip biscuit. Sometimes we can see the chocolate chips from the outside, but most of the time the chips are not visible because they are inside the biscuit.


Figure 22.14: The ore needs to be crushed, such as a choc chip biscuit, to get the minerals out.

The only way to find out how many choc chips there are is to crush the biscuit. In the same way we can sometimes see mineral crystals from the outside of the rock, but mostly we don't know what minerals there are and what concentrations are inside the rock. The only way to find out is to break the rock into smaller and smaller pieces.


Figure 22.15: The choc chips are separated from the rest of the biscuit, just as minerals are extracted from the rock.

Once we have crumbled the choc chip biscuit, the chocolate pieces can be separated from the crumbs. In the same way, in the mining process, the valuable minerals can be separated from the unwanted rock. The unwanted rock is called waste rock.



1. eccentric shaft
2. balance plate
3. fly wheel
4. toggle
5. cheek plate
6. movable jaw plate
7. crusher frame
8. fixed jaw plate
9. fixed jaw plate shim
10. retraction spring
11. toggle plate
12. toogle seat
13. adjusting wedge
14. groove block

Figure 22.16: Jaw crusher at a mine. The rocks are fed into the funnel and crushed as the two sides move back and forth.

Let's look at an example. You have learnt in the previous unit that copper minerals are found in rocks. In South Africa, the Bushveld Igneous Complex is an area which stretches across the North West and Limpopo Provinces. Igneous rock with high mineral content is found here. Here they mine for PGMs, chromium, iron, tin, titanium, vanadium and other minerals using open pit and underground mining. The rocks from the mines are transported by conveyor belts to crushers. Jaw crushers and cone crushers break the huge rocks into smaller rocks.

The smaller rocks are then moved to mills where large rod mills and ball mills grind them further into even smaller pieces until they are as fine as powder.

Figure 22.17: A ball mill.

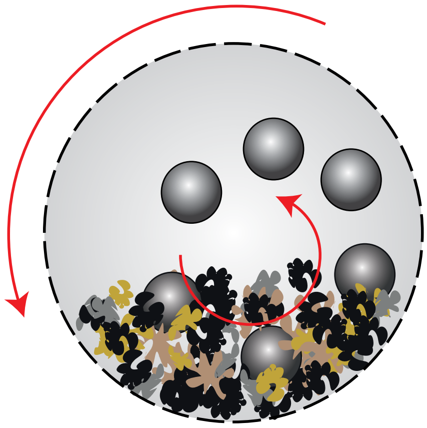


Figure 22.18: Inside a ball mill, the balls move round in a circle as the mill turns, crushing the ore into a powder.

This process of reducing the size of the rocks requires a lot of energy. Just image how hard it is to break a rock. How much more energy do you think is needed to crush a rock until it is like sand? This is one of the steps in the mining process that is very expensive because energy is needed to drive the process.

Most minerals are found as compounds in rocks. Only a few minerals are found in their pure form, in other words not bound to any other element. Examples of minerals found in their pure form are gold and diamonds (diamonds consist of the element carbon).

Figure 22.19: A rough diamond crystal embedded in rock.

Some rocks are used as is, and do not need to be crushed into powder, or involved in minerals extraction. For example, phosphate rock itself can be used as a fertiliser, or it can be used to make phosphoric acid. Sand, or the mineral silicon dioxide (SiO_2), is used in the building industry.

Figure 22.20: A gold nugget.

Coal found in sedimentary rock is crushed into the appropriate size and used as fuel for electricity generation or the iron-making process.

Figure 22.21: Lumps of coal can be used directly as a fuel. However, some coal is first washed to make it into 'high-grade coal'. It can also be sorted into various sizes, depending on what the fuel is required for.

> **Take note**
>
> An electro-magnet, as we have learnt in Energy and Change, is a type of magnet in which the magnetic field is produced by electric current.

22.3 Separating minerals from waste

Before the minerals can be used, they need to be separated from the waste rock. A number of different separation techniques are used. These techniques are based on the properties of the minerals. Different minerals are often found together, for example copper and zinc, gold and silver, or the PGMs. A combination of techniques is used to separate the minerals from the waste and then the minerals from each other.

Hand sorting

Sorting by hand is not a very effective method to separate out the minerals you want. It can be used only in exceptional situations or by individuals; for example many people mine for alluvial diamonds by hand in rivers in Angola. It is a cheap and easy process to do individually, but it is not feasible on an industrial scale.

> **Keywords**
>
> - magnetic separation
> - panning
> - size separation

Magnetic separation

Iron is a metal with magnetic properties. Iron ore can be separated from waste rock by using magnetic separation techniques. Conveyor belts carry the ore past strong electromagnets which remove the magnetic pieces (containing the iron) from the non-magnetic waste. How do you think this works? Study the following diagram:

> **Take note**
>
> You may remember some of the different methods of physical separation from previous grades. This was covered in *Matter and Materials*.

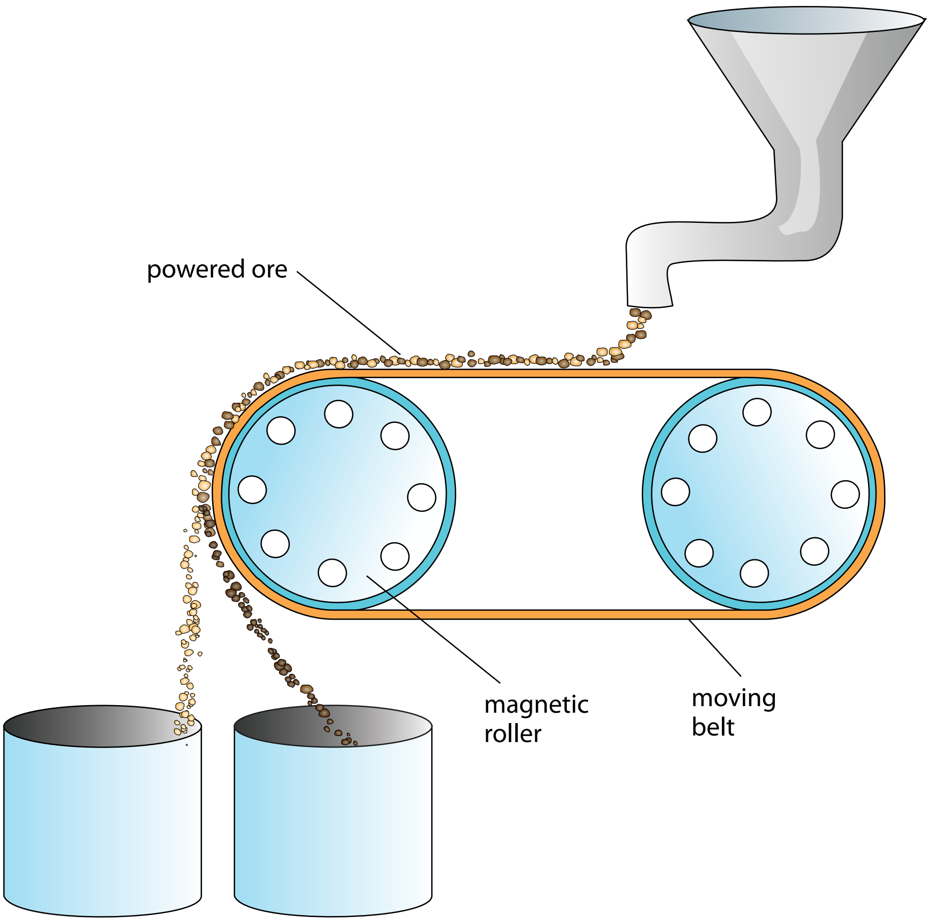


Figure 21.22

Which container, the left or the right, will contain the magnetic iron ore and which one will contain the non-magnetic waste? Copy this diagram into your exercise book, then label it providing a reason for your answer.

Density separation

One of the first methods for mining gold was that of **panning**, a technique where ore is mixed with water and forms a suspension. When it is shaken, the dense particles of gold sink to the bottom and can be removed.

Figure 21.23: Panning.

ACTIVITY Separating beads

In this activity you are going to separate beads as an analogy for separating minerals in the mining industry.

Materials

- collection of beads, different shapes, sizes, densities and magnetic properties
- paint tray
- piece of carpet
- plastic cup and mesh
- magnet
- water

Figure 21.24: Different plastic and metal beads.

Instructions

1. Work in groups of three.
2. Your teacher will indicate to you which bead is the valuable mineral. You need to design a process to separate the valuable mineral from the waste rock.
3. Draw a flow diagram for the process you have designed. Consider using a number of steps in different orders. You may use the same technique more than once.
4. Also remember that repeating a technique improves the efficiency of it. Think about changing the order in which you separate the beads to see if you can find a more efficient process.

Did you know?

When gold was discovered in Pilgrim's Rest in Mpumalanga in the 1840s, they mostly used panning to separate the gold nugget ore from sand and stones in rivers.

5. Hand sorting may NOT be used.
   Use your exercise books to draw a final flow diagram of the process your group has designed.

Questions

1. How did you sort the beads based on size?
2. How did you sort the beads based on shape?
3. How did you sort the beads based on density?
4. How did you sort the beads based on magnetic properties?

Take note

Although hand sorting is an effective method, it is very time consuming, which makes it an expensive process. So it is almost never used in the mining industry, except for diamond sorting.

As you have seen in the activity, separating a mixture can be done using different properties, depending on the different properties of the beads. There could be a number of different ways to separate the beads depending on which type of bead you want to select (considered to be the most valuable ones). Size separation is used frequently in mining to classify ore. For example, when iron ore is exported, it needs to be a certain size to be acceptable to the world market. Coal that is used in power stations also needs to be a certain size so that it can be used to generate electricity effectively.

Density separation is used widely in mining, and you will see why in the next section.

Keywords

- flotation
- physical separation methods
- slurry

Flotation

Flotation makes use of density separation, but in a special way. Chemicals are added to change the surface properties of the valuable minerals so that air bubbles can attach to them. The minerals are mixed with water to make a **slurry**, almost like a watery mud. Air bubbles are blown through the slurry and the minerals attach to the bubbles. The air bubbles are much less dense than the solution and rise to the top, where the minerals can be scraped off easily.

Figure 21.25: Separating minerals by flotation

ACTIVITY Separating peanuts and raisins

You will be working in pairs for this activity. You need to observe carefully and explain your observations.

Materials

- peanuts
- raisins
- soda water
- tap water
- two tall glasses or beakers

Instructions

1. Pour tap water into the first glass until it is about $\frac{3}{4}$ full.
2. Add a handful of the peanuts and raisins mixture to the water and note what happens.
3. Pour soda water into the second glass until it is about $\frac{3}{4}$ full.
4. Add a handful of the peanuts and raisins mixture to the soda water and note what happens.
5. Write down your observations.
6. Explain your observations.

Peanuts and raisins.

Separating peanuts and raisins.

Looking down into the water-filled beaker.

Figure 21.26

Record and explain your observations.

The methods mentioned so far are all **physical separation methods**. Sometimes they are sufficient to separate minerals for use, such as coal or iron ore. But more often the element that we are looking for is found as a chemical compound, and so will have to be separated by further chemical reactions, for example, copper in Cu_2S or aluminium in Al_2O_3.

What is the name for the force that is holding atoms together in a compound?

Once the compound is removed from the ore, the element we want needs to be separated from the other atoms by chemical means. This process forms part of refining the mineral, as you will see in the next section.

Keywords

- bloomery
- bellows
- blast furnace
- brittle
- slag

Did you know?

Iron appears to have been smelted in the West as early as 3000 BC. The start of the Iron Age in most parts of the world coincides with the first widespread use of bloomeries.

22.4 Refining minerals

There are many different methods used to concentrate and refine minerals. The choice of methods depends on the **composition** of the ore. Most of the methods, however, make use of chemistry to extract the metal from the compound or remove impurities from the final product. We will discuss the extraction of iron from iron ore as an example.

Extraction of iron

Iron atoms are found in the compounds FeO, Fe_2O_3 and Fe_3O_4 and in rocks such as haematite and magnetite. South Africa is the seventh-largest producer of iron ore in the world. Iron has been mined in South Africa for thousands of years. South African archaeological sites in KwaZulu-Natal and Limpopo provide evidence of this. Evidence of early mining activities was found in archaeological sites dating mining and smelting of iron back to the Iron Age around 770 AD.

The first iron mining techniques used charcoal which was mixed with iron ore in a **bloomery**. When heating the mixture and blowing air (oxygen) in through **bellows**, the iron ore is converted to the metal, iron. The chemical reaction between iron oxide and carbon is used here to produce iron metal. The balanced chemical equation for the reaction is:

$$2Fe_2O_3 + 3C \rightarrow 4Fe + 3CO_2$$

This extraction method is still used today using the chemistry. However, the bloomery is replaced with a **blast furnace**, a huge oven where iron ore is burned with oxygen and coal to produce the metal, iron. Iron ore, a type of coal called coke (which contains 85% carbon), and lime are added to the top of the blast furnace. Hot air provides the oxygen for the reaction. The temperature of a blast furnace can be up to 1 200 °C.

Figure 21.27: A small bloomery.

The reaction takes place inside the furnace and molten iron is removed from the bottom. Lime (calcium carbonate or $CaCO_3$) is added to react with the unwanted materials, such as sand (silicon dioxide or SiO_2). This produces a waste product called **slag**. The slag is removed from the bottom and used for building roads. Iron is used to make steel. Hot gases, mainly carbon dioxide, escape at the top of the furnace.

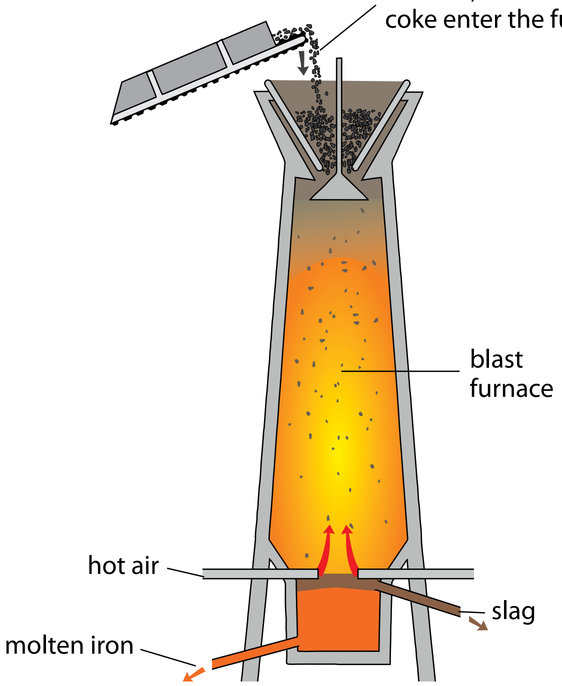


Figure 21.28: Blast furnace.

ACTIVITY Separating lead from lead(II)oxide

In this demonstration you are going to react lead(II)oxide with carbon. This is similar to the process used in iron mining where iron ore is reacted with coke (carbon) to form iron metal.

Materials

- lead(II) oxide (red)
- charcoal block
- Bunsen burner
- blowpipe
- spatula
- safety glasses

Instructions

1. Use safety glasses in this experiment.
2. Use the spatula to scrape a hollow in the charcoal block. Ensure that the loosened carbon remains in the hollow.
3. Add an equivalent amount of lead oxide to the carbon in the hollow.
4. Add a drop or two of water to make a paste.
5. Use a blowpipe to direct the flame of the Bunsen burner into the hollow where the lead(II)oxide-carbon paste is. Create a steady flow of air through the flame.
6. Keep the flame directed onto the paste for 2 – 3 minutes.
7. Observe if any changes have taken place. If not, continue blowing for another minute.

Take note

Remember from Matter and Materials that lead (II) oxide is formed from the reaction between lead (a metal) and oxygen.

Questions

1. What have you observed? Were there any colour changes? Describe your observations.
2. Write a balanced equation for the reaction that has taken place.
3. Carbon dioxide is formed in this reaction. What would the impact of this be if the reaction is done on large scale?

In this experiment carbon was used to remove the oxygen from the lead(II)oxide. The carbon and oxygen form carbon dioxide, and the lead is left behind as a metal. This is the same process that is used in iron extraction in the blast furnace that we discussed above. Coke, which is mainly carbon, removes the oxygens from the iron(III)oxide to form carbon dioxide and leaves behind the iron metal.

Refining iron

The iron that is formed in the blast furnace often contains too much carbon – about 4%, where it should contain not more than 2%. Too much carbon makes the iron **brittle**. To improve the quality of the iron, it needs to be refined by lowering the amount of carbon. This is done by melting the metal and reacting the carbon with pure oxygen to form carbon dioxide gas. In this way the carbon is burned off and the quality of the iron improves. The iron can now be used in the steel-making process. Carbon reacts with oxygen according to the following chemical equation:

$$C + O_2 \rightarrow CO_2$$

Most minerals go through chemical extraction and refining processes to purify them for use in making materials and other chemical products. These are then distributed to where they are needed, for example, coal is distributed to coal power stations and slag is distributed to construction groups for building roads. The mining industry supplies the manufacturing industry and the chemical industry with its raw materials, for example iron is distributed to steel manufacturing industries.

22.5 Mining in South Africa

Long before diamonds were discovered in the Kimberley area and the Gold Rush started in Pilgrim's Rest and the Witwatersrand areas in the late 1800s, minerals were mined in South Africa. At Mapungubwe in the Limpopo Province evidence of gold and iron mining and smelting was found which dates back to the early 11th century AD. However, it was the large-scale mining activities that accelerated the development of the country.

South Africa has a wealth of minerals. We are the world's largest producers of chromium, manganese, platinum, vanadium and alusite; and the second-largest producer of ilmenite, palladium, rutile and zirconium. We are the third-largest coal exporter, fifth-largest diamond producer and seventh-largest iron ore producer. Up to 2010 we were the world's largest gold producer, but our gold production has declined steadily over a number of years. We are currently fifth on the list of gold producers.

Figure 21.29: The Okiep Copper Mine, South Africa, established in the 1850s, is one of the richest bodies of copper ore ever found to this day.

Minerals are mostly found in the northern part of the country. They are often concentrated in specific areas which are linked to the geology of the area.

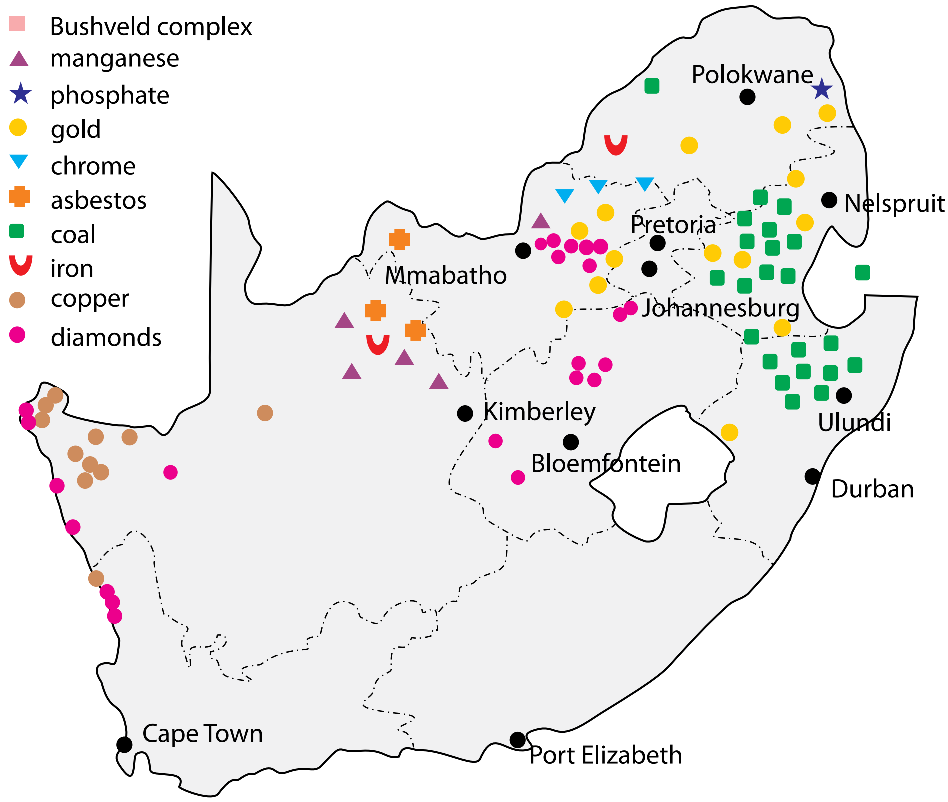


Figure 21.30: Minerals in South Africa

Did you know?

In South Africa in 2012, miners were striking because of dangerous working environments and low wages. A series of devastating and violent incidents followed, resulting in the deaths of 44 people.

The Bushveld Igneous Complex has the world's largest primary source of platinum group metals, indicated on the map in light blue. It is one of the most important mining areas in South Africa because of its abundance of minerals.

Figure 21.31: The Cullinan Premier Diamond Mine, near Pretoria, Gauteng.

ACTIVITY Create your own mining map

Instructions

1. Use the map of South Africa and the data provided in the table below to draw your own map of where mining in South Africa takes place. You will need to find out where the towns are located in South Africa and indicate them on the map.
2. Your teacher will indicate whether you must show all the locations, or a selection of the ones provided.
3. You need to decide on a key for your map and appropriate labels.
4. Also complete the table by filling in the chemical symbols or formulae and answer the questions that follow.

Mineral	Chemical symbol/formula	Where it is found
Lead		Aggenys
Andalusite	Al_2SiO_5	Namakwaland; north of Lydenburg; Eastern Bushveld Complex
Zinc		Aggenys; between Vryburg and Kuruman
Iron		Vredendal; Postmasburg; Sishen/Kathu, Thabazimbi
Salt		Port Elizabeth; Velddrif; between Prieska and De Aar; Douglas; Koffiefontein; Jacobsdal; Petrusburg, Upington

Mineral	Chemical symbol/formula	Where it is found
Limestone		Port Elizabeth; Port Shepstone; Saldanha; Lichtenburg; Mahikeng; Zeerust; between Christiana and Bloemhof; West of Thabazimbi
Vermiculite	$(Mg,Fe^{2+},Al)_3(Al,Si)_4O_{10}(OH)_2 \cdot 4(H_2O)$	Between Pietermaritzburg and Durban; east of Musina; west, south and east of Makhado; Phalaborwa
Diamonds		Kimberley; northwest of Kimberley; Alexander Bay; Luderitz; Port Nolloth; on the west coast north of Vredendal; Mahikeng; north of Ventersdorp; Cullinan; west of Musina
Titanium		West coast north of Saldanha Bay; Richards Bay
Manganese		North of Kuruman; northeast of Ventersdorp
Zirconium		West coast north of Saldanha Bay; Richards Bay
Gold		Virginia; Welkom; Stilfontein; Klerksdorp; Potchefstroom; Carletonville; Johannesburg, Vereeniging; Vryheid, Barberton; west of Phalaborwa; Evander
Chromium		Western Bushveld Complex; Eastern Bushveld Complex
PGMs		Western Bushveld Complex; Eastern Bushveld Complex; Northern Bushveld Complex
Phosphate		Phalaborwa
Coal		Virginia; Welkom; Bothaville; Kroonstad; Vereeniging; Sasolburg; Vanderbijlpark; Dundee; Newcastle; Utrecht; Vryheid; Ermelo; Standerton; Secunda; Evander; Witbank; Middleburg; Carolina; Lephalale
Nickel		West of Barberton
Copper		Aggenys/Springbok; Phalaborwa; Western Bushveld Complex; Eastern Bushveld Complex
Antimony		West of Phalaborwa

Questions

1. What mineral(s) are mined closest to where you live?
2. What do you notice about the gold mines in South Africa?
3. There are two types of diamond mining, alluvial (which is found on the coast or in inland rivers which have washed through kimberlite pipes) and kimberlite (which is found inland). What is the link between these two types of diamond mining?
4. Which mining industry do you think is the best or most important one in South Africa? Give a reason for your answer.

ACTIVITY Drawing a mining flow diagram

Instructions

1. Choose one mining industry mentioned in this unit. It can be the one you are doing your research project on.
2. Draw a flow diagram to show the different steps in the mining of the chosen mineral.
3. End your flow diagram with something or somewhere where this mineral is used in real life. Look at the beginning of the unit for a generic flow diagram for mining. Draw your flow diagrams in your books.

The impact of mining

Mining has played a major role in the history of South Africa. It accelerated technological development and created infrastructure in remote areas in South Africa. Many small towns in South Africa started because of mining activity in the area. It also created a demand for roads and railways to be built. Most importantly, it created job opportunities for thousands of people. Even today many households are dependent on mining activities for jobs and an income. Mining is an important part of our economic wealth. We export minerals and ore to many other countries in the world.

Mining activities also have a negative impact on the environment. In many cases the landscape is changed. This applies particularly to surface mines (open pit mines), where large amounts of soil and rock must be removed in order to access the minerals.

Take note

Tailings are the materials left over after separating the valuable minerals from the ore.

The shape of the landscape can be changed when large amounts of rocks are dug up from the Earth and stacked on the surface. These are called mine dumps. Open pit mines also create very large unsightly and dangerous holes (pits) in the ground that change the shape of the land.

Air and water pollution can take place if care is not taken in the design and operation of a mine. Dust from open pit mines, as well as harmful gases such as sulphur dioxide and nitrogen dioxide, could be released from mining processes and contribute to air pollution. Mining activities produce carbon dioxide. Trucks and other vehicles give off exhaust gases. If the mining process is not monitored properly, acid and other chemicals from chemical processing can run into nearby water systems such as rivers. This is poisonous to animals and plants, as well as to humans, who may rely on that water for drinking.

An example are pollutants (dangerous chemicals), called tailings, left over from gold mining, which pose a threat to the environment and the health of nearby communities. Dangerous waste chemicals can leak into the groundwater and contaminate water supplies if the tailings are not contained properly.

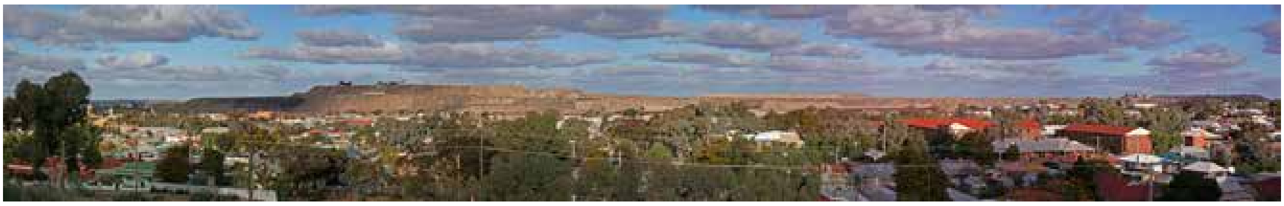


Figure 21.32: An aerial photograph of Primrose Gold Mine. Can you see the piles of gold tailings on the left of the photograph?

ACTIVITY What would we do without mining?

Instructions

1. Imagine that all the mines in South Africa close down. What do you think would be the impact on the points outlined below?
   a) Carbon emissions
   b) Jobs
   c) Economy
   d) Future of small towns
   e) Add one of your own issues here.
2. Discuss the following aspects with your group.
3. Present your discussion to the class in a few short sentences on each issue.

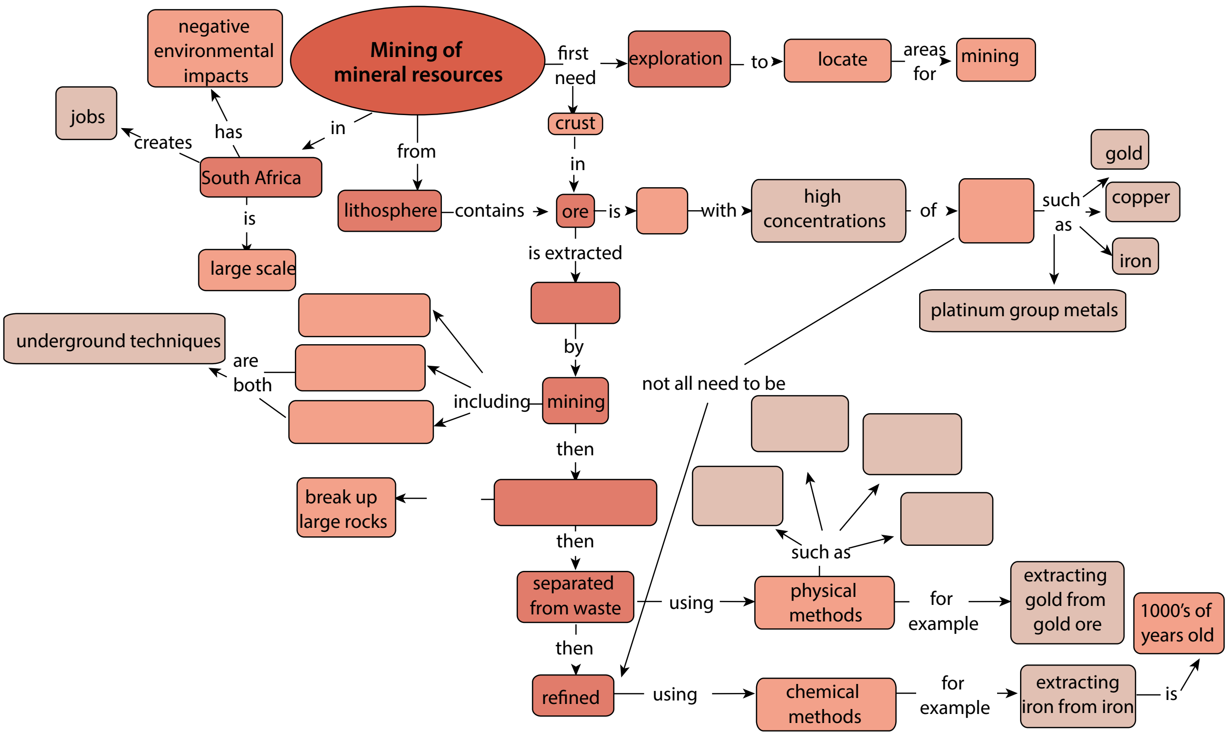
Summary

Key concepts

- People extract valuable minerals from the lithosphere.
- Rock that contains high concentrations of valuable minerals are called ore.
- Various methods are used to locate potential sites for mining.
- Ore is removed from the crust by mining, either on the surface (open-pit mining) or underground (shaft mining or room-and-pillar mining).
- Some minerals can be used in their natural form, for example sand in the building industry, phosphate rock for fertilisers, and diamonds in jewellery.
- Some minerals require a physical and/or chemical process to remove them from the ore.
- Large rocks containing minerals need to be crushed and milled.
- The valuable minerals are then separated from the rock using a variety of physical and chemical separation methods.
- People have extracted minerals, for example iron and copper, from ores for thousands of years.
- Examples of how minerals were mined long ago can be found at archaeological sites in South-Africa, such as Mapungubwe.
- Today iron is extracted using coke (carbon) to make steel.
- South Africa has a large mining industry.
- The industry creates jobs and contributes to the economy.
- The mining industry has a significant impact on the environment.

Concept map

Use the following concept map to summarise what you have learnt in this unit about mining of mineral resources. What are the three types of mining that we discussed in this unit? Fill these into the concept map. Remember that you can add in your own notes to these concept maps, for example, you could write more about the environmental impacts of mining.



Revision

1. Bauxite is an aluminium ore that contains four different minerals: Al, SiO_2, TiO_2 and Fe_2O_3.
   a) What are the chemical names of each of these minerals? [4]

Bauxite.

   b) Bauxite is found close to the surface of the Earth. What type of mining would you expect to be used in bauxite extraction? [1]
   c) What is the common name for SiO_2? [1]
   d) SiO_2 is present as unwanted material in the iron blast furnace and needs to be removed. How is it removed and what is the waste product used for? [2]
   e) Bauxite also contains iron(III)oxide. Write down the common name for iron(lll)oxide. [1]
   f) Suggest one way of separating the iron(III)oxide from the rest of the minerals in bauxite. Give a reason for your answer. [2]

2. Explain how iron is extracted from iron ore. [6]

3. Write a balanced equation for the extraction of iron from its ore. [3]

4. What is the environmental impact of the iron mining process? [2]

5. Case study: Read the following article and answer the questions that follow.

The story of Loolekop

Phalaborwa is home to one of the largest open-pit mines in the world. The original carbonate outcrop was a large hill known as Loolekop. Archaeological findings at Loolekop revealed small-scale mining and smelting activities carried out by people who lived there long ago. An early underground mine shaft of 20 metres deep and only 38 centimetres wide were also found. The shafts contained charcoal fragments dating the activities to 1 000 – 1 200 years ago.

In 1934 the first modern mining started with the extraction of apatite for use as a fertiliser. In 1946 a well known South African geologist, Dr. Hans Merensky, started investigating Loolekop and found economically viable deposits of apatite in the foskorite rock. In the early 1950s a very large low-grade copper sulfide ore body was discovered.

In 1964 the Phalaborwa Mine, an open-pit copper mine, commenced its operations. Today the pit is 2 km wide. Loolekop, the large hill, has been completely mined away over the years. A total of 50 different minerals is extracted from the mine. The northern part of the mine is rich in phosphates and the central area, where Loolekop was situated, is rich in copper. Copper with the co-products of silver, gold, phosphate, iron ore, vermiculite, zirconia and uranium are extracted from the rocks.

The open-pit facility closed down its operation in 2002 and has now been converted to an underground mine. This extended the lifetime of the mine for another 20 years. The mine employs around 2 500 people.

2 000 million years ago this area was an active volcano. Today the cone of the volcano is gone and only the pipe remains. The pipe is 19 km^2 in area and has an unknown depth, containing minerals such as copper, phosphates, zirconium, vermiculite, mica and gold.

This mine was a leader in the field of surface-mining technology with the first in-pit primary crushing facility. This meant that ore was crushed by jaw crushers before being taken out of the mine. They also used the first trolley-assist system for haul trucks coming out of the pit. Today the mine has secondary crushing facilities, concentrators, and a refinery on site.

In 1982 a series of cavities with well-crystallised minerals were discovered, for example calcite crystals up to 15 cm on edge, silky mesolite crystals of up to 2cm long and octahedral magnetite crystals of 1 – 2 cm on the edge.

Mesolite crystals

Hematite crystals

Calcite crystals

a) What type of rock would you find in the Phalaborwa mine? Give a reason for your answer. [2]
b) Why did the open pit facility closed down in 2002? [2]
c) What is phosphate rock used for? [1]
d) What has the impact of the Phalaborwa Mine been on the landscape? How was it possible for very large crystals to form? [2]
e) What are the environmental impacts of open pit mining? Name any three. [3]

Total [32 marks]

23 The atmosphere

Key questions

- What is the atmosphere?
- What makes up the atmosphere?
- Does the atmosphere change as you go further from the Earth's surface?
- Can the atmosphere be divided into different layers?
- Where does the atmosphere end?
- What important aspect of the atmosphere allows life to exist on earth?
- What is the greenhouse effect?
- How do humans contribute to the greenhouse effect?

In the first unit of *Earth and Beyond*, you learnt about the different spheres of the Earth. The atmosphere was mentioned briefly in Unit 20. In this unit we will look at the atmosphere in more detail.

Keywords

- atmosphere
- troposphere
- stratosphere
- mesosphere
- thermosphere
- exosphere
- altitude
- temperature gradient
- density

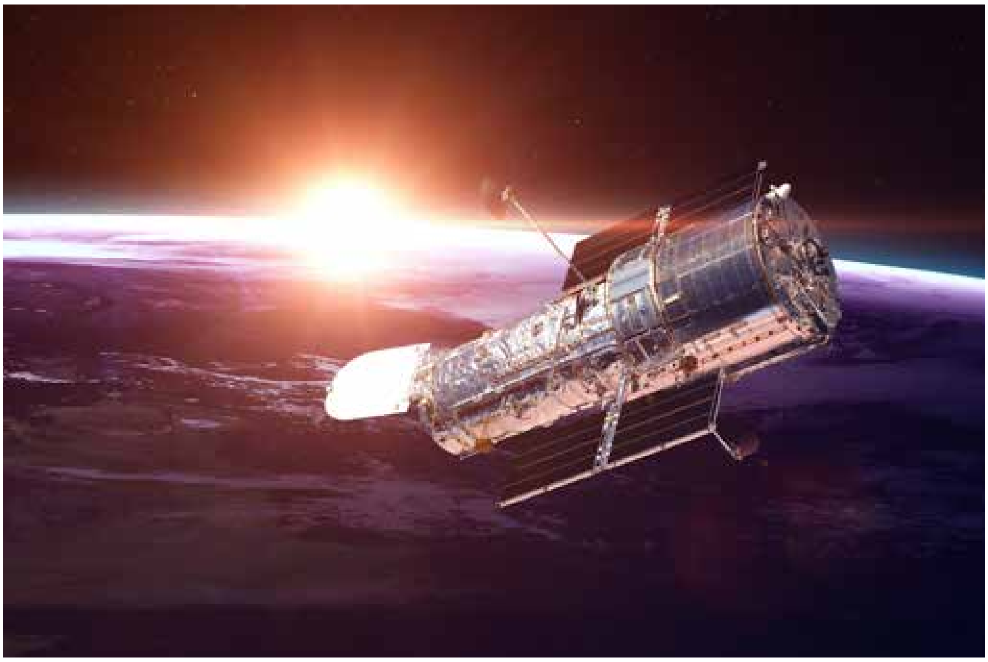


Figure 23.1: Photograph of the Earth's atmosphere taken from the International Space Station. You can see the curved Earth below the bright atmosphere – a very thin layer of gases around the Earth. Above and beyond the atmosphere is where we find outer space. The bright spot is the Sun just going below the horizon.

23.1 The atmosphere

The atmosphere is the layer of gases which surrounds the Earth. It contains the following mixture:

- nitrogen (78,08%)
- oxygen (20,95%)
- argon (0,93%)
- carbon dioxide and other trace gases (0,04%).

Take note

The exosphere is not considered part of the atmosphere because of the very low density of gases, but it is still briefly discussed in this unit. In some other resources which you may look at, the exosphere is sometimes discussed as a layer or upper limit of the atmosphere.

The gas molecules in the atmosphere are kept close to the Earth by gravity. The effect of gravity means that there will be more gas molecules closer to the Earth's surface than further away. As you move further and further away from the surface of the Earth, the gas molecules become fewer and the spaces between the molecules become larger, until there are no more gas molecules and only spaces are left. The atmosphere therefore does not have a set boundary, but rather fades away into space.

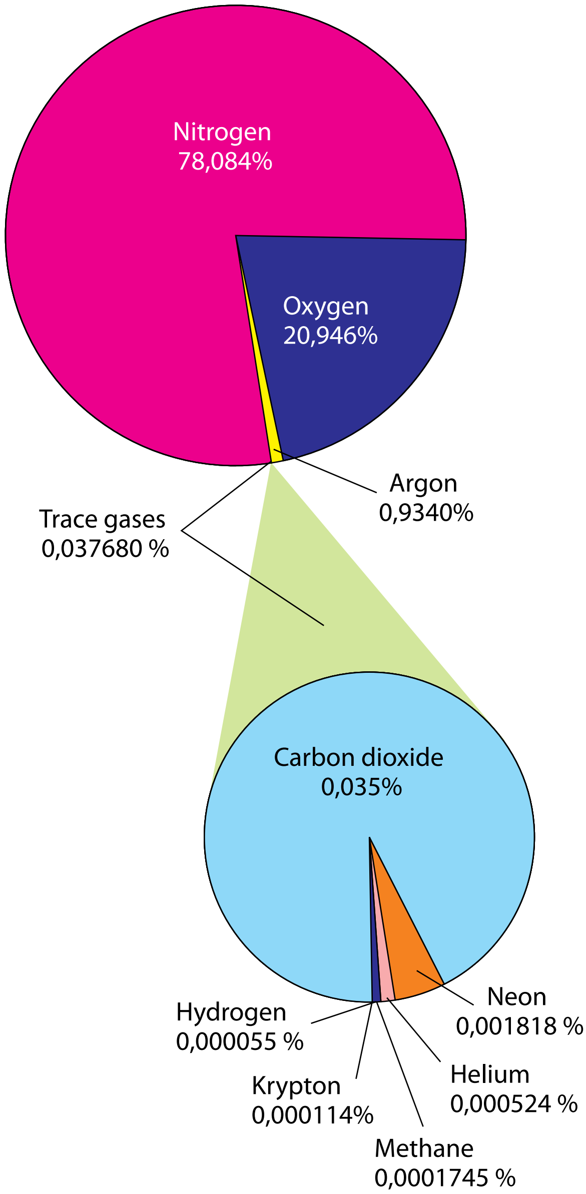


Figure 23.2: The composition of the atmosphere by volume.

Density and altitude

When we walk up a very high mountain, there is less oxygen present. We may feel out of breath. People sometimes say that the air is thinner higher up. When they say this, they mean that there is a lower concentration of oxygen molecules.

Take note

Sports teams and athletes need to acclimatise before performing when they get to a new location at altitude, so that their bodies can get used to the lower level of oxygen.

The **density** of the atmosphere decreases with an increase in the height above sea level (altitude).Density is an indication of how many particles are present in a specific volume of gas. When the density is high, there are lots of gas molecules present. If the density is low, there are fewer gas molecules present.

The atmosphere is a very important part of the Earth. It keeps the planet warm and protects us from the harmful radiation of the Sun. It also ensures a healthy balance between oxygen and carbon dioxide so that life can be sustained on the planet.

The layers of the atmosphere

The atmosphere has four main layers. We start measuring these from sea level and move towards space. The diagram alongside illustrates this. The surface of the Earth is at the bottom of the diagram, with Mount Everest drawn in. The first layer is the **troposphere**, then the **stratosphere**, the **mesosphere** and the **thermosphere**. Above the thermosphere, the atmosphere merges with outer space in the layer known as the exosphere.

The atmosphere is actually a very thin layer compared to the size of the Earth. It is almost like the skin of an orange, relative to the size of the orange.

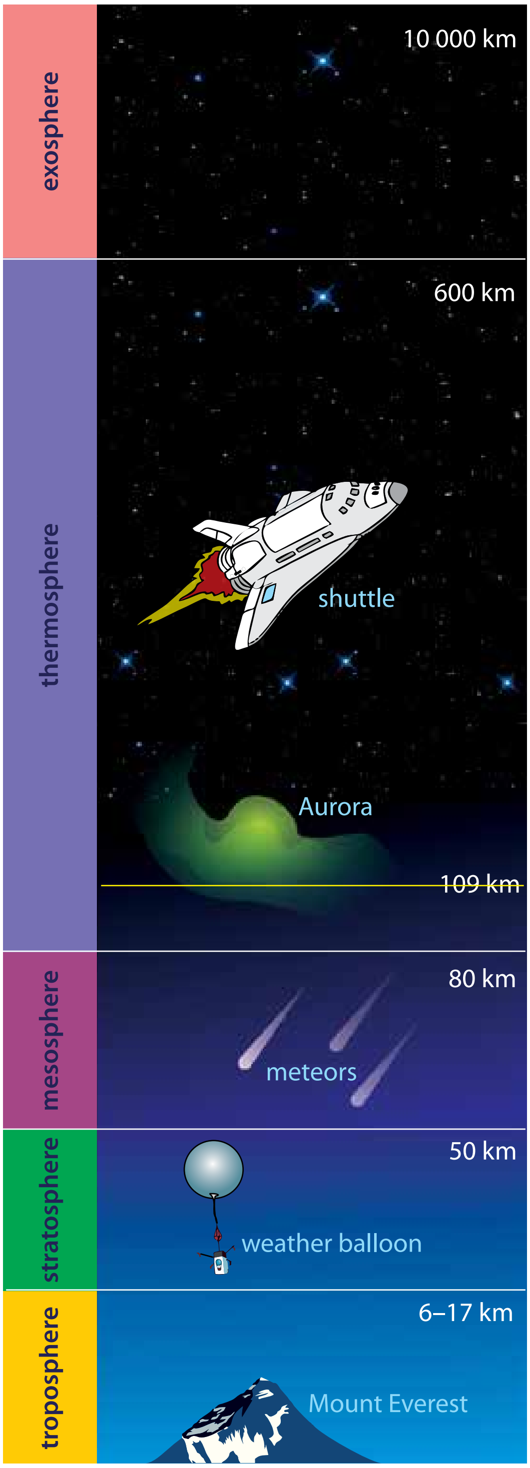


Figure 23.3: Space Shuttle Endeavour in between the stratosphere (white layer) and mesosphere (blue layer). The orange layer is the troposphere.

Figure 23.4: The layers of the atmosphere, and the exosphere.

ACTIVITY How thick is the atmosphere compared to the size of the Earth?

Questions

1. You need to draw a scale diagram to show how thick (or thin) the atmosphere is in comparison to the size of the Earth. Use graph paper.

Did you know?

Some endurance athletes spend several weeks training at high altitudes, preferably 2 400 m above sea level, so that their bodies adapt by producing more red blood cells. This gives them a competitive advantage when returning to a lower altitude to compete.

Take note

Altitude is a measure of height above sea level.

2. Answer the following questions to help you draw the diagram.
   a) What is the radius of the Earth? Choose an appropriate scale and draw a circle in your notebook to represent the Earth.
   b) How thick is the atmosphere in km? Use the same scale as above and draw the atmosphere around the Earth.
   c) Indicate the atmosphere density gradient on your diagram.

Each layer of the atmosphere has a different **temperature gradient**, in other words, the temperature changes gradually as you move through each layer. The following graph shows how the temperature changes as you move through the atmosphere. The layers of the atmosphere are also indicated on the graph.

Temperature is on the *x*-axis and altitude is on the *y*-axis. The red line shows the change in temperature. Note that as you move further to the left on the graph it is colder, dropping far below 0 °C, and further to the right is hotter, reaching over 1 000 °C.

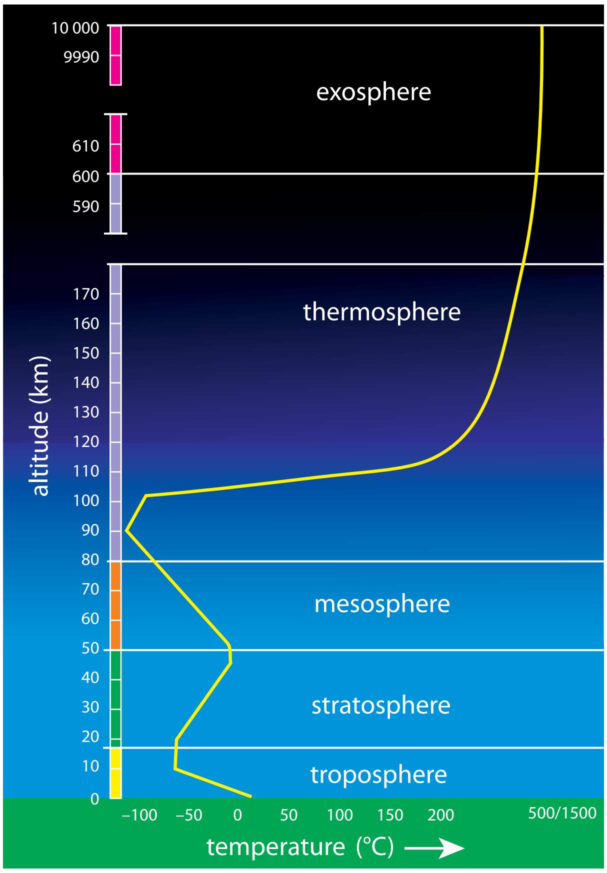


Figure 23.5: The average temperature profile of the Earth's atmosphere and the exosphere.

Now let's look at each of the layers of the atmosphere.

23.2 The troposphere

The troposphere is the lowest layer in the atmosphere. It stretches from sea level up to about 9 km at the poles and 17 km at the equator. As a result of the Earth's rotation, the atmosphere is thicker at the equator than at the poles. On average it is about 12 km thick.

The density of air decreases as you move further away from the surface of the Earth. The first two layers of the atmosphere contain most of the mass of the atmosphere. The bottom part of the troposphere has a high enough density for us to breathe and is the layer of the atmosphere in which we live.

Figure 23.6: Shown here is the orange-coloured troposphere, the lowest and most dense portion of the Earth's atmosphere above the Earth's surface, with the Moon above.

The air in the troposphere is in constant motion. As it is warmed by the Earth, the warm air moves away and gets replaced by cooler air which travels in convection currents. This is the basis for cloud formation and weather patterns.

All the Earth's weather systems take place close to the Earth in the troposphere.

Figure 23.7: Cloud formations typical of a tropical cyclone. This one was photographed approaching the south-eastern coast of Brazil.

Figure 23.8: Clouds forming in the troposphere.

The temperature in the troposphere decreases with altitude – the further you move away from the surface, the colder it becomes. The temperature decreases about 6,4 °C for every kilometre increase in altitude. In the following activity you will investigate the change in temperature as height above sea level increases.

ACTIVITY Drawing a graph of the temperature gradient in the troposphere

Questions

1. Using the information in the previous text, set up your own table displaying the temperature change in the troposphere from 0–12 km.
2. Then draw a neat, accurate graph of this data.
3. Assume that the average temperature on the surface of the Earth is 16 °C.
4. Choose an appropriate scale for the *x*- and *y*-axes of your graph.
5. Label the axes and give the graph a heading.
6. Draw a table for your data.
7. Draw your graph.

Take note

Troposphere comes from the Greek word 'tropein', meaning to change, circulate or mix.

The temperature in the troposphere decreases steadily until it reaches about –60 °C at about 10 – 12 km above sea level. The temperature here stabilises before it increases again. This is the transition zone between the troposphere and the stratosphere. This layer forms an invisible barrier which prevents the warmer moist air from escaping from the troposphere. Beyond this region air does not circulate much and weather patterns are not found any more.

Keywords

- ozone
- CFCs

23.3 The stratosphere

The stratosphere is the layer above the troposphere. It stretches from 12 km to 50 km above the surface of the Earth. 90% of the mass of the atmosphere is found in the troposphere and the stratosphere.

Aeroplanes fly in the lower stratosphere because the air is much more stable than in the troposphere. The density of the air in the stratosphere is very low and decreases with altitude.

Scientists use **weather balloons** to gather information on the temperature and pressure as they move up from the Earth's surface to the stratosphere. A weather balloons carries a small device, called a radiosonde, which sends back information on atmospheric pressure, temperature, humidity and wind speed.

Did you know?

Weather balloons were first used 70 years ago, and are still the key instrument for meteorologists to assess and predict the weather. This information is used in many ways, for example, to compile the weather report on TV or to warn of flooding or hurricanes.

Figure 23.9: A weather balloon being released from a US Navy ship.

Figure 23.10: A photograph taken by a weather balloon 30 km above Earth in the stratosphere.

Weather balloons are filled with helium or hydrogen and rise higher and higher into the atmosphere. Do you think they continue rising up for ever? What do you think happens to the balloon as it increases in altitude? Hint: Think of what happens to the gas inside the balloon as the altitude increases. Discuss this with your class and write some notes in your exercise books.

Ozone gas (O_3) is found in the stratosphere. Ozone gas is made up of ozone molecules. Each molecule consists of three oxygen atoms. Ozone plays an important role in absorbing harmful UV rays from the Sun by forming, breaking down and reforming ozone molecules over and over again. When UV light reaches the Earth, it can cause cancer and affect plant growth, and the life cycles of species.

Take note

Although ozone is considered a pollutant in the troposphere, in higher altitudes in the stratosphere, ozone is considered vital as it protects Earth from too much ultraviolet radiation.

What happens to ozone in the atmosphere?

The formation and destruction of ozone is a natural process that takes place in the stratosphere. Oxygen forms ozone, and ozone breaks apart again to form oxygen. The following diagram shows the reactions that take place.

Did you know?

The ozone hole is an annual thinning of the ozone layer over Antarctica, caused by chlorine from CFCs in the stratosphere.

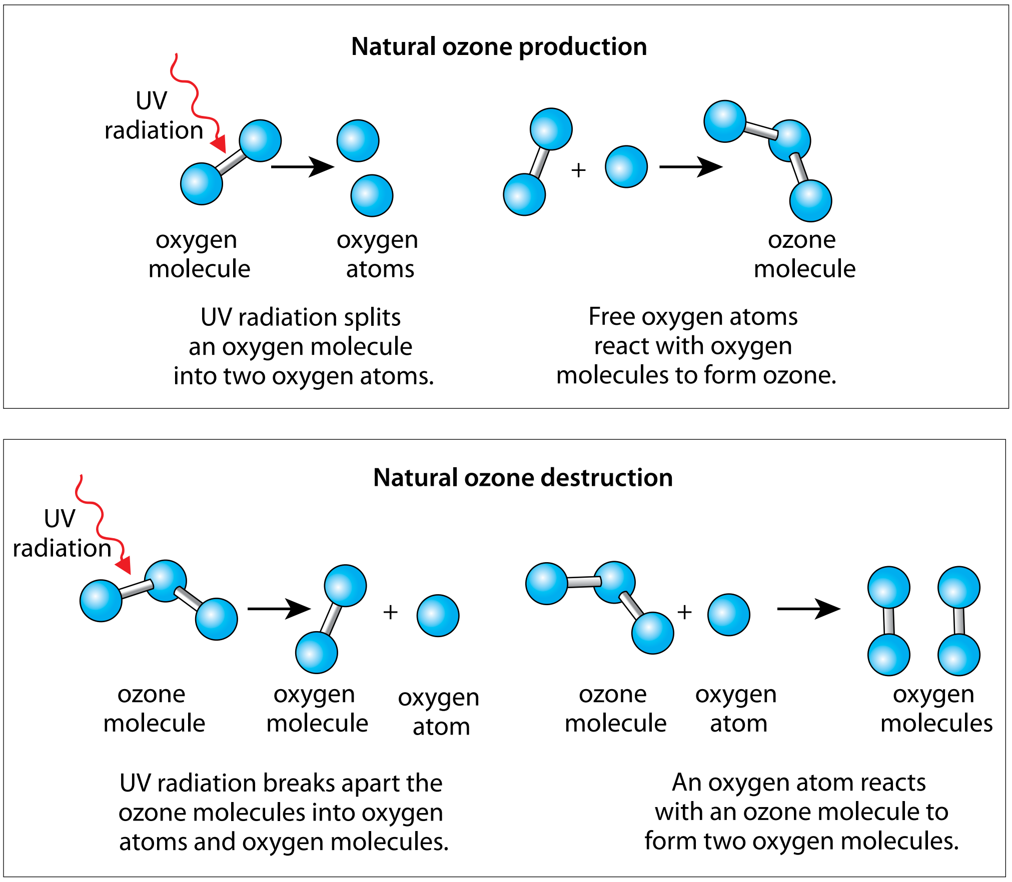


Figure 23.11: Production and destruction of ozone.

What holds the oxygen atoms together in a molecule?

What is the term given to a molecule of oxygen which consists of two atoms of the same element bonded together?

The ozone reactions lead to the heating of the stratosphere, increasing the temperature from –60 °C to about 0 °C. As a result, the air becomes warmer as you move further away from the Earth in the stratosphere. The problem arises when there are molecules present which interfere with these natural

> **Did you know?**
>
> NASA is thinking of sending high-altitude weather balloons to probe the atmosphere of Mars.

processes. Chlorofluorocarbons, or CFCs, are molecules which release chlorine atoms into the stratosphere. Chlorine atoms react with ozone, destroying it before it can absorb harmful UV rays. The following diagrams show how CFCs react with ozone.

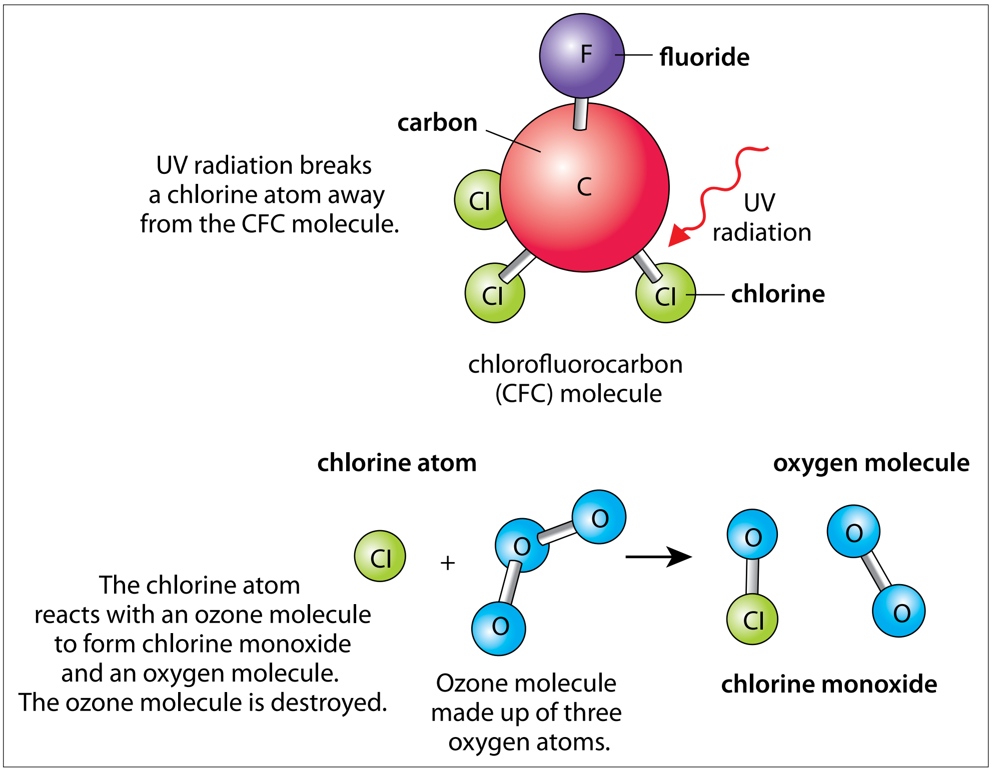


Figure 23.12: CFC's react with ozone.

CFCs used to be found in aerosols and refrigerator gas, and were given off by industrial processes. Scientists noticed that these gases interfered with ozone. This could have had a serious impact on life on Earth and the use of CFCs was banned.

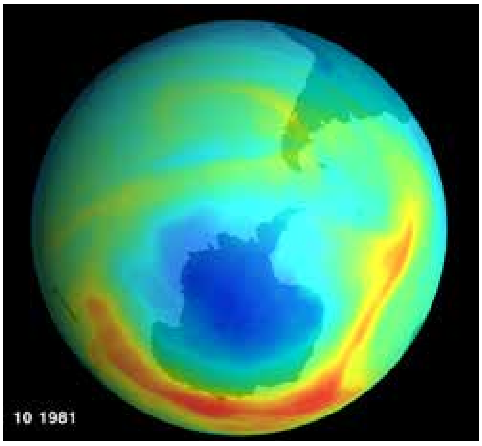


Figure 23.13: In 1985, a British scientist working in Antarctica discovered a 40 percent loss in the ozone layer over the continent. In the 1990s, this prompted a worldwide ban on CFCs. In this image, the blue/purple areas show low ozone, while the red areas indicate higher ozone levels.

> **Did you know?**
>
> On 14 October 2012 Felix Baumgartner set a world record by jumping from an altitude of about 39 km – from the stratosphere to the Earth. He is the first man to jump from the stratosphere.

23.4 The mesosphere

The mesosphere extends from around 50 km to 80 km above the Earth's surface. The atmosphere reaches its lowest temperature (–90 °C) in the mesosphere. The air density is extremely low, but there is still enough air to burn up rocks and dust entering from space.

A **meteor** is a rock that enters the atmosphere from space. It travels at extremely high speed, up to 30 000 m/s. As a meteor enters the atmosphere, the air in front of it is compressed. The air heats up and the meteor burns up as a result of heat and friction. When we look up at the night sky, we may see a streak of light flashing for a brief moment. This is commonly called a shooting star, but is in fact a meteor burning up in the mesosphere.

Figure 23.14: A meteor is a rock burning up in our atmosphere.

Most meteors are fairly small and burn up completely while whizzing through the mesosphere. Some of the larger, denser meteors can reach the Earth and are then called **meteorites**. When the meteorite strikes the ground, it kicks up dust and soil and leaves an impact crater on the Earth's surface. The size of the crater depends on the size, density and speed of the meteorite.

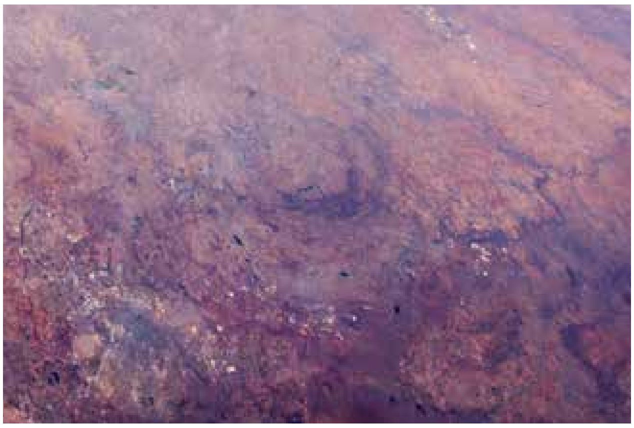


Figure 23.14: The impact crater at Vredefort in South Africa has a diameter of 300 km, and nearly fills the complete aerial photograph shown here.

Figure 23.15: The Tswaing Crater, just north of Pretoria, is 1,1 km wide and formed as the result of a meteorite impact about 200 000 years ago.

23.5 The thermosphere

The thermosphere is the layer of the atmosphere from 80 km upwards. The density of the air is extremely low. The further away you move from the Earth, the less dense the concentration of molecules becomes until the atmosphere becomes space. Most satellites that we depend on every day are in Low Earth Orbit (LEO), orbiting the Earth at an altitude between 160 km and 2 000 km. The **International Space Station** (ISS) is situated at 370 km into the thermosphere. This is an international facility in space that is used for research purposes.

Keywords

- meteor
- meteorite
- ionosphere
- aurora
- northern lights (Aurora Borealis)
- southern lights (Aurora Australis)
- International Space Station

Did you know?

In 2002, Mark Shuttleworth became the first South African in space when he launched with a Russian space mission. He spent eight days on board the International Space Station, participating in experiments related to AIDS and genome research.

Figure 23.16: The International Space Station orbits the Earth in the thermosphere.

The temperature in the thermosphere increases from –90 °C to as high as 1 500 °C. The thermosphere is very sensitive to an increase in energy, and a small change in energy results in a high temperature increase. At times of increased solar activity, the temperature can easily increase up to 1 500 °C. However, the thermosphere will feel cold as there are few particles present to collide with our skin and transfer enough energy for us to feel the heat.

High energy light (for example, UV light) can cause atoms or molecules to lose electrons, forming ions. The region where this takes place is called the **ionosphere**. The ionosphere is found mainly in the thermosphere. The Sun also gives off charged particles (the solar wind), which can enter the Earth's atmosphere (mostly near the poles) and react with the ions and electrons in the ionosphere, causing a phenomenon called an **aurora**. It is a colourful display of light in the sky at the poles. In the northern hemisphere, it is called the **northern lights** or **Aurora Borealis**, and the **southern lights** or **Aurora Australis** in the southern hemisphere.

The ionosphere reflects longer wavelength radio waves, for example the radio waves we use for radio and surface-broadcast television (not satellite television), allowing the signal to be broadcast over a larger distance. The ions in the ionosphere also absorb ultraviolet radiation and X-rays. The region beyond the thermosphere is called the **exosphere**. This layer has very few molecules and extends into space.

Figure 23.17: Sunset from the International Space Station. The troposphere is the deep orange and yellow layer. Several dark clouds are visible within this layer. The pink-white layer above is the stratosphere. Above the stratosphere, blue layers show the mesosphere, thermosphere (dark blue) and exosphere (very dark blue), until it gradually fades to the blackness of outer space.

ACTIVITY How thick are the layers of the atmosphere?

In this activity you will build a model to represent the different layers of the atmosphere. In addition to the model, you need to draw an accurate diagram in your workbook to represent the thickness of each layer. Use a ruler to draw an accurate scale diagram.

Figure 23.18: Beans, samp, popcorn and dried peas.

Materials

- large measuring cylinder or tall drinking glass
- corn kernels (popcorn)
- samp
- dried peas
- beans
- ruler

Instructions

1. Add a 0,5 cm layer of dried split peas to represent the troposphere (1 layer of peas thick).
2. Add a 1,5 cm layer of corn kernels on top of the peas to represent the stratosphere.
3. Add a 1,5 cm layer of samp on top of the corn kernels to represent the mesosphere.
4. Add a 24 cm layer of beans on top of the samp to represent the thermosphere.

Your column should look something like this.

A close-up photograph of the layers.

Figure 23.19

You will notice that the area where the two layers meet is not always clear cut. The kernels may have mixed a little bit. The atmosphere is the same. There is a not a clear line separating two layers, but they mingle in the area of contact.

Table showing the heights of the layers in Earth's atmosphere and in the model:

Layer	Represented by	Height of layer (km)	Height of layer (cm)
Troposphere	Dried split peas	10	0,5
Stratosphere	Corn kernels	30	1,5
Mesosphere	Samp	30	1,5
Thermosphere	Beans	480	24

Questions

1. Draw a labelled diagram of the model using a graph paper. Include a scale. The density of the atmosphere decreases with altitude. Show this on your diagram as well.
2. What atmospheric layers are represented by the different grains in the model?
3. In the model in the activity, how many kilometres does 1 cm represent?
4. How much thicker is the stratosphere compared to the troposphere?
5. How much thicker is the thermosphere compared to all the other layers combined?
6. Where in this model would you expect to find clouds?
7. Where in this model would you expect to find the Drakensberg Mountains?
8. Where in this model would you expect to find a satellite?
9. Where in this model would you expect to find meteors burning up?
10. In which layer is there life? What is different about this layer?

Keywords

- greenhouse gases
- greenhouse effect
- global warming
- climate change
- carbon dioxide
- methane
- water vapour
- radiation

23.6 The greenhouse effect

You have learned a lot about greenhouse gases in Natural Sciences. In this section we will be looking at how important greenhouse gases are to sustain life on Earth.

Earth's atmosphere contains mostly (99%) nitrogen and oxygen, but a small percentage (1%) of the atmosphere contains gases such as **water vapour** (H_2O), **carbon dioxide** (CO_2), and **methane** (CH_4). Carbon dioxide is a product of respiration in all organisms and also a gas given off by industrial processes and the burning of fossil fuels and vegetation.

Methane is a gas, also called natural gas, which occurs in reservoirs beneath the surface of the Earth. It is also given off by decomposing plant and animal material, and animals give off methane as part of their digestion. Water vapour is formed when water evaporates on Earth.

Water vapour, methane and carbon dioxide are gases which let through incoming visible light from the Sun. The incoming radiation from the Sun is absorbed by the Earth's surface and warms it. The Earth's surface emits **infrared radiation**. Infrared radiation is absorbed by the greenhouse gases and re-emitted in all directions. This increases the temperature of the Earth's surface and lower atmosphere, above what it would be without the gases, called the **greenhouse effect**. These gases play a very important role in regulating the Earth's temperature.

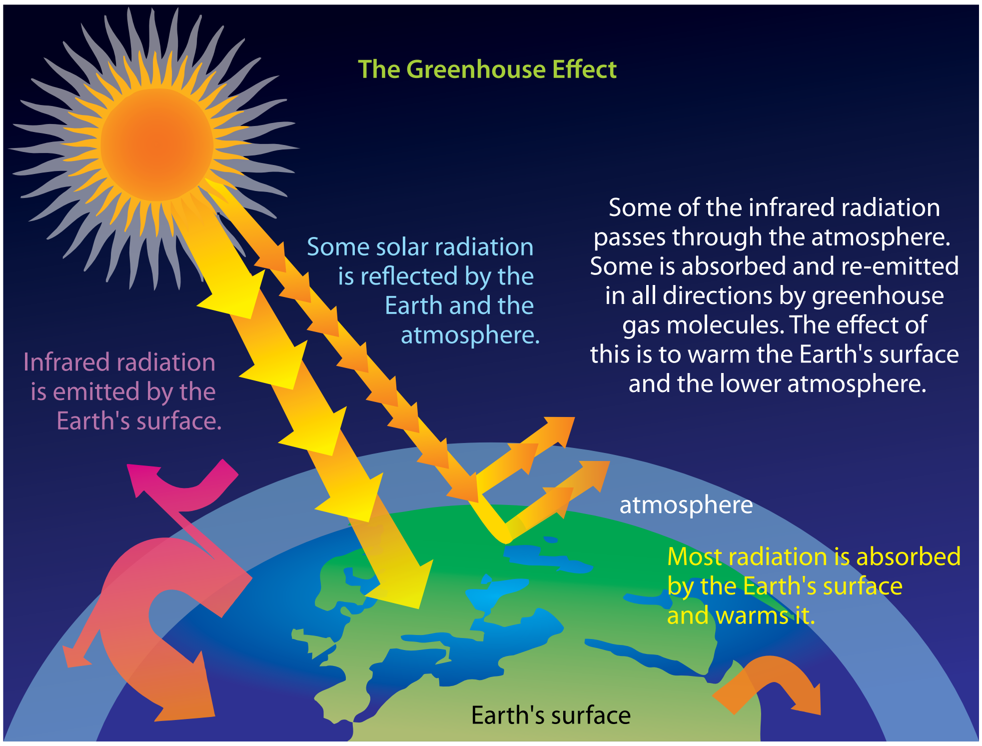


Figure 23.20: The greenhouse effect.

As you can see in the diagram, the radiation from the Sun is able to reach the Earth and warm it up. The energy that is given off by the Earth is trapped by the water vapour, carbon dioxide and methane. This ensures that the Earth stays warm. It is almost as if the gases form a blanket around the Earth keeping some of the heat inside. The gases are referred to as **greenhouse gases**. A greenhouse is a glass structure that is used to grow plants. The glass lets the heat of the Sun through, but then keeps the heat inside the structure so that the plants have a moderate climate in which to grow. Water vapour, carbon dioxide and methane act in the same way.

The Earth is a very unique planet because of the make-up of its atmosphere. In this unit you have learned about the composition of the Earth's atmosphere. Let us compare the atmosphere of Earth to its neighbouring planets, Mars and Venus.

ACTIVITY Comparing Venus, Earth and Mars

Figure 23.21: Venus, Earth and Mars.

Instructions

1. The table below gives information about the gases in the atmospheres of the three planets: Venus, Earth and Mars.
2. Study the table and answer the questions that follow. Percentage of gases making up the atmospheres of Venus, Earth and Mars.

	Venus	Earth	Mars
Carbon dioxide (CO_2)	96,5%	0,03%	95%
Nitrogen (N_2)	3,5%	78%	2.7%
Oxygen (O_2)	Trace	21%	0,13%
Argon (Ar)	0,007%	0,9%	1,6%
Methane (CH_4)	0	0,002%	0

Questions

1. Compare the data for Venus and Earth. What similarities and differences do you notice?
2. Compare the data for Venus and Mars. What similarities and differences do you notice?
3. What is the biggest difference between Earth's atmosphere and the atmospheres of the other two planets?
4. Why is the level of oxygen so much higher on Earth than on the other two planets?
5. Why do you think there is no methane gas on Venus and Mars?
6. Predict whether you think the temperature on the surface of Venus will be low or high. Give reasons for your answer.

The atmospheres of Venus and Mars are very similar. Both planets have mainly carbon dioxide in the atmosphere, and very few other gases. However, the two planets are quite different.

Did you know?

Venus is the hottest planet in the solar system; the temperature is hot enough to melt lead.

Venus has a very dense atmosphere which results in a high concentration of carbon dioxide on its surface. This causes an extreme greenhouse effect and very high temperatures on the surface of Venus. Venus has an average surface temperature of 462 °C. This is too high to sustain life as we know it.

Mars, on the other hand, has almost no atmosphere, so, although there is carbon dioxide present, the density is very low and almost no greenhouse effect takes place. Mars is also much further away from the Sun. It is a very cold planet, with an average temperature of –55 °C. This is too low to sustain life as we know it.

Earth has the right composition of atmospheric gases to sustain life. It has the right balance between oxygen and nitrogen so that plants and animals can breathe, and just enough carbon dioxide and methane to keep the planet warm enough so that life can be sustained. Many scientists think that it is the life on Earth that keeps the atmosphere in this perfect balance. Plants produce oxygen and re-circulate carbon dioxide on Earth. They believe that if life were to disappear from Earth, the atmosphere would become like Mars or Venus.

Investigation A model of the greenhouse effect

In the greenhouse effect, carbon dioxide traps the heat of the Sun. In this investigation, you will use bottles with air and carbon dioxide, respectively, to model the greenhouse effect. You are going to investigate the following question: Does air or carbon dioxide absorb more heat?

Aim

Write an aim for this investigation.

Hypothesis

Write a hypothesis for this investigation.

Materials and apparatus

- two glass bottles or clear cold drink bottles with lids
- 2 thermometers
- Prestik
- heat source (two study lamps)
- vinegar
- bicarbonate of soda
- small cold drink bottle with lid

Method

Set up the experiment as in the photograph.

Figure 23.22: Experimental set-up

1. Mark one bottle as 'Air' and the other bottle as 'CO_2'.

2. If the lids do not have the thermometers in them already, prepared by your teacher, make a hole in each of the lids. You can do this using a hammer and nail and hammering the nail through the lid into a wooden block. Secure the thermometer in each lid. You can use Prestik to do this.

3. Fill the first bottle with air, secure the thermometer, and close the lid tightly.
4. Fill the second bottle with carbon dioxide:
   a) To collect a bottle of carbon dioxide, add one tablespoon of bicarbonate of soda to the small bottle.
   b) Add 10–20 ml of vinegar and place the lid back on.
   c) Hold the mouth of the small bottle over the large CO_2 container and pour the CO_2 collecting in the small container into the large container.
   d) Hold the small bottle horizontal so that the vinegar does not spill into the bigger bottle; only the heavier carbon dioxide gas pours into the large container.
   e) Add more vinegar when the effervescence stops. Repeat 2–3 times until the bottle is full. If a burning match at the mouth of the bottle goes out immediately, the bottle is full.
   f) Secure the thermometer and close the lid tightly.

Figure 23.23: Pour carbon dioxide from the small bottle into the large bottle. Your teacher will prepare the carbon dioxide for you.

5. Measure and record the starting temperature of both bottles.
6. Switch on the heat source and measure the temperature increase in both bottles. You need to decide for yourself what time increments are appropriate and record these in the table.

Figure 23.24: The CO_2 container with the light positioned to shine on it.

Results

Copy and complete the table below in your exercise books.

Time (minutes)	Temperature of air bottle (°C)	Temperature of CO_2 bottle (°C)

Represent your results by drawing a graph for each of the experiments to show how the temperature for each bottle changed over time. You need to decide what values to use for each axis. Label the axes clearly and provide a heading for each graph.

What have you observed?

Conclusion

What do you conclude for your experiment?

Extension investigation

What factors make the temperature of the atmosphere increase faster?

Design your own investigation to answer one or more of the following questions. Use the experiment above to guide your experimental set-up.

1. Does dark soil make the temperature increase more quickly?
2. Does water vapour make the temperature increase more quickly?
3. Does the thickness of the layer of gases make the temperature increase more quickly?
4. Does the presence of dust/aerosols make the temperature increase more quickly?
5. Does the distance of the Sun make the temperature increase more quickly?

Global warming

What do you think will happen if the levels of carbon dioxide and other greenhouse gases increase? Think about what you discovered in the last investigation and look at the diagram of the greenhouse effect again. Write down your answer.

If there are more greenhouse gases in the atmosphere, more ultraviolet radiation will be trapped and the Earth will heat up. This will result in more of the polar ice melting than usual. Even a difference of one degree in the average temperature has an effect on the melting of polar ice. If more ice than usual melts, the water levels in the oceans will rise and low-lying areas could flood.

Take note

An **ice core** is a core sample from the accumulation of snow and ice over many years that have recrystallised and have trapped air bubbles from previous time periods

A change in the temperature will also result in a change in weather patterns.

More rain will fall in some areas, and less in others. If this change is permanent, it is called **climate change**, and if it occurs on a worldwide scale, it is called global climate change, which is what is being discussed here.

Global warming affects weather patterns, which in turn has a knock-on effect on agriculture and food production. The impact on food production can lead to food shortage for humans and animals. Long-term climate change can lead to the extinction of plants and animals, which are unable to adapt to changed conditions.

The levels of greenhouse gases vary naturally over time. A question that scientists often ask is whether the concentration of greenhouse gases is rising more than it would naturally as a result of human activities. How do you think this can be investigated?

Since the industrial revolution humans have burned more fossil fuels than ever before. Human activities have resulted in the increase of carbon dioxide emissions over time. Carbon dioxide is therefore the main greenhouse gas under discussion amongst scientists and environmentalists. The following investigation will look at the levels of carbon dioxide over thousands of years.

Take note

The expression 'ppm' is short for 'parts per million'.

Investigation Ice core analysis

Carbon dioxide is trapped in the ice which forms at the poles. As the ice is compacted and becomes thicker over thousands of years, the carbon dioxide remains trapped. The levels of carbon dioxide in ice can be determined by analysing the ice cores. A research team in Antarctica drilled an ice core containing ice from 160 000 years ago. They analysed the ice for carbon dioxide and presented their data in the following table.

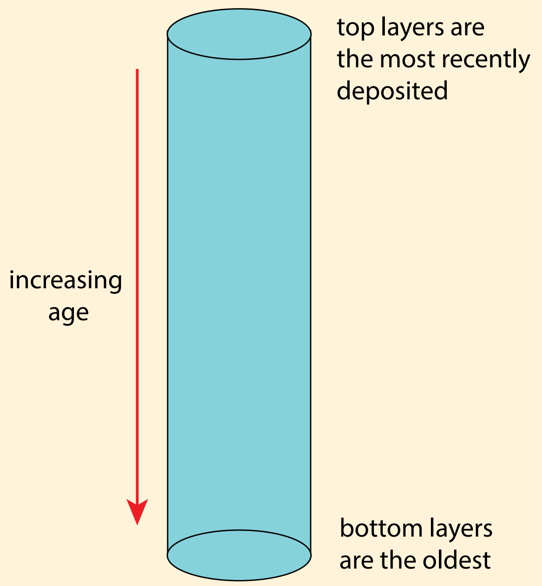


Figure 23.25: Ice cores trap carbon dioxide over time periods.

Figure 23.26: Drilling through the ice to obtain ice cores.

Figure 23.27: Sawing through the ice core to obtain samples for analysis.

Results from the ice core analysis.

Number of years ago	CO_2 levels (ppm)	Number of years ago	CO_2 levels (ppm)
160 000	190	70 000	250
150 000	205	60 000	190
140 000	240	50 000	220
130 000	280	40 000	180
120 000	278	30 000	225
110 000	240	20 000	200
100 000	225	10 000	260
90 000	230	8 160	280
80 000	220	0	387

Investigative question

Write down an investigative question for this study.

Analysis

1. Draw an accurate graph to represent your data. You need to choose your own set of axes, and label them appropriately.
2. What is the link between the levels of CO_2?

Conclusion

1. Write down a conclusion for this investigation.
2. What is the impact of global warming on the planet?

Summary

Key concepts

- The layer of gases around the Earth is called the atmosphere.
- The density of the gas molecules decreases as the distance from the Earth increases – the further away from the Earth you travel, the fewer gas molecules there are.
- The atmosphere can be divided into different layers – the troposphere, stratosphere, mesosphere and thermosphere.
- The exosphere is the uppermost layer directly above the thermosphere, where the gases thin out and the atmosphere merges with space. It is considered part of outer space.
- The troposphere is the densest layer, has the highest air pressure, and is closest to the surface of the Earth. It is on average about 12 km thick, and temperature decreases with altitude.
- Meteorites usually burn up in the mesosphere. Temperature decreases with altitude from 0 °C to –90 °C
- The thermosphere stretches up to 480 – 600 km. It absorbs ultraviolet light and X-rays. Temperature increases with altitude and can reach 1 500 °C.
- The ionosphere is the layer where molecules are ionised by the Sun's ultraviolet light. Radio waves can be transmitted and reflected as a result of the ionised layer.
- The greenhouse effect is a natural phenomenon – it warms the atmosphere sufficiently to sustain life.
- Greenhouse gases trap the re-radiation from Earth's surface and reflect it back to the Earth (like inside a greenhouse).
- The most common greenhouse gases are carbon dioxide, water vapour and methane.
- An increase in greenhouse gases leads to global warming.
- Global warming is an increase in the average temperature of the atmosphere.
- Global warming is a potentially life-threatening situation on Earth. It can lead to climate change, rising sea levels, food shortages and the extinction of organisms on Earth.
- The stratosphere stretches from 12 – 50 km and contains the ozone layer.
- Aeroplanes fly in this layer because the air is more stable. Temperature increases with altitude, from –60 °C to 0 °C.
- The mesosphere stretches between 50 – 80 km. The air is very thin.

Concept map

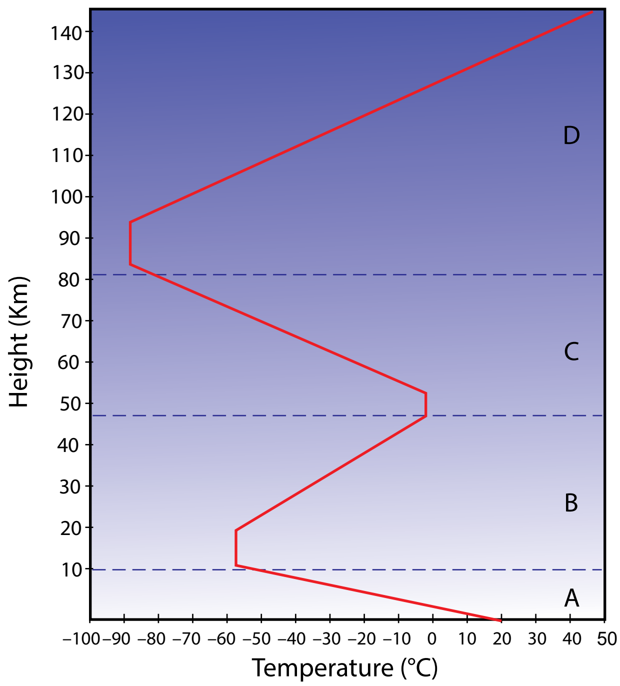
Through the past 2 – 3 years you have come across concept maps in Natural Sciences. Use what you know about concept maps and design a map for this unit. You must add terms and examples to the list. Remember to use linking words between concepts, and arrows to indicate the direction in which information is read. Plan your concept map on rough paper first before drawing the final one into your workbook. Use the following terms to help you with your map:

- atmosphere
- layers
- mesosphere
- thermosphere
- troposphere
- stratosphere
- weather
- ozone
- satellites
- radio waves
- global warming
- greenhouse gases
- greenhouse effect
- oxygen
- carbon dioxide
- water vapour

The atmosphere

Revision

1. The following graph shows the variation in temperature as you move further away from the Earth. Study it and answer the questions that follow.



a) Give labels for A-D, the layers of the atmosphere. [4]
b) Describe the temperature change in each of the layers. [4]
c) Explain why the temperature changes as you move further away from the Earth in Layer A? [2]
d) In which layer is the density of gas the highest? Give a reason for your answer. [2]
e) In which layer(s) can life survive? Give two reasons for your answer. [3]
f) In which layer are satellites found? Write only A, B, C or D. [1]
g) In which layer are meteors found? Write only A, B, C or D. [1]
h) In which layer radio waves reflected? Write only A, B, C or D. [1]
i) In which layer is weather observed? Write only A, B, C or D. [1]
j) In which layer is the aurora found? Write only A, B, C or D. [1]
k) In which layer do jet aeroplanes travel? Write only A, B, C or D. [1]
l) In which layer are lightning and storms found? Write only A, B, C or D. [1]
m) In which layer is ozone found? Write only A, B, C or D. [1]

2. Venus and Mars contains equal amounts of carbon dioxide, yet the temperature on the surfaces of these two planets are very different. Explain why. [4]

3. Earth is the only planet that we know of that sustains life. What makes Earth's atmosphere suitable to sustain life? [2]

4. Scientific evidence seems to point to the fact that carbon dioxide levels have increased steadily over the past 200 years.
   a) Why would the levels of carbon dioxide have been increasing over the past 200 years? [2]
   b) What is global warming? [1]
   c) What are the long term effects of an increase in carbon dioxide on life on Earth? [4]

Total [36 marks]

24 Birth, life and death of a star

Key questions

- Where are stars born?
- Can we talk about a star as 'living'?
- How long do stars like the Sun live?
- How do stars spend most of their life?
- Why are stars different colours?
- How do stars die?

Keywords

- stellar
- evolution
- nebula
- protostar
- constellation
- nuclear fusion
- stellar wind

Stars do not live forever, just like people. Stars are born, live their lives, changing or *evolving* as they age, and eventually they die. Often stars do this in a much more spectacular way than humans do!

Scientists speak of **stellar evolution** when talking about the birth, life and death of stars. The lifetime of individual stars is way too long for humans to observe the evolution of a single star, so how do scientists study stellar evolution? This is possible as there are so many stars in our galaxy, so we can see lots of them at different stages of their lives. In this way, astronomers can build up an overall picture of the process of stellar evolution. In this unit you will discover how stars are born, how they evolve, and how they die.

24.1 The birth of a star

Stars are born in vast, slowly rotating, clouds of cold gas and dust called nebulae (singular **nebula**). These large clouds are enormous, they have masses somewhere between 100 thousand and two million times the mass of the Sun and their diameters range from 50 to 300 light years across.

Take note

The collapse of a star can be triggered when the cloud is squeezed. For example, if a cloud passes through a spiral arm in a galaxy, it will be slowed down and compressed. This explains why lots of stars are formed in the spiral arms of galaxies.

Figure 24.1: The 'Pillars of creation'. These giant, dense dusty clouds of hydrogen gas are vast stellar nurseries where new stars are born. (NASA)

A famous example of one of these huge clouds is the Orion nebula in the constellation of Orion. It is visible with the naked eye if the sky is dark enough. These clouds are so massive that they can collapse under their own gravity if they are disturbed.

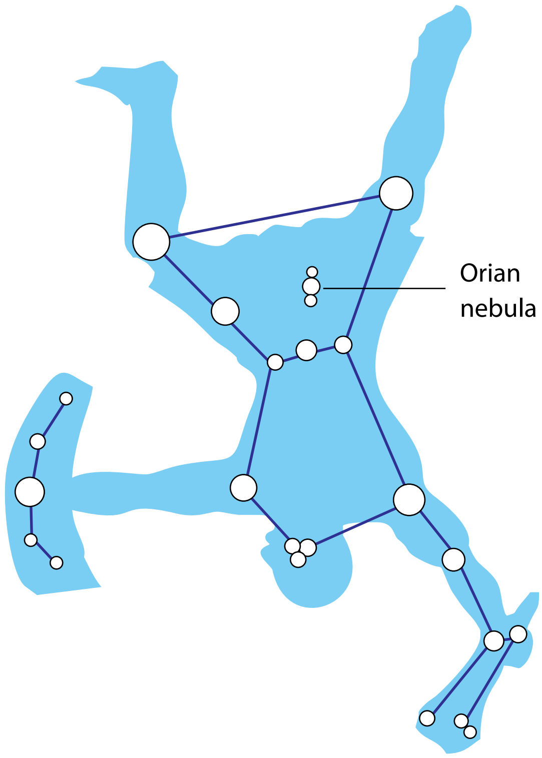


Figure 24.2: The constellation of Orion as viewed from the southern hemisphere. The Hunter Orion is 'upside down' when viewed from the south, and his sword lies above the three stars in his belt. The jewel in his sword which looks like a white-pink smudge is the Orion nebula.

Over time the clouds contract, become denser, and slowly heat up. The clouds also break up into smaller clumps. As the clumps get smaller they begin to flatten out into a disk shape. The centre of each clump will eventually contain a star and the outer disk of gas and dust may eventually form planets around the star.

This diagram shows how the stars make up the constellation of Orion, as seen in the southern hemisphere.

Figure 24.3: Hubble Space Telescope Image of the Orion Nebula showing different protostars surrounded by a dark disk of gas and dust. These disks (called protoplanetary disks) may eventually form planets around the star.

Take note

A light year is the distance that light travels in one year. Light travels extremely fast at 299 792 458 m/s. One light year is equivalent to 10 trillion kilometers

Take note

Do you remember that we learned about nuclear reactions last term in Energy and Change when looking at nuclear power plants?

As the contracting clump continues to heat up, a **protostar** is formed at the centre. A protostar is a dense ball of gas that is not yet hot enough at the centre to start nuclear reactions. This stage lasts for roughly 50 million years.

As the collapse continues, the mass of the protostar increases, squeezing it further and increasing the temperature. If the protostar is massive enough for the temperature to reach 10 million degrees Celsius, then it becomes hot enough for nuclear reactions to start and the protostar will technically be referred to as a star.

Not as well-known as its star formation cousin Orion, the Corona Australis region, with the Coronet cluster at its centre, is one one of the nearest and most active star formation regions to us. The image on the right shows the young stars at the centre, with gas and dust emissions.

Figure 24.4: The Coronet cluster.

The young star starts converting hydrogen to helium via **nuclear fusion** reactions. Nuclear reactions in stars produce vast amounts of energy in the form of heat and light, which is radiated into space. This energy production prevents the star from contracting further. As the star shines, the disk of dust and gas surrounding the star is slowly blown away by the star's **stellar wind** which leaves behind any planets if they have already formed.

Did you know?

Just as the Sun loses particles into space in the form of the solar wind, other stars also have winds, called stellar winds.

Figure 24.5: A large bubble of hot gas rising from glowing matter in a galaxy 50 million light years from Earth. Astronomers suspect the bubble is being blown by stellar winds, released during a burst of star formation.

Take note

In the upper left of the image of Large Magellanic Cloud, you can see a collection of blue and white young stars. They are extremely hot and are some of the most massive stars known anywhere in the Universe.

Figure 24.6: Star formation in the nearest galaxy outside the Milky Way, called the Large Magellanic Cloud (LMC), taken with the Hubble Space Telescope. This image shows glowing gas, dark dust clouds and young, hot stars.

24.2 Life of a star

This section covers the main stages of a star's life, from infancy to old age. Learners will also discover why stars do not all look the same and why they evolve at different rates and have different lifetimes: it is a consequence of having different masses. They will learn how important the mass of a star is in determining its evolution and observable characteristics.

A star is considered to be 'born' once nuclear fusion reactions begin at its centre. Initially hydrogen is converted to helium deep inside the star. A star that is converting hydrogen to helium is called a **main-sequence star**. Stars spend most of their lives as main-sequence stars, converting hydrogen to helium at their centres or cores. A star may remain as a main-sequence star for millions or billions of years.

Main-sequence stars are not all the same. They have different masses when they are born, depending on how much matter is available in the nebula from which they formed. These stars can range from about a tenth of the mass of the Sun up to 200 times as massive. Stars of different mass have different observable properties.

Keywords

- main-sequence star
- red giant star

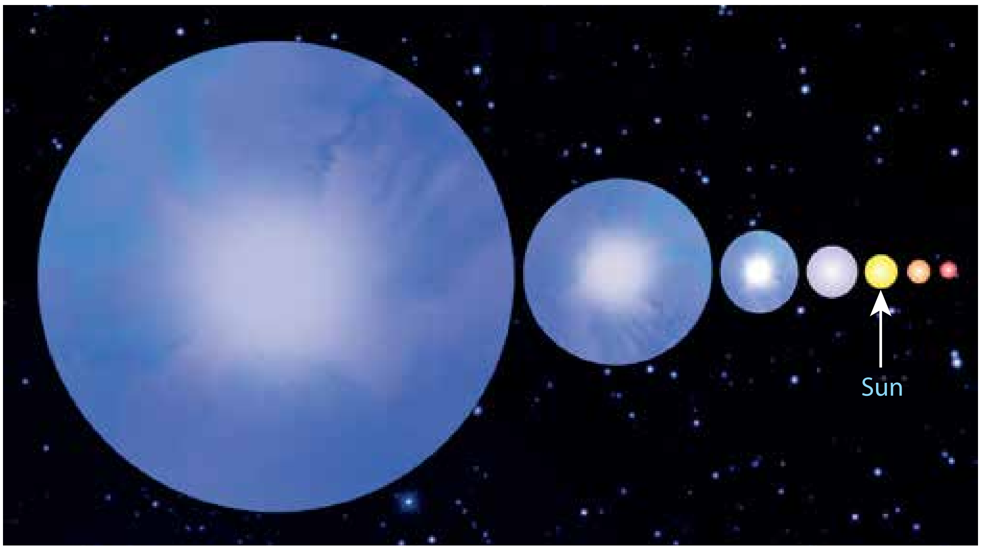


Figure 24.7: Main sequence stars come in different sizes and colours. Their sizes range from around 0,1 to 200 times the size of the Sun. Their surface temperatures determine their colours and can range from under 3 000 ˚C (red) to over 30 000 ˚C (blue).

Did you know?

Most of the stars in the Universe, about 90%, are main-sequence stars. The Sun is a main-sequence star.

Main-sequence stars also have different colours, depending on the temperatures of their surfaces.

Look at the following picture and correctly label the temperatures of all the stars using the list of temperatures below. Which star represents our Sun?

Temperature list: 3 000 ˚C, 4 500 ˚C, 6 000 ˚C, 10 000 ˚C, 40 000 ˚C

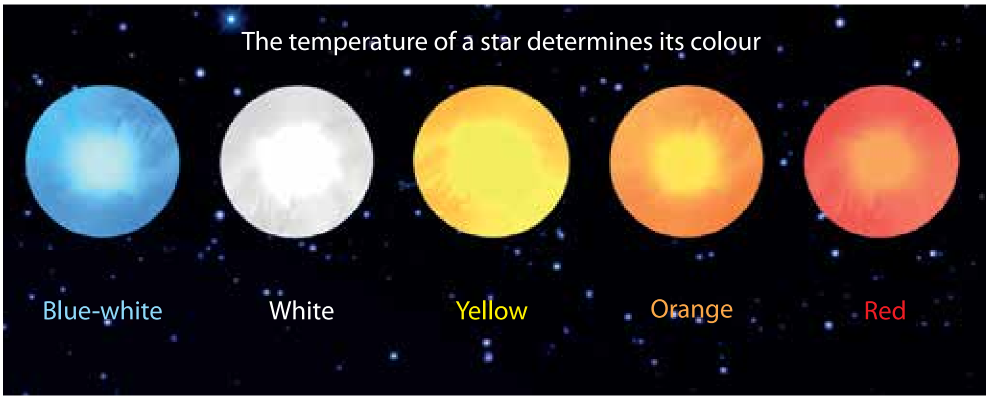


Figure 24.8: The termperature of a star determines its colour

Why are hotter stars bluer in colour? Can you remember what you learnt about the spectrum of visible light in Grade 8? The colour blue corresponds to light at shorter wavelengths (higher frequencies) than the colour red. Shorter wavelengths (higher frequencies) correspond to higher energies and thus hotter temperatures. This is also seen in the flames of a fire or candle. If you look at the flames, the central regions are bluer (and hotter) than the outer regions, which are orange and yellow.

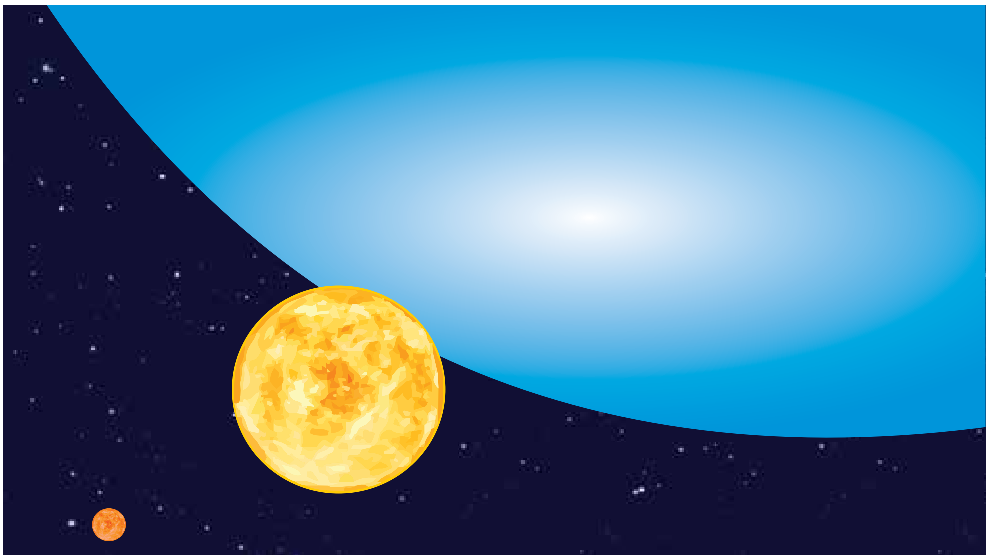


Figure 24.9: This artist's impression shows the relative sizes of young stars, from the smallest 'red dwarfs', at about 0,1 solar masses, low mass 'yellow dwarfs' such as the Sun, to massive 'blue dwarf' stars weighing eight times more than the Sun, as well as the 300 solar mass star named R136a1.

ACTIVITY Observing Orion in the spring sky

Orion is an easily recognisable constellation visible in cities as well as in dark skies. In this activity learners will have to look at the night sky to spot the constellation and identify the stars Betelgeuse and Rigel and note their difference in colour. Orion is up in the east from around 00:30 at the beginning of October, however, as the months progress, it rises earlier. By the beginning of December Orion is visible from around 20:30 in the east. If observing the constellation is unfeasible, you could ask learners to look at the image of the constellation in this unit instead.

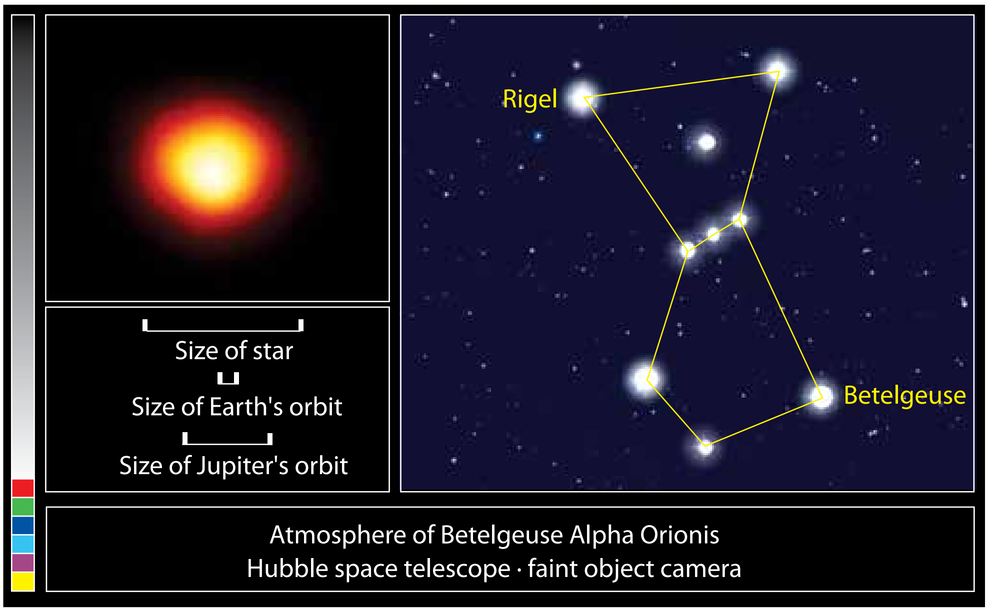


Figure 24.10: This is the first direct image of a star other than the Sun, made with NASA's Hubble Space Telescope. This is Betelgeuse, the star marking the shoulder of Orion, which we see in the bottom right of the constellation, when viewing Orion in the southern hemisphere.

Materials

- sky map

Instructions

1. A clear sky is necessary for this task. Look outside at night towards the east and identify the constellation of Orion. A photograph of the constellation is included in this unit for reference.
2. Identify the stars Betelgeuse and Rigel.

Questions

1. What did you notice about the colour of the two stars Betelgeuse and Rigel?
2. Why do you think the stars look different? Hint: Look back at the colours of stars in the diagram before this activity to see what this tells us about their temperatures.

Take note

At the beginning of October Orion is visible in the east from around 00:30 until morning. From the beginning of November Orion is visible in the east from around 22:30 and from the beginning of December it is visible in the east from around 20:30.

Did you know?

Betelgeuse is so huge that, if it replaced the Sun at the centre of our solar system, its outer atmosphere would extend past the orbit of Jupiter (see the scale at lower left of the image).

How long a main-sequence star lives depends on how massive it is. More massive stars move onto the next stages of their lives more quickly than lower mass stars. In fact, they are main-sequence stars for a shorter time than lower mass stars.

A higher-mass star might have more material, but it also uses up the material more quickly due to its higher temperature. For example, the Sun will spend about 10 billion years as a main-sequence star, but a star 10 times as massive will last for only 20 million years. A red dwarf, which is half the mass of the Sun, can last 80 to 100 billion years.

Red Giant Star

When the hydrogen in the centre of the star is depleted, the star's core shrinks and heats up. This causes the outer part of the star, the star's atmosphere, which is still mostly hydrogen, to start to expand. The star becomes larger and brighter and its surface temperature cools so it glows red. The star is now a **red giant star**. Betelgeuse, as you observed in the last activity, is a red giant star.

Why does a red giant glow red?

Why do you think red giants are called 'giant' stars?

Did you know?

Globular clusters are particularly useful for studying stellar evolution, since all of the stars in the cluster have the same age (about 10-15 billion years), but cover a range of stellar masses.

Figure 24.11: A colourful view of the globular star cluster NGC 6093 in the Milky Way, containing hundreds of thousands of ancient stars. Especially obvious are the bright red giants, which are stars similar to the Sun in mass that are nearing the ends of their lives.

Eventually the core of the star becomes hot enough for the next nuclear reaction to start: atoms of helium collide and fuse into heavier elements such as carbon and oxygen. However, eventually the helium in the core will also be depleted. From this point onwards, the fate of the star is determined by its mass.

For medium-sized stars, such as the Sun, the temperature in their centres will never get high enough to fuse the newly-formed carbon and oxygen into heavier elements and so they do not evolve much further. Following the red giant phase, the star becomes unstable and will eventually die, as you will discover in the next section.

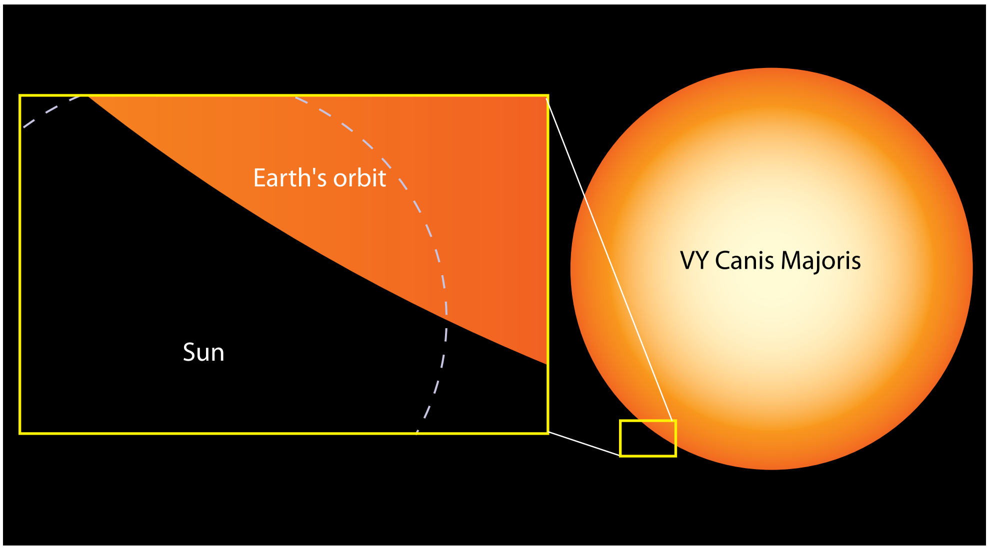


Figure 24.12: The relative sizes of the Earth, the present-day Sun, and a red super-giant star, *Canis Majoris*, in the constellation. The Sun will eventually evolve into a red giant star in about 4,5 billion years' time.

24.3 Death of a star

As a star enters the final stages of its life, after it has become a red giant, the star becomes unstable and expands and contracts over and over. This causes the star's outer layers to become detached from the central part of the star and they gently puff off into space. When the last of the gas in the star's outer layers is blown away, it forms an expanding shell around the core of the star called a **planetary nebula**. Planetary nebulae glow beautifully as they absorb the energy emitted from the hot central star. They can be found in many different shapes, as shown in the following images.

Keywords

- planetary nebula
- white dwarf
- black dwarf
- supernova
- neutron star

Figure 24.13: The beautiful Ring Nebula. The gas is lit up by the light from the central star which is the faint white dot in the centre of the nebula.

Figure 24.14: The Boomerang Nebula is a young planetary nebula and the coldest object found in the Universe so far.

Did you know?

The Butterfly Nebula is a dying star that was once five times the mass of the Sun. What resembles the butterfly wings are actually hot clouds of gas tearing across space at almost 1 million km an hour – fast enough to travel from Earth to the Moon in 24 minutes!

Figure 24.15: Kohoutek 4-55 Nebula contains the outer layers of a red giant star that were expelled into interstellar space when the star was in the late stages of its life.

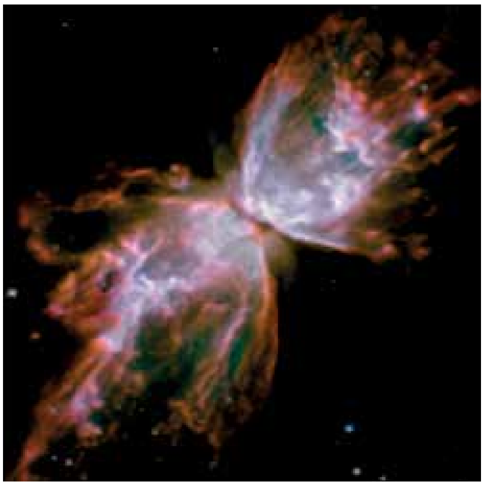


Figure 24.16: The Butterfly Nebula. The dying central star itself cannot be seen, because it is hidden within a doughnut-shaped ring of dust.

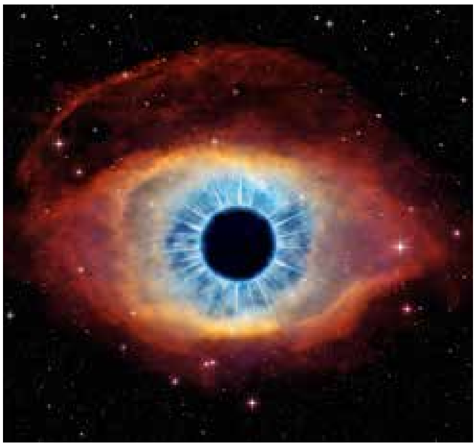


Figure 24.17: The Helix Nebula

Figure 24.18: The Dumbbell Nebula.

Some time after puffing off its outer layers, the central star will run out of fuel. When this happens, the central star begins to die. Gravity causes the star to collapse inwards, and the star becomes incredibly dense and compact, about the size of the Earth. The star has then become a **white dwarf** star.

Figure 24.19: An ultraviolet image of the Helix Nebula. As the star in the centre approaches the end of its life and runs out of fuel, it shrinks into a much smaller, hotter and denser white dwarf star.

White dwarfs have this name because of their small size and because they are so hot that they shine with a white hot light. The central parts of stars are much hotter than their surfaces, and a white dwarf is made from the remaining central parts of a star which explains why they are so hot.

The following image shows the relative size of Sirius B, a nearby white dwarf star, compared to some of the planets in our solar system. Stars and stellar remains can be smaller than planets.

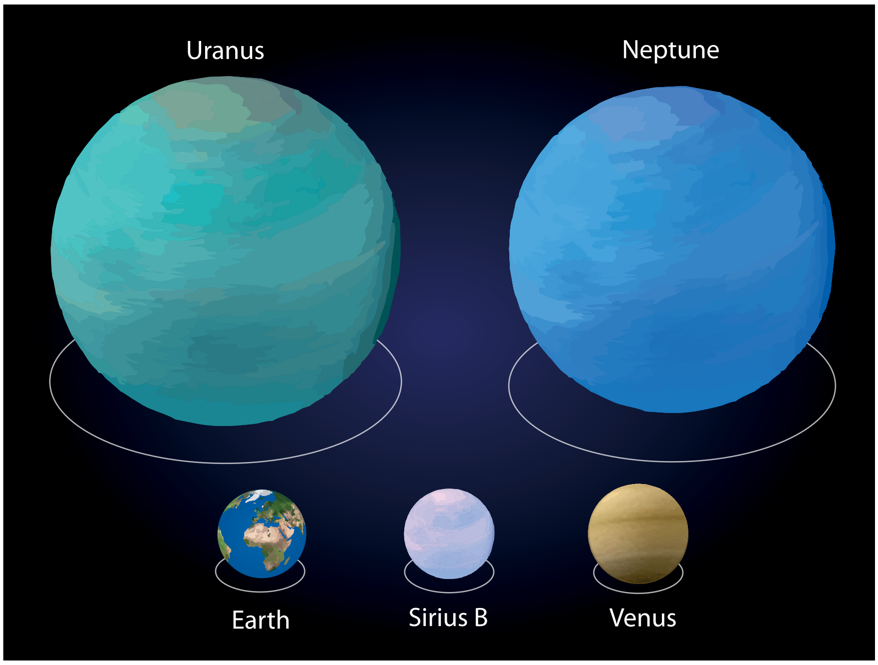


Figure 24.20: Comparitive size of Sirius B.

White dwarfs no longer produce energy via nuclear reactions so they radiate their energy into space in the form of light and heat. They slowly cool down over time. Eventually, once all of their energy is gone, they no longer emit any light. The star is now a dead **black dwarf** star and will remain like this forever.

Did you know?

White dwarf stars are so dense that one teaspoon of material from a white dwarf would weigh up to 100 000 kg.

ACTIVITY: Life cycle of a Sun-like star

Materials

- yellow round balloon – one per pair or group
- black marker
- red marker
- scissors
- 2 cm small white Styrofoam ball – one per pair

Instructions

1. In this activity you will work in pairs. One of you will instruct your partner using the instructions below. Your partner will follow your

instructions. Decide which of you will be the instructor and which of you will be the experimenter.

2. Experimenter: Insert the white Styrofoam ball into the deflated balloon
3. Instructor: Read out the step-by-step instructions from the table below (listed in order). First state the time from the star's birth which is given in the left hand column, then tell your partner what to do with the balloon.
4. Experimenter: Follow the instructions from your partner very carefully. You will be demonstrating how a Sun-like star evolves over time.

Step Number	Instructions
1) Star is born	Blow up the balloon to about 6 cm in diameter
2) 5 million years	Wait
3) 10 million years	Wait
4) 500 million years	Wait – planets are being formed around the star.
5) 1 billion years	Blow the balloon up a little bit
6) 9 billion years	Blow up the balloon some more and colour it red – it is now a red giant star
7) 10 billion years	Blow the balloon up a little bit. The outer layers are now being blown off. To simulate this, slowly allow the balloon to deflate. Cut the balloon into pieces and scatter them around the white ball. The star has now become a white dwarf (the ball) surrounded by a planetary nebula (the pieces of balloon).
8) 50 billion years	Move the planetary nebula farther away from the white dwarf.
9) 500 billion years	Remove the planetary nebula and colour the ball black – the star is now a black dwarf.

The different stages of evolution of a star like the Sun are summarised in the diagram below and compared to the lifecycle of a person.

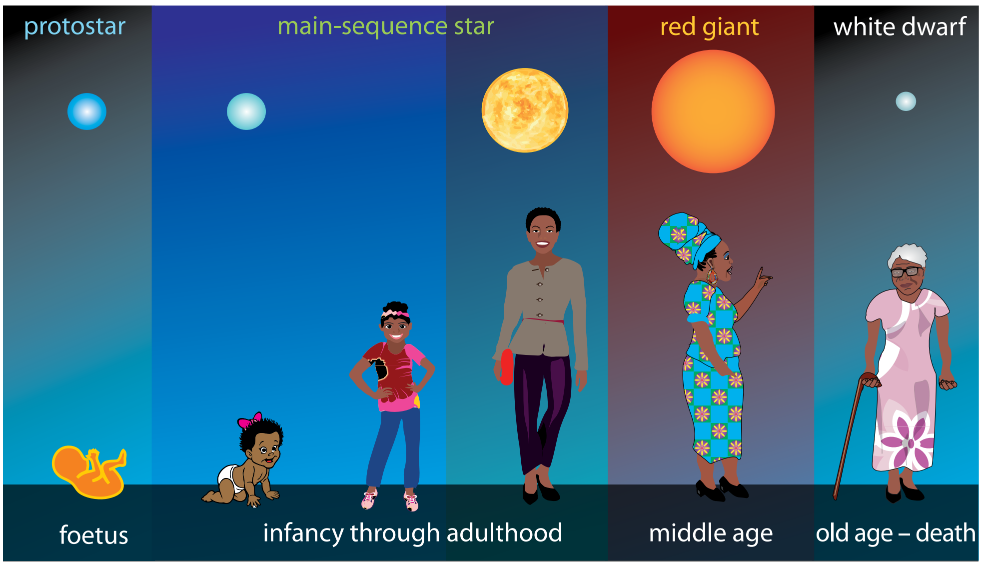


Figure 24.21: Evolution of a star.

Let's take a closer look at the life of our star, the Sun.

ACTIVITY The life cycle of the Sun

Instructions

1. The diagram below shows the life of our Sun. The Sun is a common type of star of average size and mass.
2. Copy and complete the sentences by filling in the gaps which summarise the evolution of our Sun over time.

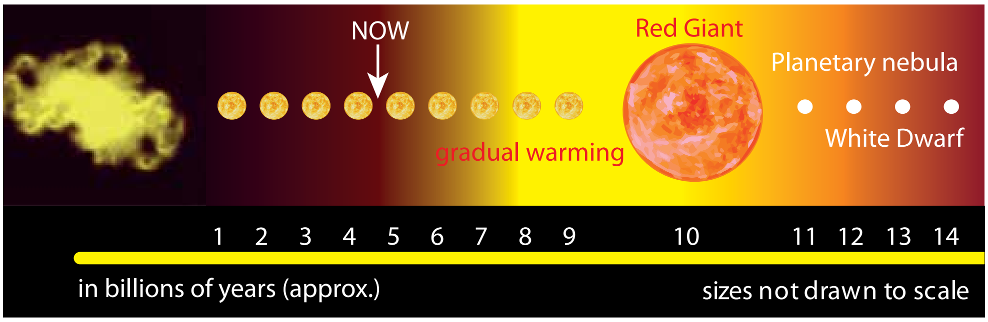


Figure 24.22: The life of our Sun.

Questions

1. The Sun is currently about halfway through its lifetime as a _______________star. In about 4.5 billion years' time the Sun will swell up to form a _______________star engulfing the Earth as it does so.
2. After the Sun has become a red giant, it will eventually become unstable and puff off its outer layers forming a beautiful _______________. The central core of the Sun will be left exposed in the centre of the planetary nebula.
3. Once the fuel runs out in the core of the Sun, nuclear reactions will_______________. The Sun will then have become a hot _______________star, left behind at the centre of the planetary nebula.
4. As there are no ongoing nuclear reactions, as the white dwarf shines, it slowly cools and will eventually form a _______________dwarf.

Take note

The plural of nebula is nebulae. Planetary nebulae have nothing to do with planets but were named like this in the 1700s because they resembled planets when observed with the telescopes of the time.

ACTIVITY Flow diagram poster showing the life of a Sun-like star

Materials

- paper or card for the poster
- pencils, crayons or paint for drawing
- printouts of photographs or pictures of the various stages in the Sun's life

Questions

1. Draw a flow diagram showing the key stages in a Sun-like star's life.
2. Include the birth, life, ageing and death of the star. If you have access to printouts of photos or drawings of the key stages, you could paste them onto the poster instead of drawing the key stages.

Take note

Here we are not talking about bigger stars, but rather stars that are more massive. It is not the size that counts, but the mass of the star.

3. Label each stage and indicate clearly with arrows the direction of flow in the evolutionary stages.
4. *Advanced*: Write down approximately how long each stage lasts. You can use the timeline of the Sun's evolution in this unit to help you.

Questions

1. Where are stars born?
2. Why is a red giant so named?
3. What kind of stellar remnant is left behind once a star like the Sun dies?
4. What is a planetary nebula?
5. How big is a white dwarf?

So far we have looked at stars that are about the same mass as our Sun. But what about stars that are more massive? How do they die?

Stars more than eight times the mass of the Sun end their lives spectacularly.

When the hydrogen at their cores becomes depleted, they swell into red super-giants which are even larger than red giants.

A red super-giant can fuse successively heavier and heavier elements for a few million years until its core is filled with iron. At this point, nuclear reactions stop and the star collapses rapidly under its own gravity. The collapsing outer layers of the star hit the small central core with such a force that they rebound and send a ripple outwards through the star, blowing the outer layers of the star into space in a huge explosion called a **supernova**.

Did you know?

Japanese and Chinese Astronomers recorded the violent supernova event that led to the Crab Nebula nearly 1 000 years ago in 1054 AD.

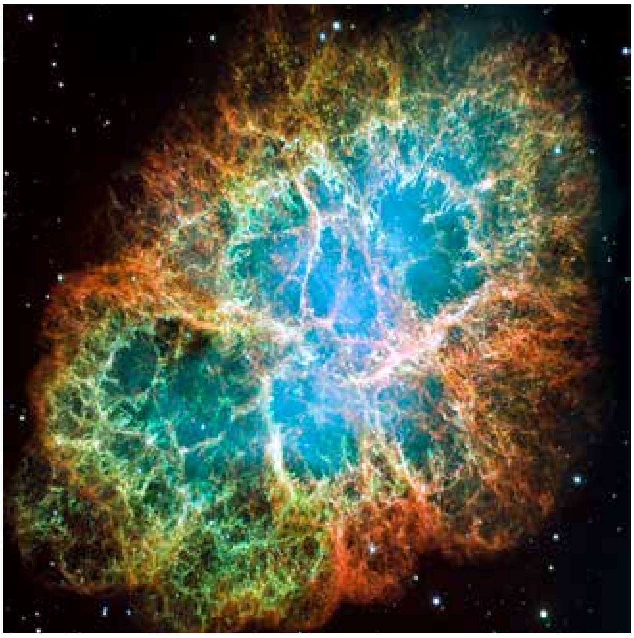


Figure 24.23: The Crab Nebula. This giant glowing cloud of gas is the remains of the outer layers of a star that exploded in a supernova explosion. In the centre is a rapidly spinning neutron star.

For a week or so, a supernova can outshine all of the other stars in its galaxy.

However, they quickly fade over time. The central star left behind is either made of neutrons and it is called a **neutron star**, or if the initial star was really massive, a **black hole** forms. The leftover neutron star or black hole is surrounded by an expanding cloud of very hot gas.

In February 1987, astronomers observed a supernova explosion, called Supernova 1987A. It is one of the brightest stellar explosions observed since the invention of the telescope 400 years ago. The supernova belongs to the Large Magellanic Cloud, a nearby galaxy about 168 000 light-years away. Even though the stellar explosion took place around 166 000 BC, its light arrived here less than 25 years ago.

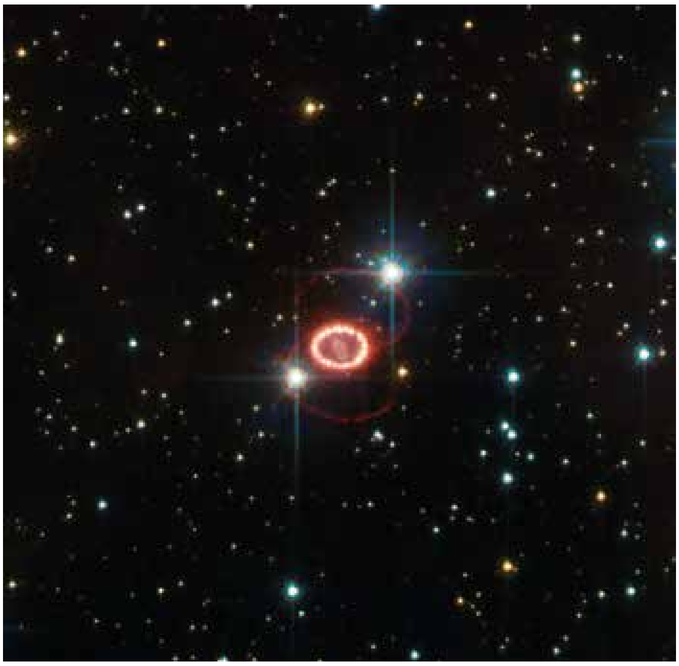


Figure 24.24: An image of the supernova called Supernova 1987A. The outer layers of the star have formed beautiful rings expanding into space.

Supernovae have also been observed previously with the naked eye before the invention of the telescope. On 9 October 1604, sky watchers, including astronomer Johannes Kepler, spotted a 'new star' in the sky. Now, we have images of the remnants of the supernova and know that it is not a new star, but rather the death of a massive star.

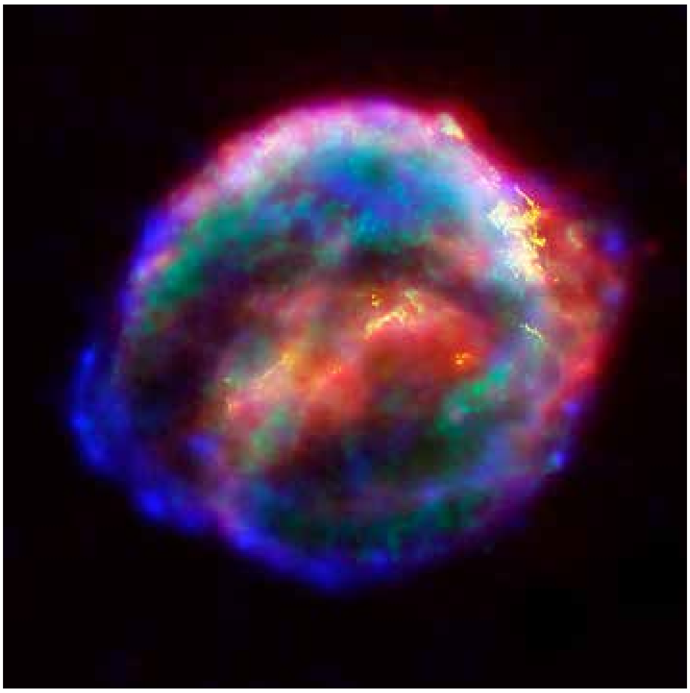


Figure 24.25: The remnants of Kepler's supernova. The explosion was observed in 1604.

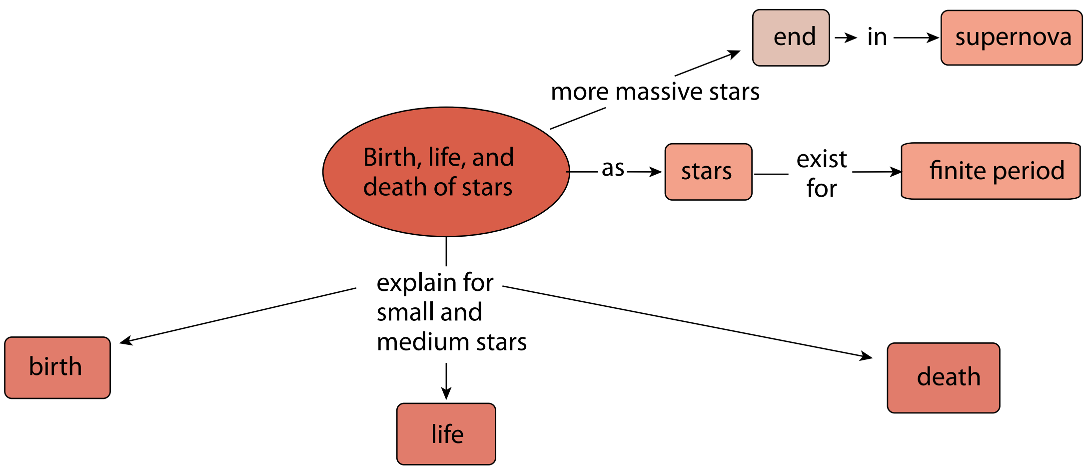
Summary

Key concepts

- Stars are born in giant, cold clouds of gas and dust called nebulae.
- A star is born once it becomes hot enough for fusion reactions to take place at its core.
- Stars spend most of their lives as main-sequence stars fusing hydrogen to helium in their centres.
- The Sun is halfway through its life as a main-sequence star and will swell up to form a red giant star in around 4,5 billion years.
- Stars similar to the Sun end their lives as planetary nebulae and leave behind a small hot white dwarf star at the centre of the planetary nebula.

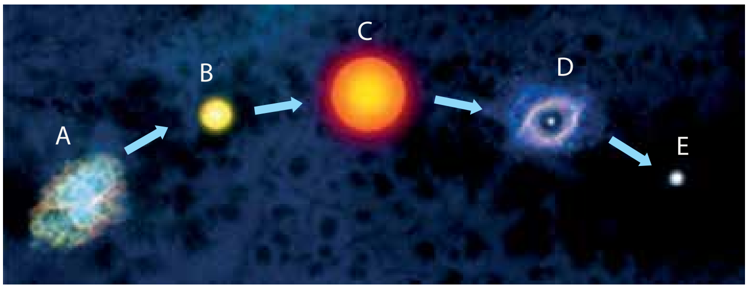
Concept map

The concept map on the life cycle of stars has been started, but you need to finish it by summarising the concepts for each stage, namely birth, life, and death of a star.



Revision

1. What is the name of the giant clouds where stars are formed? [1]
2. In the human life cycle, a foetus is the unborn baby in a mother's womb. What is the equivalent stage in a star's life called? [1]
3. Under what conditions do astronomers technically say a star has been born? [1]
4. Which star colour is hotter, white or yellow? [1]
5. What nuclear reaction does a main-sequence star undergo? [2]
6. Once the Sun has exhausted its hydrogen fuel supply, it will swell up to form what type of star? [1]
7. Low mass stars like the Sun eject their outer layers. What is the name of the object they form when they do this? [1]
8. What kind of star is left behind after a planetary nebula? [1]
9. What is the difference between a stellar nebula and a planetary nebula? [2]
10. Study the following diagram showing a star's evolution.



a) Copy the table below in your exercise books and provide labels for the different stages. [5]

Label	Stage
A	
B	
C	
D	
E	

b) What changes occur from stage B to form C? [2]
c) Sometime after puffing off its outer layers at stage D, the fuel of the central star will have become depleted. What causes the star to collapse inwards to become E? [1]
d) What eventually happens to the star after stage E? [1]

11. Massive stars die in powerful explosions. What are these explosions called? [1]

Total [21 marks]

Glossary

altitude height above sea level
atmosphere the layer of gases surrounding the Earth and held in place by gravity
aurora a natural phenomenon whereby charged particles from the Sun interact with atmospheric particles; this is observed as bright, coloured 'lights' in the sky, mostly in polar regions

bellows a device that produces a stream of air when it is squeezed
biosphere the part of the Earth and its atmosphere in which living organisms exist or that is capable of supporting life
black dwarf a white dwarf that has sufficiently cooled and used up all of its energy so that it no longer emits any heat or light; the star is now dead and will remain like this
blast furnace used in the extraction of iron from iron-ore, a high temperature oven in the form of a tower into which compressed air can be introduced from below
bloomery a type of oven used for purifying iron from iron ore
brittle hard, but can break or shatter easily

carbon dioxide a gas with the chemical formula CO_2
cementation the process of solidifying sediments by chemical compounds acting as glue
CFCs chlorofluorocarbons are molecules which release chlorine atoms due to solar radiation in the stratosphere
climate change a significant and lasting change in weather patterns; if there is a change in the world's weather patterns, it is a global climate change
compaction an increase in the density of something
composition what makes up a substance or a mixture
constellation a group of stars in a recognisable pattern
continental crust the thick part of the Earth's crust that forms the continents
cooling lowering the temperature
core innermost layer of the Earth
crust the thin, solid, outermost layer of the Earth
cycle a continuous process where the last step feeds into the first again

density separation a separation method where the differing densities of particles are used to separate them out
deposition the process where sediments, rocks and and sand are deposited (laid down) by wind or water

electromagnets a soft metal core made into a temporary magnet by passing current through a coil surrounding it
erosion the breakdown and movement of the Earth's surface by natural agents such as wind and water
evolution (of stars) the changes a star undergoes as it is born, lives and dies
excavation the process of removing rock containing ore from the surrounding rock
exosphere considered part of outer space; the uppermost layer directly above the thermosphere, where the gases thin out and the atmosphere merges with space
exploration the process of finding out where profitable mineral deposits are located
extrusive rock igneous rock which forms when magma flows out onto the surface of the Earth as lava

flotation a separation method by which hydrophobic particles are separated from hydrophilic particles by blowing air through the mixture

geochemical methods exploration methods using knowledge of geology and the chemistry of minerals
geophysical methods exploration methods using knowledge of geology and the physical properties of minerals
geosphere the core, mantle and crust of the Earth
global warming a gradual increase in the temperature of the Earth's atmosphere

greenhouse effect the trapping of the Sun's energy in the lower part of the atmosphere due to the presence of greenhouse gases

greenhouse gases gases such as water vapour, carbon dioxide and methane, which let through sunlight but reflect ultraviolet radiation

hydrosphere all the water, in all its forms, found on Earth

igneous rock a rock type formed by magma or lava

International Space Station a multinational space station, used for research purposes, which orbits the Earth at 370 km above the surface

intrusive rock igneous rock which forms from magma deep below the surface of the Earth

ionosphere the region mainly in the thermosphere where high energy light (UV light) can cause atoms, molecules or substances to form an ion or ions, typically by removing one or more electrons

lithosphere the outer part of the Earth consisting of the crust and the upper part of the mantle; it includes all rock, soil and minerals found on Earth

magnetic separation a separation method based on the magnetic properties of the mixture

main-sequence star a star that has hydrogen undergoing nuclear fusion reactions into helium in its core

mantle the middle layer of the Earth

melting the change from a solid to a liquid as a result of heating

mesosphere the layer of the Earth's atmosphere above the stratosphere, extending to about 80 km above the surface of the Earth

metamorphic rock a rock type formed through the transformation, or metamorphosis, of other rock types

meteor a small body of mass entering the Earth's atmosphere from space which emits light as a result of friction and heat, and appears as a streak of light

meteorite a meteor which has collided with the Earth

methane a gas with the chemical formula CH_4

mineral a natural compound formed through geological processes; the term 'mineral' includes both a material's chemical composition and its structure

nebula a vast cloud of gas and dust in space

neutron star an extremely dense star made of neutrons about the size of a small town in diameter

northern lights the aurora in the northern hemisphere; also called the Aurora Borealis

nuclear fusion process in which two light nuclei of atoms combine to produce a heavier single nucleus, with a total mass slightly less than that of the total initial material, the difference in mass is radiated as energy

oceanic crust the thinner part of the Earth's crust that underlies the oceans

ore a naturally occurring solid material from which a metal or valuable mineral can be extracted

overburden the layer of rock or sand overlying a mineral deposit

ozone a gas molecule found in the stratosphere consisting of 3 oxygen atoms (O_3)

panning a separation method based on the density gradient of the mixture

PGM platinum group metals, which includes ruthenium, rhodium, palladium, osmium, iridium and platinum, and are elements on the Periodic Table

planetary nebula a cloud of gas (the remains of the original star's atmosphere) surrounding an old star; these have a confusing name because they actually have nothing to do with planets at all

protostar a contracting mass of gas that will become a star once it is hot enough for nuclear fusion to start

radiation the transfer of energy from a source that does not require physical contact or movement of particles

red giant star an old, bright, very big, cool star; main-sequence stars evolve to become red giant stars once the hydrogen in their cores has been depleted

rehabilitation an area is restored to certain specifications, for example an area that has been mined is rehabilitated by planting trees or grass

remote sensing gathering information from a distance, without making physical contact

sediment particles, for example those that arise from erosion and weathering, settle in layers

sedimentary rock a rock type formed from solidifying sediment

sedimentation the deposition and solidification of sediment

size separation a separation method based on the size of the particles

slag the waste product extracted from a blast furnace after extracting iron from iron-ore

slurry a watery mixture of solids and liquids

solidify becoming a solid

southern lights the aurora in the southern hemisphere; also called (Aurora Australis)

stellar of stars, such as a stellar nebula

stellar wind a flow of neutral or charged gas ejected from a star (the solar wind refers specifically to the stellar wind of our Sun)

stratosphere the layer of the Earth's atmosphere above the troposphere, extending to about 50 km above the surface of the Earth

supernova an explosion in a high mass star where the outer layers of the star are flung off into space

temperature gradient how much the temperature changes as height above sea level (altitude) increases

thermosphere the layer of the Earth's atmosphere above the mesosphere, extending from about 480 to 600 km above the surface of the Earth

topsoil the upper surface of the Earth consisting of a layer of vegetation and soil

troposphere the lowest layer of the Earth's atmosphere, extending from sea level to about 9 – 17 km

water vapour a gas with the chemical formula H_2O; water in its gaseous form

weathering the wearing away of rocks as a result of exposure to wind, water and ice

white dwarf a small, hot, very dense star that is the size of a planet